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φ, π, e & i

David Perkins



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Preface

This book is about the numbers φ , π , e , and i and the connections among them. My goals are to illustrate to students of calculus how the techniques that they learned can be used to discover intriguing and useful results, and to compile many of the most elementary proofs related to these numbers into a resource for teachers.

Reading the entire book requires knowledge of integration techniques (especially integration by parts), but I try to provide plenty of algebraic and geometric assistance so that an undergraduate student of calculus can read the book solo. With this in mind, I provide endnotes throughout, in case the reader is unable to work out some of the missing steps. They appear in the final section, which is called Extra Help.

Each chapter concludes with a series of exercises, all of which introduce new historical figures or content. The exercises are neither simply routine practice, nor are they intended to be overly puzzling. I outline the more challenging exercises in the form of guided tours with the hope that any motivated student who has read the text carefully will move fluidly through them.

In short, my hope is that this book will help students see how calculus leads to remarkable and elegant truths.

My thanks to the members of the Spectrum Board, led by Jim Tattersall and Jerry Alexanderson. While I was writing, Jim Tattersall worked through everything that I sent to him, “pencil in hand”, strengthening my resolve. I am deeply grateful to him.

My thanks as well to hard-working and highly organized Beverly Ruedi and Carol Baxter at the MAA, and to Woody Dudley, one of my writing heroes, who thoroughly copy edited the final draft.

In 2004, Richard Jacobson, chair of the Houghton College mathematics department, approved my special topics course that proved to be the

origin point of this book. Jake then approved two more goes in the ensuing two years. He trusted me so benevolently. I am not only thankful to him, but to the dozens of Houghton students who enrolled in the course. Without them, this dream would have died in its infancy.

About a decade later, Jay Stine, chair of the mathematics department at Misericordia University, allowed me the flexibility to teach a special topics course during which we read through my first draft. Like Jake, he trusted me, but with far less history to base his trust upon. And as at Houghton, I am not only grateful to Jay, but also to the ten students who took the course: Kyle Kogoy, Nicole Hunsinger, Alec Garrity, Malachi Treaster, Austin Galiardo, David Rudalavage, Melissa Bostjancic, Robert Knipper, and finally Lydia Watkins, who not only read through the document line by line, spending hours and hours talking to me about the details, but who also gave me a gigantic chocolate chip cookie to celebrate the book's acceptance by the Spectrum Board.

I owe a great debt to Misericordia students Mike Gottstein, Taylor Rupp, and Craig Siekierka, three students who were funded by the university to read through my second draft over the summer. In her double role as writing major and mathematics major, Taylor was particularly helpful, and I enjoyed talking to her about the proper use of commas. Hamilton College students Chenchen Zhao, Alexander Black, and Benjamin Oltsik also contributed when I asked for help, proofreading and writing programs and finding alternate solutions. My friend Patrick Barringer read my entire first draft, and gave me several key ideas that ultimately made this book much better. My thanks to all of you.

I dedicate this book to Michelle LaBarre, my spouse of eight years, who chuckled when I said that I was going to describe her as "my constant". It was a pleasant chuckle, but still, I won't pluck this low-hanging fruit. Instead, I will tell you what she often tells her riders as she trains them in the art of dressage: if the horse and rider are not generating energy, then there is no way to steer. I find that this is true in teaching and learning as well. I hope above all that this book brings energy to each one of its readers. Thank you, Michelle, for being an ever-present source of energy to me.

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CHAPTER 1



Timaeus (a character invented by the philosopher Plato) spoke to Socrates at great length in pursuit of the answer to, “Of what is everything made?” Two thousand years later, we still struggle with this question. At times, there seems to be no limit to what “everything” is, as we split the smallest particles into yet smaller.

Lacking a microscope, Timaeus appealed to the elegance of proportion and the evidence of his senses. “Separated from fire nothing would ever become visible,” he began, “nor would it become touchable without something solid, or solid without earth.” Having posited the fundamental nature of fire and earth, Timaeus immediately linked the two via mathematics. “It’s not possible for two things alone to be beautifully combined apart from some third: some bond must get in the middle and bring them both together. And the most beautiful of these bonds is that which, as much as possible, makes itself and the things bound together one, and proportion is suited by nature to accomplish this most beautifully.”

In fact, Timaeus argued that fire and earth should be bound not by just one other element but by two, air and water, and thus by two proportions:

$$\frac{\text{fire}}{\text{air}} = \frac{\text{air}}{\text{water}} \quad \text{and} \quad \frac{\text{air}}{\text{water}} = \frac{\text{water}}{\text{earth}} .$$

These mathematical relationships, Timaeus suggested, brought harmony to the four building blocks of creation.

Today we see that Timaeus was onto something; the stuff of which everything is made does manifest harmonic mathematical unity. One feature of the unity is the appearance, in multiple settings, of certain num-

bers that we call *constants*. We will discover our first constant, the golden ratio φ , by pursuing Timaeus's line of reasoning.

1.1 Of what is everything made?

Timaeus's link between the four basic elements and proportions prompted him to search for geometric solids that could represent the elements. He settled on four solids that can be built from exact duplicates of regular polygons so that all faces meet at a uniformly constant angle. The *tetrahedron's* four faces are equilateral triangles, and the faces of the *cube* are squares (Figure 1.1). Timaeus assigned these solids to fire and earth, respectively.

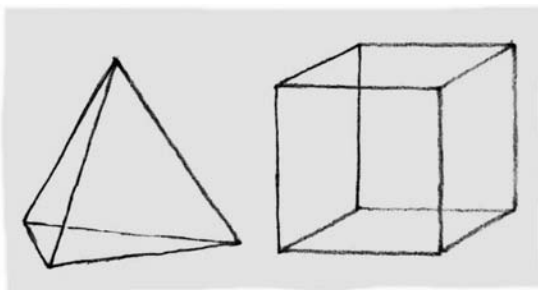


Figure 1.1

Earth gives rise to air in a simple but ingenious way. Locate the center point of each face of the cube, and connect each pair of neighboring centers with an edge, as in Figure 1.2. The *octahedron* thus formed has

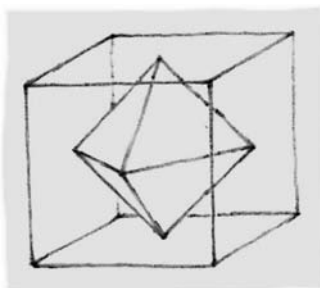


Figure 1.2

six corners, twelve edges, and eight triangular faces. (If we carry out this same process inside the octahedron, a new cube is born, and so on, forever.) Timaeus associated the octahedron with air.

Fire's tetrahedron spawns a smaller version of itself using this technique, so Timaeus was forced to search from scratch for a solid to represent water. The only other solid with triangular faces that meet at a uniformly constant angle has twenty faces. Called the *icosahedron*, this nearly spherical solid inspired Timaeus to associate it with a droplet of water. (Skip ahead to Figure 1.5 for a glimpse.)

Connecting the center of each face of the icosahedron with its five neighbors creates the last solid that has the properties of the other four, the *dodecahedron*, a twelve-sided solid with pentagonal faces (Figure 1.3). As with the cube-octahedron pair, the dodecahedron and icosahedron nest inside one another in an infinite embrace.

Having already linked the elements fire, earth, air, and water to the tetrahedron, cube, octahedron, and icosahedron, Timaeus assigned the dodecahedron to represent the entire cosmos. Such a metaphorically important shape surely merits closer investigation.

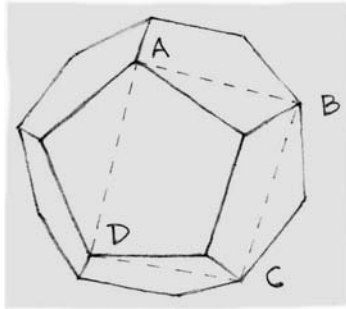


Figure 1.3

If we connect the centers of adjacent pentagonal faces of the dodecahedron, we create one edge of the icosahedron within. But if we instead connect a pair of nonadjacent corners *A* and *B* of a pentagonal face, as in Figure 1.3, then we create one edge of what will be a different solid contained within the dodecahedron. Continue connecting nonadjacent corners of the faces to create quadrilateral *ABCD*. The symmetry of the dodecahedron implies that *ABCD* is a square. Five more such identical squares follow by the same technique, resulting in a cube. Thus, the

dodecahedron contains both the icosahedron and the cube, or, poetically, from the cosmos we create water and earth.

It seems esthetically appropriate to stipulate that the length of each edge in the cosmos (that is, each edge of the pentagonal faces of the dodecahedron in Figure 1.3) is 1 unit. Now it is only natural to inquire about the edge lengths of the cube and the icosahedron that are inscribed within the dodecahedron. Earth first: each edge of the inscribed cube is a diagonal of one of the pentagonal faces (as Figure 1.3 makes clear), so consider the length of AD in Figure 1.4. Diagonal AD is parallel to

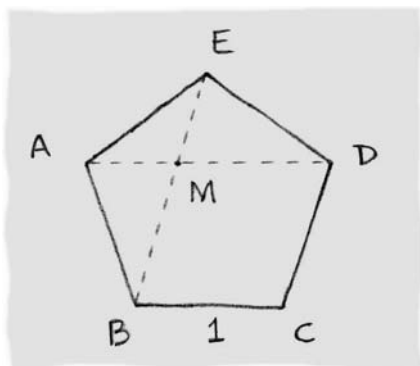


Figure 1.4

side BC because the pentagon is regular^[1] and so also is diagonal BE parallel to side CD . (As mentioned in the Preface, all marks such as [1] refer to the endnotes, which fill in mathematical details.) Thus $BCDM$ is a rhombus with sides of length 1. Because $MD = 1$, then, we know that $AM = AD - 1$. Now $AM = ME$ and $\angle BCD = \angle AME = \angle AED$, so

$$\triangle AME \sim \triangle AED, \text{ yielding } \frac{AM}{1} = \frac{1}{AD}.$$

Thus

$$AD \cdot AM = 1 \quad \text{or} \quad AD(AD - 1) = 1, \quad (1.1)$$

which by completing the square^[2] gives us

$$AD = \frac{1 + \sqrt{5}}{2}.$$

This number is the ratio of the cube's edge length to that of the dodecahedron that encloses it, and as such reflects the mathematical relationship of

earth to the cosmos in Timaeus's metaphor. This fact alone would make this number poetically intriguing, but this fact is by no means alone.

Before we investigate this constant further, let's name it. The sculptor Phideas, born one generation before Plato, is said to have made use of this pentagonal proportion in his works, which include the statue of Zeus at Olympia, one of the ancient world's wonders. The Greek letter phi (the first letter of Phideas) denotes this constant:

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.61803.$$

1.2 The golden rectangle

Consider the icosahedron (Figure 1.5). Five equilateral triangles meet at each vertex, and their bases form a regular pentagon. Choose any pair of neighboring vertices, such as A and B in Figure 1.5, and their opposites D and C . Set the length of each edge to 1 unit. Both AD and BC are diagonals of pentagons, and we saw in Figure 1.4 that if a regular pentagon has sides of length 1, then φ is the length of each of its diagonals. So the shape $ABCD$ in the figure is a rectangle with sides proportional to each other in the same ratio as the diagonal and the side of a pentagon. The icosahedron contains 15 such *golden rectangles*, each one meeting two others at right angles.

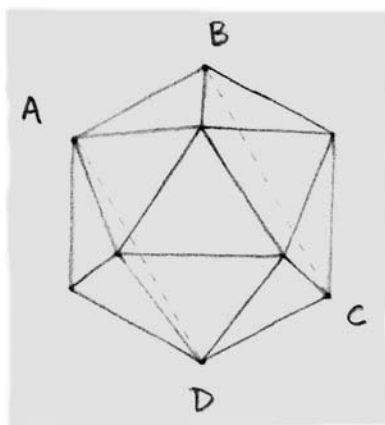


Figure 1.5

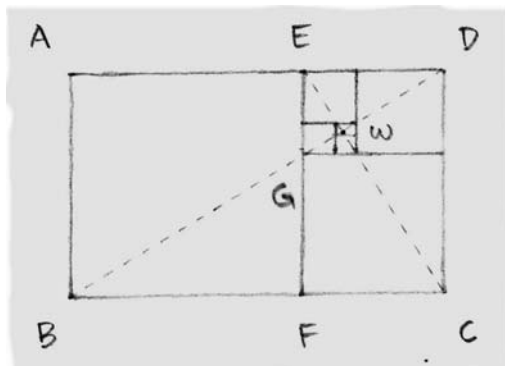


Figure 1.6

Lay rectangle $ABCD$ flat as in Figure 1.6. Let $AB = 1$, so $AD = \varphi$. Locate EF so that $ABFE$ is a square. From (1.1), we know that

$$\varphi(\varphi - 1) = 1, \quad \text{which may be written} \quad \frac{\varphi}{1} = \frac{1}{\varphi - 1}. \quad (1.2)$$

But this proportion applied to the figure yields

$$\frac{AD}{AB} = \frac{CD}{DE},$$

revealing that rectangles $ABCD$ and $FCDE$ are similar. We can now apply what we just did to the original rectangle to the smaller one, with similar results, and this effect will continue as the process is repeated. The figure thus created (Figure 1.6) is a sort of spiral composed of squares, curling inward toward a point. This intriguing point, which we will label ω (the Greek letter omega), must lie at the intersection of diagonals BD and CE thanks to the repetitive similarity^[3] of rectangles $ABCD$, $FCDE$, and so on. This point has been enthusiastically called “the Eye of God” by some of its admirers.

1.3 The Eye, and the arithmetic of φ

As ω is the fulcrum of spiraling squares inscribed on a golden rectangle that spans an icosahedron embedded in a dodecahedron that symbolizes the cosmos, it would be a shame not to spend some time thinking about its qualities. Along the way, we will use some interesting arithmetic properties of φ .

Sides AD and BC of rectangle $ABCD$ (in Figure 1.6) are parallel, so $\angle CBD = \angle BDA$, and $\angle B\omega C = \angle D\omega E$ as vertical angles. Thus we can deduce that $\triangle B\omega C \sim \triangle D\omega E$, which leads to

$$\frac{B\omega}{\omega D} = \frac{BC}{DE} = \frac{\varphi}{\varphi - 1}. \quad (1.3)$$

Back at (1.1), where we first revealed φ in the form of the pentagonal diagonal AD , we saw that

$$\varphi^2 - \varphi = 1, \quad \text{or} \quad \varphi^2 = \varphi + 1. \quad (1.4)$$

Dividing by φ yields

$$\varphi = 1 + \frac{1}{\varphi}, \quad (1.5)$$

so φ is one greater than its reciprocal. Now (1.3) may be rewritten

$$\frac{B\omega}{\omega D} = \frac{\varphi}{(1 + 1/\varphi) - 1} = \frac{\varphi}{1/\varphi} = \varphi^2.$$

So the fulcrum point ω cuts the diagonal of the golden rectangle into segments proportional to the square of φ .

How far is it from B to ω ? From $\omega D = BD - B\omega$, we have

$$\frac{B\omega}{BD - B\omega} = \varphi^2, \quad \text{or} \quad B\omega = BD \cdot \frac{\varphi^2}{1 + \varphi^2}.$$

By the Pythagorean theorem, we may replace BD with $\sqrt{1 + \varphi^2}$, so

$$B\omega = \sqrt{1 + \varphi^2} \cdot \frac{\varphi^2}{1 + \varphi^2} = \frac{\varphi^2}{\sqrt{1 + \varphi^2}} = \sqrt{\frac{\varphi^4}{1 + \varphi^2}}. \quad (1.6)$$

We pause to consider φ^4 . Starting with (1.4) and repeatedly multiplying both sides by φ gives

$$\begin{aligned} \varphi^2 &= \varphi + 1, \\ \varphi^3 &= \varphi^2 + \varphi, \\ \varphi^4 &= \varphi^3 + \varphi^2, \\ \varphi^5 &= \varphi^4 + \varphi^3, \end{aligned} \quad (1.7)$$

and so on. Each power of φ is the sum of the two previous powers. Now we use this list of observations to create a new list of identities^[4] in which

each is the sum of the previous two:

$$\begin{aligned}
 \varphi &= \varphi, \\
 \varphi^2 &= \varphi + 1, \\
 \varphi^3 &= 2\varphi + 1, \\
 \varphi^4 &= 3\varphi + 2, \\
 \varphi^5 &= 5\varphi + 3,
 \end{aligned} \tag{1.8}$$

and so on. Using this list, we may continue with (1.6) to find that

$$B\omega = \sqrt{\frac{\varphi^4}{1 + \varphi^2}} = \sqrt{\frac{3\varphi + 2}{\varphi + 2}}.$$

With the help of (1.8), we may multiply the numerator and denominator by $3 - \varphi$ to convert^[5] the denominator to a whole number:

$$B\omega = \sqrt{\frac{4\varphi + 3}{5}}. \tag{1.9}$$

While this is a concise expression for the distance to ω from the farthest corner of the golden rectangle, we can do better. Focus on $\triangle ACD$ in Figure 1.7, which depicts a regular pentagon. Earlier (see Figure 1.4), we concluded that if side $CD = 1$, then diagonal $AC = \varphi$. Drop altitude AM to bisect CD . Its length is

$$AM = \sqrt{\varphi^2 - (1/2)^2}$$

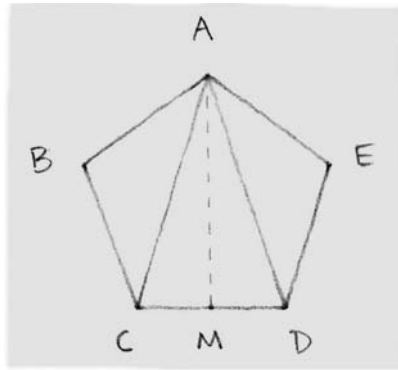


Figure 1.7

by the Pythagorean theorem. Simplifying, we see^[6] that

$$AM = \sqrt{(1/4)(4\varphi^2 - 1)} = \frac{1}{2}\sqrt{4\varphi + 3}.$$

Now AM is the side opposite $\angle ACM$ in $\triangle ACM$, so

$$\tan \angle ACM = \frac{AM}{MC} = \sqrt{4\varphi + 3}.$$

Because the pentagon is regular, we know that $\angle ABC = 108^\circ$, and thus that $\angle ACM = 72^\circ$. Back to (1.9), then, we see that

$$B\omega = \frac{\tan 72^\circ}{\sqrt{5}}.$$

While many of us learned early on the exact values of trigonometric quantities such as $\cos 30^\circ$ and $\sin 45^\circ$, not so many have gone on to discover such delights as $\tan 72^\circ = \sqrt{4\varphi + 3}$.

1.4 The Fibonacci (Hemachandra) sequence

Let's take a look at the (extended) sequence of coefficients

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots \quad (1.10)$$

on the right side of the expressions in (1.8). Each entry is the sum of the previous two. This sequence is named for **Leonardo Pisano** (Italy, born c. 1170), known in modern times by the nickname Fibonacci. As a boy, Fibonacci traveled to North Africa, where he learned the Hindu-Arabic numeral system, which used ten digits (much like 1, 2, 3, ..., 9, 0) as well as the concept of place value (so that, for example, 23 equals two tens and three ones). Fibonacci argued that this approach should replace the Roman numeral system that was in wide use in Europe. In a book on these topics, Fibonacci introduced the sequence (1.10) as the answer to a word problem.

Perhaps less well known is the description of this sequence by **Ācārya Hemachandra** (India, born c. 1088) about fifty years prior to Fibonacci's publication. Hemachandra discussed the sequence in connection with his interest in Sanskrit poetry. Sanskrit belongs in the family of the most ancient languages, with origins as distant as 1500 BCE. It is a highly structured language; the name "Sanskrit" translates as "refined".

Sanskrit syllables are either *long* or *short*, depending on a handful of rules defined around vowels. The vowel *a* has long and short versions, for example, and the Romanized transliterations of Sanskrit words use the usual bar over a long vowel, as in *āsana*. When speaking Sanskrit poetry, one draws out a syllable with a long vowel for two beats, so *āsana* is pronounced “aasana”, not quite as four distinct syllables, but instead as two beats and one beat and one beat. We can write

2 1 1
ā sa na

to show the number of beats above each syllable. So writing

1 1 2 2 2
ku ṭi lā kā rā

indicates that the first two syllables of the word *kuṭilākārā* are each spoken with the usual one beat, but the last three syllables are drawn out for two beats each when the word is recited aloud during a poem. So the word *kuṭilākārā* has five syllables but is spoken with eight beats, as in “kutilaakaaraa”.

Of course, this rhythmic consideration is of utmost interest to poets. A poet like Hemachandra, with a bent to mathematics, cannot help but ask how many ways that a line of poetry with a fixed number of syllables can be built of two-beats and one-beats. For example, the 16 syllables in the line

2 1 2 1 1 2 2 2 2 1 2 1 1 2 1 2
Kuṇ ḍa li ku ṭi lā kā rā sar pa vat pa ri kīr ti tā

(which, as an aside, translates “Kundali is said to be coiled like a serpent”) are spoken with a total of 25 beats. A poet who chooses to use this 25 beat structure must, by the formal rules of Sanskrit poetry, use it throughout the poem.

So Hemachandra asked himself: how many different 25 beat lines of poetry (like the one above) are possible? Listing them all would be tedious. Instead, Hemachandra reasoned that every 25 beat line must begin with either a 1 beat syllable or a 2 beat syllable, as in

1 + 24 more beats
2 + 23 more beats.

So if he knew how many 24 beat patterns and how many 23 beat patterns there were, then he could add the two totals to get the number of 25 beat patterns. Inspired by this insight, Hemachandra began with the simplest cases and built toward the goal of counting 25 beat patterns:

- 1 beat pattern: 1
 2 beat patterns: 11 and 2
 3 beat patterns: 111 and 12 and 21
 4 beat patterns: 1111, 112, 121, 211, 22.

To illustrate Hemachandra's strategy, note that if we ignore the first symbol in each of the 4 beat patterns, then what remains are exactly those patterns in the previous two cases (2 and 3 beats). So we can add the number of 2 beat patterns and 3 beat patterns to find the number of 4 beat patterns. In this way, Hemachandra produced the sequence

$$1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots$$

where the n th number in the list is the total number of n beat patterns. The 25th such entry is 121393, so this is the desired number of 25 beat patterns.

Exercise 2 explains a shortcut that allows us to calculate the number of n beat patterns directly, without having to calculate the number of $n - 1$ and $n - 2$ beat patterns. (Exercises are found at the end of each chapter.)

1.5 A continued fraction for φ

The numbers in the Fibonacci sequence appear as bit players in an infinitely nested fraction that equals the golden ratio. That *continued fraction* is

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

and we will warm up to its infinite aspect step by step.

If we begin to write the continued fraction, stop after one plus sign, and simplify, we find that

$$1 + \frac{1}{1} = \frac{2}{1}.$$

If we stop after two plus signs, we obtain

$$1 + \frac{1}{1 + \frac{1}{1}} = 1 + \frac{1}{2/1} = \frac{3}{2}.$$

Halting at the third plus sign yields

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}} = 1 + \frac{1}{3/2} = \frac{5}{3} .$$

These truncated fractions are called the *convergents* of the continued fraction. In each case, the convergent is the ratio of two consecutive Fibonacci numbers.

We may confirm this link between the convergents and the Fibonacci numbers using a *proof by induction*. Here we will not state the proof formally, but will instead explore our way into the heart of it. Some notation will be helpful: Let F_n be the n th number in the Fibonacci sequence

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots$$

and note that the Fibonacci numbers obey the rule

$$F_{n+2} = F_{n+1} + F_n \quad \text{for } n \geq 1. \quad (1.11)$$

Now define

$$S_0 = 1 \quad \text{and} \quad S_n = 1 + \frac{1}{S_{n-1}} \quad \text{for } n \geq 1.$$

This definition of S_n is *recursive*, meaning that we repeat the rule to generate a sequence of results. Here, it begins with

$$S_1 = 1 + \frac{1}{S_0} = 1 + \frac{1}{1} = \frac{2}{1} = \frac{F_3}{F_2} .$$

Next, we calculate that

$$S_2 = 1 + \frac{1}{S_1} = 1 + \frac{1}{2/1} = \frac{3}{2} = \frac{F_4}{F_3} \quad \text{and}$$

$$S_3 = 1 + \frac{1}{S_2} = 1 + \frac{1}{3/2} = \frac{5}{3} = \frac{F_5}{F_4} ,$$

whereupon we state a conjecture that captures this emerging pattern:

$$S_n = \frac{F_{n+2}}{F_{n+1}} . \quad (1.12)$$

Now if we assume that

$$S_{n-1} = \frac{F_{n+1}}{F_n}, \quad (1.13)$$

we can check the truth of (1.12) via

$$\begin{aligned} S_n &= 1 + \frac{1}{S_{n-1}} && \text{by our original recursive rule} \\ &= 1 + \frac{1}{F_{n+1}/F_n} && \text{by our assumption (1.13)} \\ &= \frac{F_{n+1} + F_n}{F_{n+1}} \\ &= \frac{F_{n+2}}{F_{n+1}} && \text{by rule (1.11).} \end{aligned}$$

So by assuming (1.13) we can prove (1.12). Such an argument is called a proof by induction, and it is valid because (1.13) is a safe assumption; after all, we verified it for several cases when we began.

We may now untether our imaginations and ponder

$$S_\infty = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \cdots}}} \quad (1.14)$$

where the subscript ∞ indicates that there are infinitely many plus signs. Now when one steps into the infinite, one must tread carefully. After all, in what sense could the number of plus signs be infinite? A rigorous way to approach (1.14) is given in the endnotes^[7] but here we simply appeal to our intuitive grasp of the infinite. The equation

$$\varphi = 1 + \frac{1}{\varphi} \quad (1.15)$$

from (1.5) contains a curious self-reference, in that the denominator φ on the right is the very same quantity that appears on the left. We may thus replace φ on the right with a copy of the entire right side of (1.15) to obtain

$$\varphi = 1 + \frac{1}{1 + \frac{1}{\varphi}},$$

whereupon we can for the same reason replace φ on the right with an entire copy of the right side. Nothing stops us from making this move repeatedly, or even forever (as long as your intuition allows for a process to loop forever). Doing this, we conclude that

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}} \quad (1.16)$$

gives us a continued fraction equal to the golden ratio.

Earlier in the section, we concluded that the convergents of this continued fraction are ratios of consecutive Fibonacci numbers. This does not guarantee, however, that the convergents actually converge to a limit. But exercise 4 provides a lovely graphical demonstration that the limit of the sequence $\{S_n\}$ of convergents exists.

1.6 φ is irrational

Now that we have encountered φ and seen how it may be represented, we might ask what kind of a number it is. For example, we noted that φ is the limit of successive ratios of Fibonacci numbers

$$\frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \dots$$

in the previous section. Is φ perhaps itself a rational number?

Suppose it is, so $\varphi = m/n$ where m and n are natural numbers and the ratio is reduced (so n is the smallest appropriate whole number). Because $1 < \varphi < 2$, we can be sure^[8] that $0 < m - n < n$. Now from (1.5) we have

$$\frac{1}{\varphi} = \varphi - 1 = \frac{m}{n} - 1 = \frac{m - n}{n}, \quad \text{or} \quad \varphi = \frac{n}{m - n}.$$

We have just created a ratio for φ with a smaller numerator than the original m/n , which we supposed to be reduced. This contradiction implies that we could not have hypothesized $\varphi = m/n$ in the first place. So φ is irrational.

And now we approach the claim that φ is irrational from a different direction. We know that

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \text{so} \quad 2\varphi - 1 = \sqrt{5}.$$

Thus if φ is rational, then so is $\sqrt{5}$. But $\sqrt{5}$ is not rational (as we will verify now), so φ cannot be rational^[9] either.

Claim. $\sqrt{5}$ is irrational.

As we did just above in showing that φ is irrational, we will argue by contradiction. Suppose that $\sqrt{5} = m/n$ where the ratio is reduced. Squaring, we get

$$5 = \frac{m^2}{n^2}, \quad \text{which can be written} \quad \frac{m}{n} = \frac{5n}{m}.$$

Because $\sqrt{5} > 2$, we know that $m > n$, so the fraction $5n/m$ has a larger denominator than does m/n . This makes m/n a reduced version of $5n/m$. We can therefore assume the existence of a natural number k such that $mk = 5n$ and $nk = m$. Combining these observations yields $nk^2 = 5n$ or $k^2 = 5$, an impossible conclusion, as 5 is not the square of any natural number. So $\sqrt{5}$ cannot be rational as originally assumed.

The exercises provide a second algebraic argument (exercise 5) and an elegant geometric demonstration (exercise 6) of the irrationality of $\sqrt{5}$. We may also use the continued fraction (1.16) to find that φ is irrational, as exercise 7 demonstrates. Finally, anyone familiar with how natural numbers can be written in their *base three* (or *ternary*) representation will find the proof in exercise 8 intriguing.

Ever since the discovery of irrational numbers fractured the Greek belief that all numbers were proportions, mathematicians have sorted numbers into categories and hunted for numbers that defied existing categories. Irrationality has long been a central focus of mathematics. In this section, and in the exercises, are a half dozen ways to prove the irrationality of the golden ratio. But if a proof is an airtight, logical demonstration of truth, then why prove a result in more than one way?

Mathematicians linger on cherished topics, illuminating them from a variety of viewpoints, much like artists and poets try over and over to capture truths about the human condition. Re-proving something important in a new way brings joy to both the discoverer and the audience. Every result in this book, not just the irrationality of φ , can be proved in more than one way. Imagine that each proof in this book is like a painting that one sees upon entering a gallery full of artwork, in which each work presents an artist's unique vision of the same theme.

1.7 The arithmetic geometric mean inequality

As this book opened, we saw how Timaeus used the proportions

$$\frac{\text{fire}}{\text{air}} = \frac{\text{air}}{\text{water}} \quad \text{and} \quad \frac{\text{air}}{\text{water}} = \frac{\text{water}}{\text{earth}}$$

to link the four elements. Each of these proportions is of the form

$$\frac{p}{q} = \frac{q}{r}$$

and if we stipulate that $p = q + r$ as well (as in Figure 1.8), we can calculate^[10] that $q = \varphi r$ and $p = \varphi q$. Thus we say that the point that separates r and q cuts the line segment in the *golden mean*, or we say that q is the golden mean of the lengths p and r . Two other well known proportions highlight this section, and their relationship will prove useful to us throughout the rest of the book.

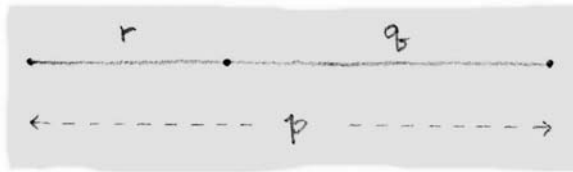


Figure 1.8

The *arithmetic mean* of a set of n positive real numbers answers the question, “What single amount may we sum n times to arrive at the total of the n original numbers?” Let A_n denote the arithmetic mean of the numbers $a_1, a_2, \dots, a_n$. To calculate A_n , we split the sum equally:

$$A_n = \frac{a_1 + a_2 + \dots + a_n}{n}.$$

The *geometric mean* of such a set of numbers answers an analogous question, but about multiplication, not addition. For example, to find the final cost of a \$100 item that is on sale for 20% off and then an additional 10% off, we multiply

$$100(.80)(.90) = 72.$$

If we ask, “What unique percentage may we use twice in a row so that the \$100 item again reduces to \$72?”, the answer is not the arithmetic mean of 80% and 90%. Rather, because we seek a percent P so that

$$100(.80)(.90) = 100(P)(P),$$

we find that

$$P = \sqrt{(.80)(.90)} \approx 0.8485.$$

This “average” of the percentages may be generalized to define the geometric mean G_n of $a_1, a_2, \dots, a_n$ as

$$G_n = (a_1 a_2 \cdots a_n)^{1/n}.$$

The geometric mean of .80 and .90 is less than the arithmetic mean. Figure 1.9 shows why the geometric mean of any pair of positive numbers is never larger than their arithmetic mean. Lay segments of length $a > b$ end to end as the diameter of a semicircle with center M . Then altitude MN has length equal to the arithmetic mean $(a+b)/2$. Now angle EBF is a right angle because^[11] it is inscribed in the semicircle, so both $\triangle EAB$ and $\triangle BAF$ are similar to $\triangle EBF$ (and therefore to each other). Thus,

$$\frac{a}{AB} = \frac{AB}{b} \quad \text{or} \quad AB = \sqrt{ab},$$

which is the geometric mean of a and b . The figure not only shows that the geometric mean cannot exceed the arithmetic mean, but also re-

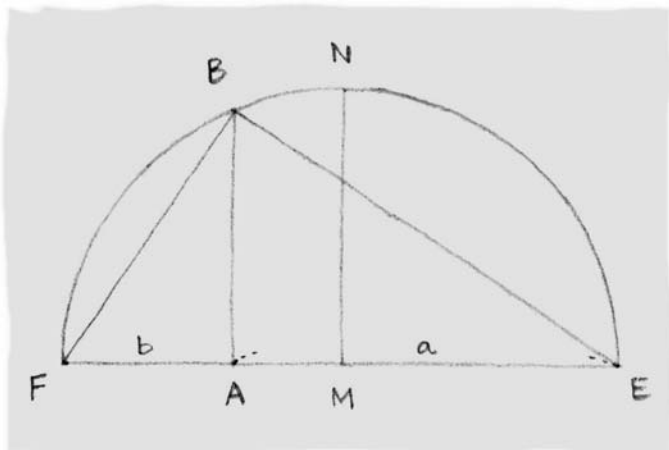


Figure 1.9

veals that the geometric mean equals the arithmetic mean precisely when $a = b$.

In fact, the geometric mean of any set of positive real numbers never exceeds the arithmetic mean, and this relationship will support other proofs throughout this book. This result may be stated,

$$\frac{1}{n}(a_1 + a_2 + \cdots + a_n) \geq (a_1 a_2 \cdots a_n)^{1/n} \quad (1.17)$$

with equality^[12] precisely when $a_1 = a_2 = \cdots = a_n$. This claim is called the *arithmetic geometric mean inequality*. The following equivalent statement allows a simpler proof: let $a_i = (A_i)^n$ in (1.17) and multiply both sides by n to obtain

$$(A_1)^n + (A_2)^n + \cdots + (A_n)^n \geq nA_1 A_2 \cdots A_n. \quad (1.18)$$

We will prove (1.18) for $n = 3$ by way of indicating how a full induction proof would proceed. First we prove a claim that will be useful in the main argument.

Claim. If a, b, c, d are positive real numbers with $a \geq b$ and $c \geq d$, then

$$ac + bd \geq ad + bc. \quad (1.19)$$

In words, when the numbers c and d swap places, the total decreases. The short proof begins with

$$(a - b)(c - d) \geq 0,$$

which follows immediately from $a \geq b$ and $c \geq d$. Distributing, we have

$$a(c - d) - b(c - d) \geq 0, \quad \text{and so} \quad (ac + bd) - (ad + bc) \geq 0,$$

from which the claim follows. Armed with this maneuver, we tackle (1.18) in the case where $n = 3$, letting x, y, z denote the three numbers (rather than A_1, A_2, A_3) for the sake of clarity.

Claim. For positive numbers x, y, z , we have

$$x^3 + y^3 + z^3 \geq 3xyz. \quad (1.20)$$

Assume that $x \geq y \geq z$ (just relabel the numbers if necessary) and begin with the left side of (1.20). Observe that

$$x^3 + y^3 + z^3 = x^3 + y^2 y + z^2 z$$

and mentally isolate the last two terms. Because $y^2 \geq z^2$ and $y \geq z$, these terms fit the pattern on the left of (1.19). Thus, we can swap y and z and introduce the inequality from (1.19) to reach

$$x^3 + y^2y + z^2z \geq x^3 + y^2z + z^2y.$$

Again we may use (1.19), this time on the first two terms, to write

$$x^3 + y^2z + z^2y \geq x^2z + y^2x + z^2y.$$

With these two moves, we have promoted a z from the original third term up to the first term. Two more judicious swaps bring us to

$$\begin{aligned} x^2z + y^2x + z^2y &\geq xyz + x^2y + z^2y \\ &\geq xyz + xyz + xyz \\ &= 3xyz, \end{aligned}$$

whereupon we have satisfied ourselves that (1.20) is true. A full induction proof of (1.18) may mimic this same sort of shuffling.

1.8 Further content

1.1 Constructing φ geometrically. For millennia, mathematicians have limited the tools with which they investigate geometry, often to just two, the compass (for drawing circles) and straightedge (for drawing lines). The construction outlined in this exercise, in which we split a line segment proportionally into the golden ratio, may be performed with these two tools.

On a line, as in Figure 1.10, construct equilateral triangle ABC . We stipulate (for convenience only) that AB has length 2. From side BC , build square $BCDE$. Locate point F on the original line such that $CE = CF$.

Explain first why $BF = \sqrt{5} - 1$. (Hint: draw a perpendicular line from C to AB .) Then explain why

$$\frac{AF}{AB} = \varphi.$$

Thus AF is split by B into two lengths that are proportional to the golden ratio, and bisecting AF yields a segment of length φ .

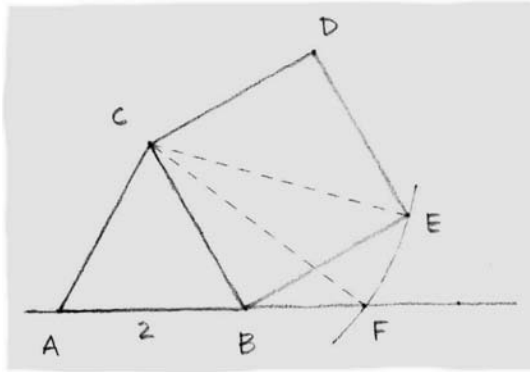


Figure 1.10

2.2 Binet's formula. Here we derive a closed formula (commonly named Binet's formula, although the result had already been discovered by Abraham de Moivre, who we will meet again in Chapter 4) for the n th Fibonacci number, that is, an expression that can be evaluated for any whole number n to obtain the associated Fibonacci number.

- (a) We have seen at (1.4) that φ satisfies the equation $\varphi^2 = \varphi + 1$. Show that

$$\tau = \frac{1 - \sqrt{5}}{2}$$

satisfies the similar equation $\tau^2 = \tau + 1$.

- (b) We defined the Fibonacci number F_n to be the n th entry in the sequence $1, 1, 2, 3, 5, 8, \dots$. With this in mind, consider how the argument leading to (1.8) can establish the truth of the identities

$$\varphi^n = \varphi F_n + F_{n-1} \quad \text{and}$$

$$\tau^n = \tau F_n + F_{n-1},$$

expressing the required details in your own words.

- (c) Subtract the bottom expression from the top and manipulate the result to obtain the desired result

$$F_n = \frac{1}{\sqrt{5}}(\varphi^n - \tau^n). \quad (1.21)$$

Check the formula by calculating any two Fibonacci numbers you wish.

- (d) A slightly faster way to calculate F_n relies on an observation about (1.21). The closest integer to $\varphi^n/\sqrt{5}$ equals the n th Fibonacci number; for example,

$$\frac{\varphi^5}{\sqrt{5}} \approx 4.96 \approx 5,$$

$$\frac{\varphi^{10}}{\sqrt{5}} \approx 55.004 \approx 55,$$

and so on. What feature of (1.21) makes this shortcut possible?

- 3.3 The harmonic-geometric-arithmetic right triangle.** We discussed the geometric mean and arithmetic mean of a set of positive numbers in section 1.7, concluding that their geometric mean never exceeds their arithmetic mean. This exercise introduces a third mean, and from the three we will build a triangle that features the golden ratio.

Like the arithmetic and geometric means, the *harmonic mean* was known to Greek geometers in the time of Plato. In *Timaeus* (the dialogue featured in the introduction to this chapter), Timaeus hypothesizes that the Creator designed the human soul in much the same way that we calculate the harmonic mean of two numbers.

- (a) In Timaeus's words, the harmonic mean is "the one exceeding and exceeded by equal parts of its extremes (as for example 1, $4/3$, 2, in which the [harmonic] mean $4/3$ is one-third of 1 more than 1, and one-third of 2 less than 2)." Let's formulate this concept. Denote the "extremes" by a and b with $a < b$. (So in Timaeus's example $a = 1$ and $b = 2$.) Let x be the "part" of each extreme that Timaeus references. Then the harmonic mean H of a and b satisfies both

$$H = a + ax \quad \text{and} \quad H = b - bx.$$

Set these equal and solve for x . Then, setting H equal to either expression above, conclude that

$$H = \frac{2ab}{a+b}. \tag{1.22}$$

- (b) Check (using algebra) that the square of the geometric mean G of a and b equals the product of their arithmetic mean A and their harmonic mean. That is, show that

$$G^2 = AH. \tag{1.23}$$

- (c) In section 1.7, we used Figure 1.9 to argue that the geometric mean G of two positive numbers is always less than or equal to their arithmetic mean A . Draw segment BM in Figure 1.9, then draw a segment perpendicular to BM from point A to a point we will call C on BM . Using $\triangle BCA \sim \triangle BAM$, argue that BC equals the harmonic mean (1.22) of a and b .
- (d) Explain how the figure thus implies that

$$H \leq G$$

and give the conditions on which these two means are equal.

- (e) So $H \leq G \leq A$ (for a pair of positive numbers). Suppose that there is a right triangle with these three side lengths. Then A (because it exceeds H and G) must be the length of the hypotenuse. By the Pythagorean theorem, we have

$$A^2 = G^2 + H^2.$$

Use (1.23) to prove that

$$\frac{A}{H} = \varphi. \quad (1.24)$$

- (f) And so the golden ratio arises in the unique right triangle that features three of the most important means. Now as the dessert course, use (1.24) to show that $a/b = \varphi^3$.

4.4 The continued fraction for φ converges (via graphing). In section 1.5, we created the continued fraction

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

but did not prove that its convergents approach a limit. It so happens that continued fractions with all numerators equal to 1 (these are called *simple* continued fractions) have this convergence property. Here we will not prove this general result, but instead will see a graphical demonstration of the convergence of the continued fraction for φ .

Recall from section 1.5 that we defined the convergents

$$S_0 = 1 \quad \text{and} \quad S_n = 1 + \frac{1}{S_{n-1}} \quad \text{for } n \geq 1$$

so that

$$S_1 = \frac{2}{1}, S_2 = \frac{3}{2}, S_3 = \frac{5}{3}, S_4 = \frac{8}{5}, \text{ and so on,}$$

and then proved that they are indeed ratios of consecutive Fibonacci numbers. Now let's turn to Figure 1.11, which shows the curves $y = x$ and $z = 1 + 1/x$ and their intersection at the point (φ, φ) .

- Explain why the curves intersect at (φ, φ) .
- Substitute $S_2 = 3/2$ (marked on the horizontal axis at A) into z to find that

$$AB = z(S_2) = z(3/2) = \frac{5}{3} = S_3.$$

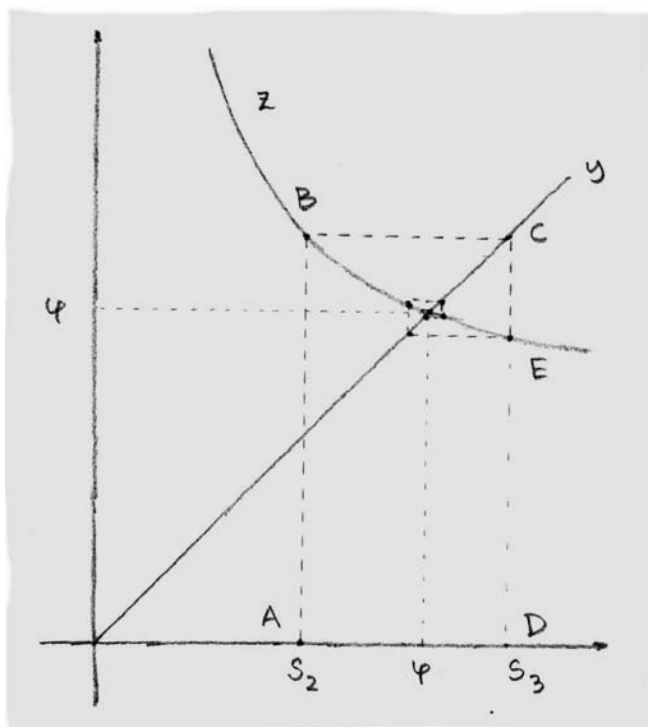


Figure 1.11

Move horizontally from B to intersect $y = x$ at C . Explain why point C is (S_3, S_3) .

- (c) Point D indicates where $x = S_3$. Mimic the first substitution $z(S_2)$, this time with S_3 , to explain why $DE = S_4$.
- (d) Fully explain now how we can continue following this path, which is a sort of rectangular spiral, to the point (φ, φ) .

5.5 $\sqrt{5}$ is irrational (via algebra). In section 1.6, we saw a proof by contradiction that proves the irrationality of $\sqrt{5}$. Here is an alternative.

Note that $2 < \sqrt{5} < 3$, and let $\alpha = \sqrt{5} - 2$. If α is irrational then so is $\sqrt{5}$. Suppose, though, that α is rational, so $\alpha = p/q$ where the ratio is reduced (so q is the smallest appropriate whole number).

- (a) Multiply α by $(\sqrt{5} + 2)/(\sqrt{5} + 2)$ and explain why

$$\alpha = \frac{1}{\alpha + 4}, \quad \text{so} \quad \frac{p}{q} = \frac{1}{\alpha + 4}.$$

- (b) Solve for α and use the solution to find a contradiction to how we defined q .

6.6 $\sqrt{5}$ is irrational (via geometry). We show that φ is irrational by way of proving that $\sqrt{5}$ is as well. In exercise 5, we saw an algebraic proof. Here we sidestep algebra with the help of geometry.

Right triangle ABC in Figure 1.12 has sides 1, 2, and $\sqrt{5}$. Point D sits where $CD = CB = 1$. We draw ED perpendicular to AC at D , making $ED = EB$.

- (a) Why is $\triangle ABC \sim \triangle ADE$?
- (b) Deduce that

$$\sqrt{5} = \frac{AC}{CB} = \frac{AE}{DE}.$$

In conclusion: Suppose that $\sqrt{5} = AC/CB$ is rational. Then AE/DE itself is both rational and a reduced version of AC/CB . But then we can reapply the argument with $\triangle ADE$ in the role of $\triangle ABC$, finding another version of AC/CB that is yet further reduced. Because fractions may not be reduced infinitely, we will have reached a contradiction.

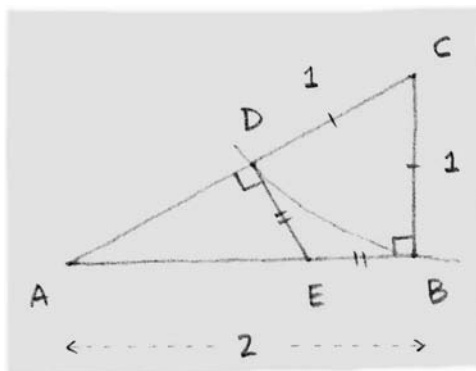


Figure 1.12

7.7 φ is irrational (via its continued fraction). In section 1.6, we saw that φ is irrational by proving that $\sqrt{5}$ is irrational. Here we prove the same truth but by analyzing the continued fraction

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}} \quad (1.25)$$

that we first described at (1.16). As in section 1.6, we define S_n to be the convergent of (1.25) that is truncated after the n th plus sign. So

$$S_0 = 1, \quad S_1 = 1 + \frac{1}{1}, \quad S_2 = 1 + \frac{1}{1 + \frac{1}{1}}, \quad \text{etc.}$$

Let F_k be the k th Fibonacci number from the sequence 1, 1, 2, 3, 5, 8, ... that arises from the property $F_{n+2} = F_{n+1} + F_n$. Recall that we showed at (1.12) that

$$S_n = \frac{F_{n+2}}{F_{n+1}}. \quad (1.26)$$

(a) The square of any Fibonacci number differs from the product of the numbers on either side of it by one. For example,

$$F_1 F_3 - (F_2)^2 = 1, \quad F_2 F_4 - (F_3)^2 = -1,$$

$$F_3 F_5 - (F_4)^2 = 1, \quad \text{etc.}$$

The conjecture

$$F_{n-1}F_{n+1} - (F_n)^2 = (-1)^n \quad (1.27)$$

therefore seems viable. Prove (1.27) by induction.

- (b) With (1.26) and (1.27) in hand, explain why the difference between consecutive convergents S_{n+1} and S_n is

$$S_{n+1} - S_n = \frac{(-1)^n}{F_{n+2}F_{n+1}}. \quad (1.28)$$

- (c) From (1.28), deduce that

$$\varphi \in (S_{2k}, S_{2k+1}) \quad \text{for all } k. \quad (1.29)$$

- (d) Put (1.28) and (1.29) together to explain why

$$|\varphi - S_k| \leq \frac{1}{F_k F_{k+1}} \quad \text{for all } k. \quad (1.30)$$

- (e) Now we can prove that φ is irrational. Suppose, on the contrary, that φ is rational and may therefore be written $\varphi = m/n$ for some whole numbers m and n . Now if we substitute m/n for φ in (1.30), we obtain

$$\left| \frac{m}{n} - S_k \right| \leq \frac{1}{F_k F_{k+1}}.$$

Use (1.26) and the fact that the right side of the above is less than 1 to find a contradiction. (Once you do this, you will have demonstrated that φ cannot be written as m/n , implying that φ is irrational.)

8.8 φ is irrational (via ternary arithmetic). This proof requires knowledge of modular arithmetic and base three (ternary) representation of numbers. But anyone armed with such knowledge should find this proof appealing.

Claim. $\sqrt{5}$ is irrational.

First, we argue that every perfect square has 1 as its final nonzero digit when represented in base three. For any natural number q we must have

$$q \equiv 1 \pmod{3} \quad \text{or}$$

$$q \equiv 2 \pmod{3} \quad \text{or}$$

$$q \equiv 0 \pmod{3}.$$

We study each case in turn.

If $q \equiv 1 \pmod{3}$ then multiplying each side of the congruence by q yields

$$q^2 \equiv q \pmod{3} \quad \text{or} \quad q^2 \equiv 1 \pmod{3}.$$

So if q in base three has 1 as its final digit, then q^2 does as well.

- (a) In like manner, argue that $q \equiv 2 \pmod{3}$ implies $q^2 \equiv 1 \pmod{3}$.
- (b) The final case requires more thought. If $q \equiv 0 \pmod{3}$ then we know that q is divisible by some power of 3. Let k be the largest natural number such that 3^k divides q and let r denote the remaining factor, so $q = 3^k r$. Now because r is not divisible by 3, we know that r is congruent to either 1 or 2, and the previous cases apply with r in the role of q .

Put these ideas together to conclude that $q^2 \equiv 1 \pmod{3}$.

- (c) Now on to the proof that $\sqrt{5}$ is irrational. Suppose $\sqrt{5}$ is rational, so $\sqrt{5} = m/n$. Then $5n^2 = m^2$. Show that this equation is inconsistent.

CHAPTER 2

π

Bees have populated the earth for tens of millions of years, about as long as the land has supported flowering plants. Honey bees gather nectar from flowers and produce honey as a food source and beeswax as a storage material. They create honeycombs from the wax, a process streamlined by natural selection to a mathematically perfect solution.

To create the hexagonal tiling that is characteristic of a honeycomb, a bee secretes wax and fashions it into a cylinder around her body. She softens the walls with her own heat and, when she is finished, the surface tension acting on the cooling walls pulls them straight. Because the most efficient way to pack circles together arranges the circles into a hexagonal grid, the resulting polygons are six-sided.

Only three regular polygons can tile a surface, and among these three, the hexagon more efficiently encloses storage area (e.g., for honey) than the equilateral triangle or square. To illustrate this point, consider an equilateral triangle with perimeter 6 centimeters. Each side is of length 2 cm, so by the Pythagorean theorem, the triangle's height is $\sqrt{3}$ cm. The triangle thus has area $(1/2)(2)(\sqrt{3}) \approx 1.73$ square centimeters. A square with perimeter 6 cm has sides of length 1.5 cm, so it encloses $(1.5)^2 = 2.25$ square centimeters of area, making it more efficient than the equilateral triangle. By contrast, a hexagon with perimeter 6 cm can store even more honey. Each side has length 1 cm, and we can cut the hexagon into six equilateral triangles as in Figure 2.1. The area of each triangle is $(1/2)(1)(\sqrt{3}/2)$, so the area of the hexagon is $6\sqrt{3}/4 \approx 2.60$ square centimeters.

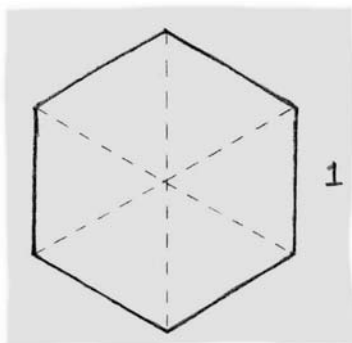


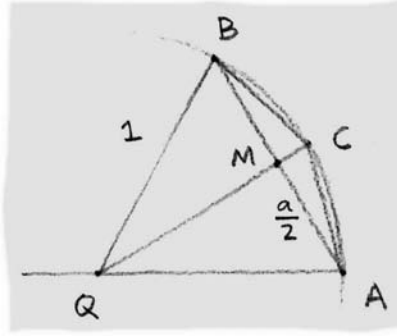
Figure 2.1

2.1 Liu Hui approximates π using polygons

Hexagons are but a byproduct of the process of making honeycombs; bees, in fact, exhibit the finest efficiency by crafting circular chambers. The mathematical community grasped the import of circles early, including the hunch that circles optimize storage area. In the third century, **Liu Hui** (China) explained how one might explore this unique feature of circles by repeatedly doubling the sides of a polygon, creating a sequence of polygons that ever more closely resembled a circle. (Five hundred years earlier, Greek mathematician **Archimedes** captured the value of π in much the same way.)

Liu Hui's strategy may be phrased thus: inscribe a hexagon in a unit circle and find both the area and the side length of the hexagon. Use this information to find the area and side length of a 12-gon inscribed in the same circle, then repeat to reach a 24-gon, then a 48-gon, and so on. Each polygon has double the number of sides as the last, so the sequence of areas will grow ever closer to the exact area of the unit circle, which has area π square units.

This geometric approximation of π iterates fairly easily thanks to a formula that calculates the side length of a $2n$ -gon given the side length of an n -gon. In his solution, Liu Hui does not use trigonometry, which had yet to be developed in China. In Figure 2.2, the circle with center Q has radius 1 and therefore area equal to π square units. Inscribed in the circle sits an n -gon (a hexagon, for example) with just one of its sides AB shown. We bisect AB at M and draw a line from Q through M to

**Figure 2.2**

locate a new vertex C for our $2n$ -gon. (In the same way, we would bisect every other side of the n -gon.)

We wish to find the area of the $2n$ -gon in terms of the n -gon's features. If we let $AB = a$, and note that QC is perpendicular to AB , then we can calculate that the area of $\triangle QAC$ (by using its altitude $AM = a/2$ and base $QC = 1$) is $a/4$. Because $2n$ such triangles tile the $2n$ -gon as a whole, we conclude

$$\text{area of the } 2n\text{-gon} = \frac{an}{2}. \quad (2.1)$$

So from a and n (the side length and number of sides of our n -gon), we can find the area of the $2n$ -gon. If we wish to iterate this process, however, then we need to calculate the side length $AC = b$ of the $2n$ -gon as well.

We may connect a from the n -gon to b from the $2n$ -gon by means of the Pythagorean theorem, starting with

$$(QM)^2 + \left(\frac{a}{2}\right)^2 = 1^2$$

from $\triangle QMA$. Combining^[13] this observation with $QC = QM + MC = 1$ yields

$$MC = 1 - \frac{1}{2}\sqrt{4 - a^2}. \quad (2.2)$$

Applying this result to $\triangle AMC$ (and simplifying^[14]), we find that

$$b^2 = 2 - \sqrt{4 - a^2} \quad (2.3)$$

as desired.

Let's return to the hexagon with sides of length 1 and area 2.60 from earlier in this section, and use its information to show (2.1) and (2.3) at work. The hexagon's six sides make $n = 6$ and we have $a = 1$ as the length of each side, so from (2.1) we know that the area of the associated dodecagon (that is, 12-gon) is 3. Now if we wish to again double the number of sides of our polygon to more closely approximate the area of a circle, we can use (2.3) to obtain

$$b = \sqrt{2 - \sqrt{4 - 1^2}} = \sqrt{2 - \sqrt{3}} \quad (2.4)$$

as the length of one side of the 12-gon. If we wished to repeat the process^[15] to find the area and side length of a 24-gon, we could do so, and so on for as long as we wish. Each polygon's area would approximate π even more closely than the area of the previous polygon.

This approach allows us to approximate the area of a circle with radius 1 as closely as we wish. Liu Hui doubled the sides of his original hexagon five times to determine that the area of a 192-gon is a bit more than 3.1410. He also repeated the analysis, this time with the unit circle inscribed within his polygons, to obtain an upper bound of 3.1427.

About fourteen centuries later, the scientific community adopted the Greek letter π to represent this value that Liu Hui had ensnared with his clever geometry. During those many years, other minds joined the search for a closer approximation of Liu Hui's

$$3.1410 < \pi < 3.1427.$$

For a long while, thinkers used versions of Liu Hui's idea. In the 1400s, for example, **Jamshīd al-Kāshī** (Persia) found the equivalent of 17 digits of π by calculating the perimeter of a polygon having 3×2^{28} sides. And while these ingenious and persistent efforts bear mention, our focus now turns away from this tale of polygons to consider an approximation strategy with a wholly different genesis.

2.2 Nilakantha's arctangent series

One of the earliest alternatives to using polygons to approximate π originated in the work of **Nilakantha Somayaji** (India, born 1444). In part inspired by his interest in astronomy, Nilakantha discovered what we now call infinite series for trigonometric functions. His series for the inverse

tangent function $\arctan x$, in particular, leads to a wholly new method for approximating π .

Nilakantha's development of the series for $\arctan x$ predated any deep understanding of integration by two centuries, so the rest of this section, while straightforward, does not follow his path.

We take the geometric series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots \quad (2.5)$$

as our starting point. The series converges^[16] for $x \in (-1, 1)$. If we replace x with $-x^2$, we establish the result

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + x^8 - \dots \quad (2.6)$$

as a lemma for what follows.

Consider $y = \arctan x$ and its equivalent identity $x = \tan y$. Implicit differentiation with respect to x yields

$$1 = \sec^2 y \cdot \frac{dy}{dx}$$

and, using the identity $\sec^2 y = 1 + \tan^2 y$,

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y}.$$

Because $x = \tan y$, we have

$$\frac{dy}{dx} = \frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

and can apply our lemma (2.6) to the right side, then integrate both sides, obtaining

$$y = \int (1 - x^2 + x^4 - x^6 + x^8 - \dots) dx$$

or

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots \quad (2.7)$$

as promised.

As with series (2.6) that triggered this approach, series (2.7) converges on the interval $x \in (-1, 1)$. Happily, the series also converges for $x = 1$, which produces

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots \quad (2.8)$$

from (2.7). This formula for π was not only discovered by Nilakantha but also independently by **James Gregory** (England, born 1638) and **Gottfried Leibniz** (Germany, born 1646), among others.

Now while formula (2.8) elegantly expresses π as an easily deciphered pattern, it approximates π at a crawl. Achieving $\pi \approx 3.1417$ requires about ten thousand terms. In response to this inefficiency, Nilakantha examined the partial sums of (2.8), and their errors, to craft the far more efficient

$$\frac{\pi}{4} = \frac{4}{1^5 + 4 \cdot 1} - \frac{4}{3^5 + 4 \cdot 3} + \frac{4}{5^5 + 4 \cdot 5} - \cdots,$$

which with just five terms trumps the first ten thousand of (2.8). While we will not plunge into the details of this result here, it bears mentioning as testimony to the power of pre-calculus mathematics and the ingenuity of those who mastered it.

2.3 Machin's arctangent formula

A few decades after James Gregory rediscovered Nilakantha's formula (2.8), **John Machin** (England, born 1686) found another way to coax the arctangent function into approximating π . In just three terms, Machin's formula reaches $\pi \approx 3.1417$, thus generating digits of π much more quickly than Nilakantha's (2.8). After we follow Machin's path to his formula, we will return to this amazing fact.

In Figure 2.3, the unit circle with center Q has radius QA and tangent AE , making $\angle QAE$ a right angle. We place B so that $\angle AQB$ measures $\pi/4$ radians. As Machin himself did, we observe that $\tan \pi/16 \approx 0.1989 \approx 1/5$. Now we locate point C such that $AC = 1/5$. Let α denote $\angle AQC$. Because the circle has radius 1, we know that $\tan \alpha = 1/5$. This means that α is slightly larger than $\pi/16$.

Now double α to locate point D . By the trigonometric identity

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}, \quad (2.9)$$

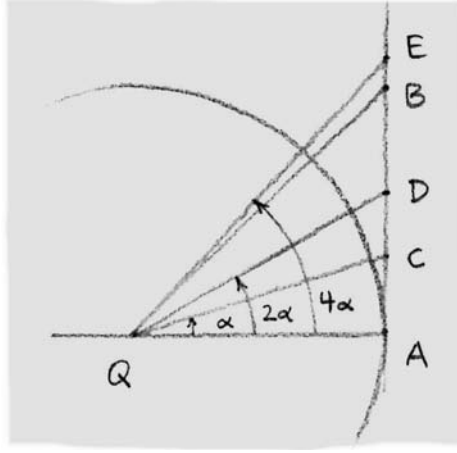


Figure 2.3

we calculate^[17] that $\tan(2\alpha) = \tan(\alpha + \alpha) = 5/12$. A second application of the identity returns $\tan(4\alpha) = 120/119$, which is slightly larger than 1. Note that therefore $AE = 120/119$.

Applying (2.9) to

$$AE = \tan(\pi/4 + \angle BQE),$$

we obtain

$$\frac{120}{119} = \frac{\tan(\pi/4) + \tan(\angle BQE)}{1 - \tan(\pi/4) \tan(\angle BQE)}.$$

Solving^[18] for $\tan \angle BQE$ yields $\tan(\angle BQE) = 1/239$. Finally, because

$$\frac{\pi}{4} = 4\alpha - \angle BQE,$$

we have Machin's formula

$$\frac{\pi}{4} = 4 \arctan(1/5) - \arctan(1/239). \quad (2.10)$$

Let's return to the claim, made at the beginning of this section, of the efficiency of Machin's formula. Using just the first three terms of (2.7) to calculate

$$\arctan 1/5 \approx 1/5 - \frac{(1/5)^3}{3} + \frac{(1/5)^5}{5} \approx 0.197397,$$

$$\arctan 1/239 \approx 1/239 - \frac{(1/239)^3}{3} + \frac{(1/239)^5}{5} \approx 0.004184,$$

we find that

$$\pi \approx 4(4(0.197397) - 0.004184) = 3.141621,$$

which differs from π by only about 0.00003.

Intrigued by this approach, other scholars since Machin have continued to create formulas similar to his, including

$$\frac{\pi}{4} = 44 \arctan \frac{1}{57} + 7 \arctan \frac{1}{239} - 12 \arctan \frac{1}{682} + 24 \arctan \frac{1}{12943},$$

which was used in 2002 (about a century after its discovery) to set a (then) world record for calculating digits of π . In exercise 1, we will see a derivation of a formula similar to Machin's that uses only basic geometry.

2.4 Wallis's formula for $\pi/2$ (via calculus)

The area of a quarter of a unit circle is $\pi/4$, and with today's calculus^[19] we find it via the integral

$$\int_0^1 (1 - x^2)^{1/2} dx.$$

When **John Wallis** (England, born 1616) turned his attention to π , integral notation had yet to be invented, such was the infancy of calculus.

In 1655, Wallis discovered a remarkable formula that could compute the digits of π in a straightforward way. His formula, which segregates the even and odd natural numbers, is

$$\frac{\pi}{2} = \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \frac{6}{7} \times \cdots \quad (2.11)$$

Wallis's original derivation rests only on algebra, geometry, and intuitive pattern seeking. His efforts appear not here but in Appendix A, simply due to the sheer amount of work involved. (Exercise 2 gives a modern proof that uses a similarly elementary approach.) Here we follow a shorter path as provided by calculus via the family of integrals

$$a_n = \int_0^{\pi/2} \sin^n x \, dx.$$

The first two integrals evaluate to

$$a_0 = \int_0^{\pi/2} \sin^0 x \, dx = \int_0^{\pi/2} 1 \, dx = \frac{\pi}{2} \quad \text{and}$$

$$a_1 = \int_0^{\pi/2} \sin x \, dx = \left[-\cos x \right]_0^{\pi/2} = 1.$$

To calculate

$$a_2 = \int_0^{\pi/2} \sin^2 x \, dx \quad \text{we use} \quad \sin^2 x = \frac{1 - \cos(2x)}{2}$$

to get

$$a_2 = \int_0^{\pi/2} \frac{1 - \cos(2x)}{2} \, dx = \frac{\pi}{4}.$$

To calculate a_3 , we split $\sin^3 x$ as in

$$a_3 = \int_0^{\pi/2} \sin^3 x \, dx = \int_0^{\pi/2} \sin^2 x \sin x \, dx$$

and then apply integration by parts with

$$u = \sin^2 x \quad \text{so} \quad du = 2 \sin x \cos x \, dx,$$

$$dv = \sin x \, dx \quad \text{so} \quad v = -\cos x,$$

which leads to

$$a_3 = \left[-\cos x \sin^2 x \right]_0^{\pi/2} + 2 \int_0^{\pi/2} \sin x \cos^2 x \, dx.$$

The first expression equals 0, and we use the identity

$$\cos^2 x = 1 - \sin^2 x$$

to write

$$a_3 = 2 \int_0^{\pi/2} \sin x (1 - \sin^2 x) \, dx,$$

or

$$\frac{a_3}{2} = \int_0^{\pi/2} \sin x \, dx - \int_0^{\pi/2} \sin^3 x \, dx.$$

The first integral is a_1 and we calculated that $a_1 = 1$ above. The second is a_3 , so

$$\frac{a_3}{2} = a_1 - a_3 \quad \text{or} \quad a_3 = \frac{2}{3}.$$

This process hints at a recursion, so we apply these moves to the general case, namely

$$a_n = \int_0^{\pi/2} \sin^n x \, dx \quad \text{or} \quad a_n = \int_0^{\pi/2} \sin^{n-1} x \sin x \, dx.$$

Integration by parts with

$$u = \sin^{n-1} x \quad \text{so} \quad du = (n-1) \sin^{n-2} x \cos x \, dx,$$

$$dv = \sin x \, dx \quad \text{so} \quad v = -\cos x,$$

leads to

$$a_n = [-\cos x \sin^{n-1} x]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx.$$

As before, we find that the first term equals 0. We can replace $\cos^2 x$ in the integral by using the identity $\cos^2 x = 1 - \sin^2 x$, then distribute to split the integral, obtaining

$$\frac{a_n}{n-1} = \int_0^{\pi/2} \sin^{n-2} x \, dx - \int_0^{\pi/2} \sin^n x \, dx.$$

Now the first integral is just a_{n-2} by definition, and the second is a_n , so

$$\frac{a_n}{n-1} = a_{n-2} - a_n, \quad \text{or} \quad a_n = \frac{n-1}{n} a_{n-2}. \quad (2.12)$$

Recursion (2.12) yields

$$\begin{aligned} a_0 &= \frac{\pi}{2} & a_1 &= 1 \\ a_2 &= \frac{1}{2} \times \frac{\pi}{2} & a_3 &= \frac{2}{3} \times 1 \\ a_4 &= \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} & a_5 &= \frac{4}{5} \times \frac{2}{3} \times 1 \\ a_6 &= \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} & a_7 &= \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \times 1 \end{aligned} \quad (2.13)$$

and so on, leading to the general terms

$$\begin{aligned} a_{2n} &= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \cdots \times \frac{1}{2} \times \frac{\pi}{2}, \\ a_{2n+1} &= \frac{2n}{2n+1} \times \frac{2n-2}{2n-1} \times \cdots \times \frac{2}{3} \times 1. \end{aligned} \quad (2.14)$$

Now we will show that the limiting ratio of these two general terms equals 1. The function $\sin x$ is bounded by 0 and 1 on the interval $[0, \pi/2]$, so we know^[20] that

$$\sin^{2n+2} x \leq \sin^{2n+1} x \leq \sin^{2n} x.$$

Integrating each from 0 to $\pi/2$ gives

$$a_{2n+2} \leq a_{2n+1} \leq a_{2n}.$$

But with (2.12) we can replace a_{2n+2} to obtain

$$\frac{2n-1}{2n} a_{2n} \leq a_{2n+1} \leq a_{2n},$$

whereupon we divide by a_{2n} and then take the limit throughout:

$$\lim_{n \rightarrow \infty} \frac{2n-1}{2n} \leq \lim_{n \rightarrow \infty} \frac{a_{2n+1}}{a_{2n}} \leq 1.$$

Because the left limit equals 1, we conclude that

$$\lim_{n \rightarrow \infty} \frac{a_{2n+1}}{a_{2n}} = 1 \quad \text{or} \quad \lim_{n \rightarrow \infty} a_{2n+1} = \lim_{n \rightarrow \infty} a_{2n}.$$

The symbol $\asymp$ commonly denotes that two expressions are approaching each other in their limits, so we may write

$$a_{2n} \asymp a_{2n+1},$$

or, from (2.14),

$$\frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \cdots \times \frac{1}{2} \times \frac{\pi}{2} \asymp \frac{2n}{2n+1} \times \frac{2n-2}{2n-1} \times \cdots \times \frac{2}{3}.$$

Solving for $\pi/2$, we find that

$$\frac{\pi}{2} \asymp \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \cdots \times \frac{2n}{2n-1} \times \frac{2n}{2n+1}. \quad (2.15)$$

Now letting $n \rightarrow \infty$, we have captured Wallis's formula.

2.5 A connection to probability

With Wallis's formula fresh in our minds, we will observe a connection between it and a simple probability question with an answer that plays a role in deeper results. The probability question is:

If a fair coin is flipped $2n$ times, what is the probability P_{2n} that the coin comes up heads and tails an equal number of times?

To calculate this probability, we divide the number of outcomes that result in n heads and n tails by the total number of possible outcomes. (The endnotes^[21] provide a numerical example, if needed.) Because the n heads might appear as any of the $2n$ flips, there are

$$\binom{2n}{n} = \frac{(2n)!}{n!(2n-n)!} = \frac{(2n)!}{n!n!}$$

outcomes with n heads (and therefore n tails). We divide this quantity by the 2^{2n} possible outcomes to conclude that

$$P_{2n} = \frac{(2n)!}{n!n!} \div 2^{2n} = \frac{(2n)!}{n!n!2^n2^n} = \frac{(2n)!}{(n!2^n)^2}. \quad (2.16)$$

Because^[22] $n!2^n = (2 \times 4 \times 6 \times \cdots \times 2n)$, we can write (2.16) as

$$P_{2n} = \frac{(2n)!}{(2 \times 4 \times 6 \times \cdots \times 2n)^2}. \quad (2.17)$$

Now we will use Wallis's formula in (2.17). First, we rewrite^[23] (2.15):

$$\frac{\pi}{2} \asymp \left[\frac{2 \times 4 \times \cdots \times (2n)}{3 \times 5 \times \cdots \times (2n-1)} \right]^2 \frac{1}{2n+1}. \quad (2.18)$$

If we multiply the numerator and denominator of the fraction inside the brackets by the numerator itself, as in

$$\frac{2 \times 4 \times \cdots \times (2n)}{3 \times 5 \times \cdots \times (2n-1)} \cdot \frac{2 \times 4 \times \cdots \times (2n)}{2 \times 4 \times \cdots \times (2n)},$$

then we may write (2.18) as

$$\frac{\pi}{2} \asymp \left[\frac{(2 \times 4 \times 6 \times \cdots \times 2n)^2}{(2n)!} \right]^2 \frac{1}{2n+1}.$$

Using (2.17), we get

$$\frac{\pi}{2} \asymp \left[\frac{1}{P_{2n}} \right]^2 \frac{1}{2n+1},$$

which simplifies to

$$P_{2n} \asymp \frac{1}{\sqrt{\pi(n+1/2)}}. \quad (2.19)$$

Thus, as we flip a fair coin, the probability of flipping exactly as many heads as tails is approximated by the expression containing π on the right. As we might expect, this probability vanishes to zero as the number of flips grows arbitrarily large.

In section 3.7, we will use this approximation to answer a less specific, and therefore deeper, question.

2.6 Wallis's formula for $\pi/2$ via $(\sin x)/x$

Eighty years after Wallis published his formula

$$\frac{\pi}{2} = \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \frac{6}{7} \times \cdots,$$

the Swiss mathematician **Leonhard Euler** (Switzerland, born 1707) discovered the infinite product

$$\frac{\sin x}{x} = \left(1 - \frac{x}{\pi}\right) \left(1 + \frac{x}{\pi}\right) \left(1 - \frac{x}{2\pi}\right) \left(1 + \frac{x}{2\pi}\right) \left(1 - \frac{x}{3\pi}\right) \cdots \quad (2.20)$$

and, by substituting $\pi/2$ for x , rediscovered^[24] Wallis's formula.

Euler arrived at (2.20) via a risky hypothesis about the zeros of the curve $y = \sin x$. To prepare for his hypothesis, we recall how the zeros of a polynomial relate to its factors. To take an example, the polynomial

$$3x^3 - 21x^2 + 42x - 24 \quad \text{factors as} \quad 3(x-1)(x-2)(x-4),$$

so its zeros are the solutions to

$$3(x-1)(x-2)(x-4) = 0,$$

which clearly are $x = 1, 2, 4$. Changing the constant 3 to another constant has no effect on the zeros. Reversing this process, we can build a polynomial that has zeros at (for example) $x = -1, 0, 3, 5$ by writing

$$A(x+1)(x-0)(x-3)(x-5) = 0$$

and then choosing any constant A .

Euler's risky hypothesis extends this idea to the curve $y = \sin x$, which has infinitely many zeros (at all of the multiples of π , including 0). So Euler guessed that he could write

$$\sin x = A(x-0)(x-\pi)(x+\pi)(x-2\pi)(x+2\pi)(x-3\pi)\cdots \quad (2.21)$$

in a sort of infinite factored form. (This audacity, which is a hallmark of Euler's scholarship, often characterizes the work of people who we recognize as geniuses.) The constant A can be determined by dividing through by x and taking the limit of the result, as in

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x}{x} \\ = \lim_{x \rightarrow 0} A(x-\pi)(x+\pi)(x-2\pi)(x+2\pi)(x-3\pi)(x+3\pi)\cdots. \end{aligned}$$

The limit on the left equals 1. On the right, we may directly substitute 0 for x . Thus we reach

$$1 = A(-\pi)(\pi)(-2\pi)(2\pi)(-3\pi)(3\pi)\cdots$$

and can solve for A . Substituting this solution into (2.21) leads to^[25] the desired identity

$$\frac{\sin x}{x} = \left(1 - \frac{x}{\pi}\right)\left(1 + \frac{x}{\pi}\right)\left(1 - \frac{x}{2\pi}\right)\left(1 + \frac{x}{2\pi}\right)\left(1 - \frac{x}{3\pi}\right)\cdots \quad (2.22)$$

mentioned at the opening of this section. This infinite product is the genesis of many intriguing corollaries, not just Wallis's formula. Exercise 3, for example, explains how we can convert an infinite product such as (2.22) to a continued fraction. Applying the conversion to (2.22) yields the lovely

$$\begin{aligned} \frac{\pi}{2} = 1 + \frac{1}{1 + \frac{1}{1/2 + \frac{1}{1/3 + \frac{1}{1/4 + \cdots}}}} \end{aligned}$$

as Exercise 4 explains.

2.7 The generalized binomial theorem

Throughout this chapter we will require a result called the *generalized binomial theorem* that is due to **Isaac Newton** (England, born 1642).

His original derivation appears in Appendix B. The theorem provides an expansion for expressions of the form

$$(a + b)^n$$

where n is not restricted to whole numbers.

The theorem states that we may expand $(a + b)^n$ using

$$(a + b)^n = a^n + \alpha_1 a^{n-1} b + \alpha_2 a^{n-2} b^2 + \alpha_3 a^{n-3} b^3 + \dots \quad (2.23)$$

where the coefficient α_m equals the product of the first m terms of

$$\frac{n}{1} \cdot \frac{n-1}{2} \cdot \frac{n-2}{3} \cdot \frac{n-3}{4} \cdot \dots$$

For example, to expand $(a + b)^3$ (where $n = 3$), we first find that

$$\alpha_1 = \frac{3}{1} = 3,$$

$$\alpha_2 = \frac{3}{1} \cdot \frac{2}{2} = 3,$$

$$\alpha_3 = \frac{3}{1} \cdot \frac{2}{2} \cdot \frac{1}{3} = 1,$$

$$\alpha_4 = \frac{3}{1} \cdot \frac{2}{2} \cdot \frac{1}{3} \cdot \frac{0}{4} = 0,$$

and so on, yielding

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

as expected. When n is a natural number, the coefficients α_m are the binomial coefficients

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}.$$

Newton argued that this same approach works for exponents that are not whole numbers. For example, to expand $(a + b)^{1/2}$ (where $n = 1/2$),

we find that

$$\begin{aligned}\alpha_1 &= \frac{1/2}{1} &&= \frac{1}{2}, \\ \alpha_2 &= \frac{1/2}{1} \times \frac{-1/2}{2} &&= -\frac{1}{8}, \\ \alpha_3 &= \frac{1/2}{1} \times \frac{-1/2}{2} \times \frac{-3/2}{3} &&= \frac{1}{16}, \\ \alpha_4 &= \frac{1/2}{1} \times \frac{-1/2}{2} \times \frac{-3/2}{3} \times \frac{-5/2}{4} &&= -\frac{5}{128},\end{aligned}$$

and so on, so

$$(a+b)^{1/2} = a^{1/2} + \frac{a^{-1/2} b}{2} - \frac{a^{-3/2} b^2}{8} + \frac{a^{-5/2} b^3}{16} - \frac{5a^{-7/2} b^4}{128} + \dots$$

We can similarly calculate that

$$(1+x)^{1/3} = 1 + \frac{x}{3} - \frac{x^2}{9} + \frac{5x^3}{81} - \frac{10x^4}{243} + \frac{22x^5}{729} - \dots$$

Later, we will want to expand expressions such as $(1-x)^n$ where the exponent is a variable. To do this, we substitute $a = 1$ and $b = -x$ into (2.23) while leaving each n untouched. Thus the expansion is

$$\begin{aligned}(1-x)^n &= 1^n + \frac{n}{1}(1)^{n-1}(-x) + \frac{n}{1} \cdot \frac{n-1}{2}(1)^{n-2}(-x)^2 + \dots \\ &= 1 - \frac{n}{1}x + \frac{n(n-1)}{1 \cdot 2}x^2 - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}x^3 + \dots\end{aligned}$$

This brief training in the generalized binomial theorem is no substitute for a proof. Knowing this, Newton tested his theory in various ways before releasing it to the world. Appendix B describes his original discovery.

2.8 Euler's $(1/2)! = \sqrt{\pi}/2$

Sometimes a mathematician sets out to express a constant such as π in an intentionally elegant way, perhaps as a continued fraction or as an infinite product. Occasionally a problem that initially seems unrelated to a specific constant reveals itself to be a path directly to it. Euler's search for a general expression that would yield the partial sums of

$$1! + 2! + 3! + 4! + 5! + \dots$$

stands as a classic example of such serendipity.

The partial sums of the series are

$$1, 3, 9, 33, 153, \text{ etc.}$$

Euler set out to find a formula that would generate any entry of this sequence. In this search, he found the family of integrals

$$\int_0^1 x^m (1-x)^n dx$$

particularly useful. To evaluate this family of integrals, we may expand

$$(1-x)^n = 1 - \frac{n}{1}x + \frac{n(n-1)}{1 \cdot 2}x^2 - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}x^3 + \dots$$

by using the generalized binomial theorem (2.23). We multiply by x^m to get

$$\begin{aligned} x^m(1-x)^n &= x^m - \frac{n}{1}x^{m+1} + \frac{n(n-1)}{1 \cdot 2}x^{m+2} \\ &\quad - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}x^{m+3} + \dots \end{aligned}$$

and then integrate:

$$\int x^m(1-x)^n dx = \frac{x^{m+1}}{m+1} - \frac{nx^{m+2}}{1(m+2)} + \frac{n(n-1)x^{m+3}}{1 \cdot 2(m+3)} - \dots$$

As a definite integral, this becomes

$$\int_0^1 x^m(1-x)^n dx = \frac{1}{m+1} - \frac{n}{1(m+2)} + \frac{n(n-1)}{1 \cdot 2(m+3)} - \dots \quad (2.24)$$

Let's work on rewriting the right side. If we successively evaluate the right side at $n = 0, 1, 2, 3, \dots$ and simplify, we find^[26] that

$$\begin{aligned} n = 0 & \text{ produces } \frac{1}{m+1} \\ n = 1 & \text{ produces } \frac{1}{(m+1)(m+2)} \\ n = 2 & \text{ produces } \frac{1 \cdot 2}{(m+1)(m+2)(m+3)} \\ n = 3 & \text{ produces } \frac{1 \cdot 2 \cdot 3}{(m+1)(m+2)(m+3)(m+4)} \end{aligned}$$

and so on. In general (Euler wrote), the n th term appears to produce

$$\frac{n!}{(m+1)(m+2)(m+3)\cdots(m+n)(m+n+1)}.$$

We can simplify this a bit if we multiply both numerator and denominator by $m!$ to get

$$\frac{m!n!}{(m+n+1)!}. \quad (2.25)$$

Now recall that (2.25) simplified the right side of (2.24), so we can write

$$\int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!}.$$

Something wonderful happens when we specifically choose $m = 1/2$ and $n = 1/2$ in this identity. Substituting, we get

$$\int_0^1 x^{1/2} (1-x)^{1/2} dx = \frac{(1/2)!(1/2)!}{(1/2 + 1/2 + 1)!},$$

which may be written

$$2 \int_0^1 \sqrt{x-x^2} dx = [(1/2)!]^2. \quad (2.26)$$

The factorial on the right is quirky. Recall that we define $n!$ by

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$$

when n is a natural number. The unusual number $(1/2)!$ defies this definition. But as we will now see, the integral on the left of (2.26) equals a real number, allowing us to calculate $(1/2)!$ despite its strange nature. The integral

$$2 \int_0^1 \sqrt{x-x^2} dx \quad (2.27)$$

on the left of (2.26) may be evaluated by trigonometric substitution, but it is much simpler to recognize the integrand $\sqrt{x-x^2}$ as representing a semicircle, as follows. Let

$$y = \sqrt{x-x^2} \quad \text{or} \quad y^2 = x-x^2$$

and shuffle the terms to get

$$x^2 - x + y^2 = 0.$$

By adding $1/4$ to each side, we complete the square, so

$$x^2 - x + 1/4 + y^2 = 1/4 \quad \text{or} \quad (x - 1/2)^2 + y^2 = (1/2)^2.$$

This is the equation of a circle with center $(1/2, 0)$ and radius $1/2$. Now the limits in (2.27) are $x = 0$ and $x = 1$, which perfectly capture the upper half of this circle. The area of a semicircle is $(1/2) \pi r^2$ so with $r = 1/2$ we can conclude from (2.26) that

$$2 \int_0^1 \sqrt{x - x^2} dx = 2 \cdot (1/2) \pi (1/2)^2 = \frac{\pi}{4}.$$

Returning with this result to (2.26), we see that

$$(1/2)! = \frac{\sqrt{\pi}}{2} \approx 0.88623.$$

Euler did not content himself with this peculiar result, but we will depart from him here, now that the constant π has manifested itself in the pursuit of a seemingly unrelated inquiry.

2.9 The Basel problem: $\sum 1/k^2 = \pi^2/6$

In 1644, mathematician **Pietro Mengoli** (Italy, born 1625) posed a question that he could not answer: what is the sum of the reciprocals of the perfect squares? In symbols, Mengoli asked to what did the series

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots \quad (2.28)$$

converge. Almost a century passed before Euler solved this puzzle. One of his solutions uses trigonometric substitution and the power series for $\arcsin x$, and it testifies not only to the creative power of calculus but also to Euler's ingenuity. We begin with the definite integral

$$\int_0^1 \frac{\arcsin x}{\sqrt{1 - x^2}} dx \quad (2.29)$$

and evaluate it using the substitution

$$x = \sin \theta, \quad \text{so} \quad dx = \cos \theta d\theta \quad \text{and} \quad \arcsin x = \theta. \quad (2.30)$$

The identity $\cos \theta = \sqrt{1 - \sin^2 \theta}$ gives us

$$\cos \theta = \sqrt{1 - x^2}$$

and the limits $x = 0$ and $x = 1$ become $\theta = 0$ and $\theta = \pi/2$, so we may write (2.29) as

$$\int_0^1 \frac{\arcsin x}{\sqrt{1-x^2}} dx = \int_0^{\pi/2} \frac{\theta \cos \theta d\theta}{\cos \theta} = \left[\frac{\theta^2}{2} \right]_0^{\pi/2} = \frac{\pi^2}{8}. \quad (2.31)$$

Now we evaluate the original integral (2.29) a second way, by replacing^[27] $\arcsin x$ with its power series

$$\arcsin x = x + \frac{1}{2 \cdot 3} x^3 + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5} x^5 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 7} x^7 + \cdots$$

to reach

$$\int_0^1 \frac{x dx}{\sqrt{1-x^2}} + \frac{1}{2 \cdot 3} \int_0^1 \frac{x^3 dx}{\sqrt{1-x^2}} + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5} \int_0^1 \frac{x^5 dx}{\sqrt{1-x^2}} + \cdots \quad (2.32)$$

Each of these integrals is of the form

$$\int_0^1 \frac{x^{2n+1} dx}{\sqrt{1-x^2}},$$

and via the substitution (2.30) from earlier, we may rewrite them as

$$\int_0^{\pi/2} (\sin^{2n+1} \theta) d\theta. \quad (2.33)$$

They belong to the family of integrals that we used in section 2.4 to prove Wallis's formula. Specifically, our integrals here in (2.33) are the odd members of that family, and are evaluated in the right column of (2.13). Using those results to simplify (2.32), we obtain

$$1 + \frac{1}{2 \cdot 3} \cdot \frac{2}{3} + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5} \cdot \frac{4}{5} \cdot \frac{2}{3} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 7} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} + \cdots$$

or, simplifying and then linking the result to (2.31), we have

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots \quad (2.34)$$

This answers the question, "What is the sum of the reciprocals of the *odd* perfect squares?" Not quite Mengoli's query, but close enough that with a cute trick we can bridge the gap. Let S denote the sum of Mengoli's series (2.28), so

$$S = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

or, sorting odds from evens,

$$S = \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \right) + \left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \cdots \right).$$

By (2.34) we know that the expression in the first set of brackets equals $\pi^2/8$, and, factoring $1/4$ from each term in the second set of brackets, we have

$$S = \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \right) + \frac{1}{4} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \right)$$

or simply

$$S = \frac{\pi^2}{8} + \frac{1}{4} S.$$

Solving for S yields

$$S = \frac{\pi^2}{6} \tag{2.35}$$

and Euler's solution to Mengoli's vexing puzzle is complete.

Before Euler solved the problem, **Jacob Bernoulli** (Switzerland, born 1655) published some partial results in Basel, Switzerland, and thus "the Basel problem" became its nickname. Coincidentally, Basel was Euler's birthplace. In exercise 3, we will see how complex numbers allow for a more streamlined solution to the Basel problem.

2.10 π is irrational

In section 1.6, we classified the golden ratio φ as an irrational number (by several different arguments, including a proof that $\sqrt{5}$ is irrational). The number π is likewise irrational, as we will see in a moment. As an aside, we note that although both φ and π are irrational, we say that π is *transcendental* while φ is not. A transcendental number cannot be the zero of a polynomial with integer coefficients. Because φ is a solution to

$$x^2 - x - 1 = 0, \tag{2.36}$$

as discussed at (1.1), it is therefore not transcendental. It can be shown that π cannot ever be the solution to an equation of the form (2.36), where the coefficients are integers, so π is transcendental. Such a proof is beyond the reach of this book, but with some clever integration, we can at least show that π is irrational.

We begin by evaluating

$$\int_0^\pi f(x) \sin x \, dx,$$

where $f(x)$ is a function to be named later, using integration by parts with

$$\begin{aligned} u &= f(x) & \text{so} & \quad du = f'(x) \, dx, \\ dv &= \sin x \, dx & \text{so} & \quad v = -\cos x. \end{aligned}$$

This yields

$$\int_0^\pi f(x) \sin x \, dx = \left[-f(x) \cos x \right]_0^\pi + \int_0^\pi f'(x) \cos x \, dx.$$

The first expression on the right simplifies to $f(\pi) + f(0)$. The integral on the right, with

$$\begin{aligned} u &= f'(x) & \text{so} & \quad du = f''(x) \, dx, \\ dv &= \cos x \, dx & \text{so} & \quad v = \sin x, \end{aligned}$$

may be written

$$\int_0^\pi f'(x) \cos x \, dx = \left[f'(x) \sin x \right]_0^\pi - \int_0^\pi f''(x) \sin x \, dx.$$

This time, the first expression on the right simplifies to zero. Now all of these pieces together give us

$$\int_0^\pi f(x) \sin x \, dx = f(\pi) + f(0) - \int_0^\pi f''(x) \sin x \, dx. \quad (2.37)$$

This last integral matches the original (on the left) with f'' in the role of f , so we can predict that

$$\int_0^\pi f''(x) \sin x \, dx = f''(\pi) + f''(0) - \int_0^\pi f^{(4)}(x) \sin x \, dx$$

and so on. Thus from (2.37) and this prediction we conclude that

$$\begin{aligned} \int_0^\pi f(x) \sin x \, dx &= \left[f(\pi) - f''(\pi) + f^{(4)}(\pi) - f^{(6)}(\pi) + \cdots \right] \\ &\quad + \left[f(0) - f''(0) + f^{(4)}(0) - f^{(6)}(0) + \cdots \right]. \end{aligned} \quad (2.38)$$

If we choose $f(x)$ to be a polynomial of degree m , then every derivative after $f^{(m)}$ in each bracket will equal zero, and (2.38) will become

$$\int_0^\pi f(x) \sin x \, dx = \left[f(\pi) - f''(\pi) + f^{(4)}(\pi) - \cdots \pm f^{(m)}(\pi) \right] \\ + \left[f(0) - f''(0) + f^{(4)}(0) - \cdots \pm f^{(m)}(0) \right]. \quad (2.39)$$

With (2.39) established, we begin a proof by contradiction by assuming that π is rational, so

$$\pi = \frac{p}{q}$$

for some natural numbers p and q . Now the heart of this proof hinges on the discovery of a polynomial $f(x)$ of degree m such that

- (1) For every m , every term on the right of (2.39) is an integer, yet
- (2) For some m , the integral on the left of (2.39) is not an integer.

The polynomial

$$f(x) = \frac{x^n(p - qx)^n}{n!} \quad (2.40)$$

(where n is a natural number) does the trick. If we were to expand the numerator, we would find that the degree of each term ranges from x^n to x^{2n} . This has two^[28] implications: the first $n - 1$ derivatives of $f(x)$ equal zero when evaluated at $x = 0$, and the denominator $n!$ will be fully canceled as of the n th derivative. So, because p and q are natural numbers, then

$$f^{(k)}(0) \quad \text{will be an integer for } k = n, n + 1, \dots, 2n.$$

Every term within the second pair of brackets in (2.39) is therefore an integer. To show that all terms within the first pair of brackets are also integers, we appeal to the symmetry within $f(x)$ by noting^[29] that

$$f\left(\left(\frac{p}{q}\right) - x\right) = f(x).$$

That is, the polynomial $f(x)$ is identical to the polynomial $f\left(\left(\frac{p}{q}\right) - x\right)$, and so their respective derivatives match as well. Because $f(0)$ is an integer (as we have just seen), then

$$f\left(\left(\frac{p}{q}\right) - 0\right) = f\left(\frac{p}{q}\right) = f(\pi)$$

is an integer (and the same holds for their derivatives). These observations establish the first of our two claims.

Next (and finally) we wish to prove the second claim: that for some m , the definite integral on the left of (2.39) is not an integer. With our polynomial (2.40) replacing $f(x)$, the integral becomes

$$\int_0^\pi \frac{x^n (p - qx)^n \sin x}{n!} dx, \quad (2.41)$$

and (as previously noted) the degree m of the polynomial is $2n$. Let's maximize each of the three factors in the numerator

$$x^n (p - qx)^n \sin x$$

one at a time on the interval $x \in (0, \pi)$. We know that $0 < \sin x \leq 1$ when $x \in (0, \pi)$, and clearly the factor x^n is maximized at $x = \pi$, so

$$x^n (p - qx)^n \sin x \leq \pi^n (p - qx)^n.$$

The curve $p - qx$ is a straight line with y -intercept p and negative slope, so its maximum equals p on our interval. Thus,

$$\pi^n (p - qx)^n \leq \pi^n p^n = (\pi p)^n.$$

We now have an upper bound for the numerator in the integral (2.41). All three factors in the numerator of the integral are positive for $x \in (0, \pi)$.

We can conclude, therefore, that

$$0 < \int_0^\pi \frac{x^n (p - qx)^n \sin x}{n!} dx \leq \int_0^\pi \frac{(\pi p)^n}{n!} dx = \pi^n \int_0^\pi \frac{p^n}{n!} dx. \quad (2.42)$$

The integrand $p^n/n!$ is positive and, for a suitably large value of n , it can be made^[30] as small as we wish. In particular, we will choose n so that $p^n/n!$ is less than $1/\pi^{n+1}$ so that the last expression in (2.42) is less than 1, and therefore not an integer. The second claim is thereby proved.

So, under the assumption that $\pi = p/q$ we showed that there is a polynomial $f(x)$ that creates a contradiction in (2.39). That is, there exists an n so that the left side is not an integer while the right side is. Our assumption that π is rational must therefore be incorrect, and we are finished.

2.11 Further content

2.1 A geometric derivation of a formula for π . Machin's formula

$$\frac{\pi}{4} = 4 \arctan(1/5) - \arctan(1/239) \quad (2.43)$$

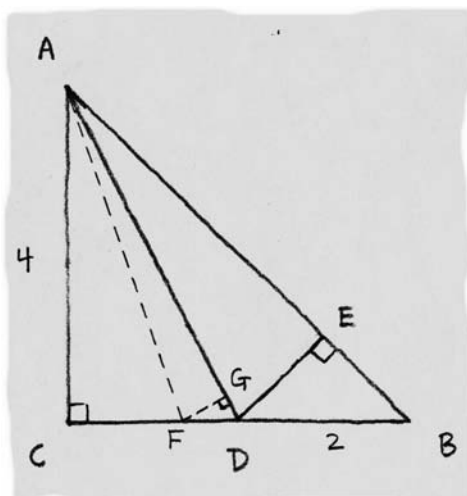


Figure 2.4

was verified in section 2.3 with the help of a trigonometric identity. We can prove the similar formula

$$\frac{\pi}{4} = \arctan(1/2) + \arctan(1/3) \quad (2.44)$$

using just basic geometry. Triangle ABC in Figure 2.4 is an isosceles right triangle with legs of length 4 (so by the Pythagorean theorem, the hypotenuse AB has length $4\sqrt{2}$). Point D bisects leg CB , and segment DE is perpendicular to AB .

(a) Because

$$\tan \angle CAD = \frac{CD}{AC} = \frac{2}{4},$$

we have $\angle CAD = \arctan(1/2)$. By finding the lengths of DE and AE , show that

$$\angle DAB = \arctan(1/3). \quad (2.45)$$

- (b) The results in the previous question are enough to establish formula (2.44). As we did in section 2.3, use the first three terms of the series (2.7) for arctangent to approximate π .
- (c) Formula (2.43) approximates π more quickly than (2.44) does because of its larger denominators. So let's improve on (2.44)

by embellishing Figure 2.4 a bit. Locate point F on CB so that $\angle CAF = \angle DAB$. Now use (2.45) to argue that

$$FC = \frac{4}{3}, \quad \text{and thus} \quad FD = \frac{2}{3}.$$

- (d) Use the Pythagorean theorem to show that $AD = 2\sqrt{5}$.
 (e) Locate point G on AD so that FG is perpendicular to AD . Because $\triangle ACD$ and $\triangle FGD$ each contain a right angle and also share $\angle ADC$, we know that they are similar. Use this to establish that

$$GD = \frac{2}{3\sqrt{5}} \quad \text{and} \quad FG = \frac{4}{3\sqrt{5}}. \quad (2.46)$$

- (f) From the figure it is clear that

$$\tan \angle FAD = \frac{FG}{AG}.$$

Use (2.46) to show that this ratio is $1/7$.

- (g) So we have shown that

$$\frac{\pi}{4} = 2 \arctan(1/3) + \arctan(1/7).$$

Use this formula, and the first three terms of the series for arctangent, to approximate π .

2.2 Wallis's formula for $\pi/2$ (via algebra). Section 2.4 presented a proof of Wallis's formula

$$\frac{\pi}{2} = \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \frac{6}{7} \times \cdots \quad (2.47)$$

that depended heavily on calculus. The alternative argument outlined here requires only algebra.

Let $s_0 = 0$, $s_1 = 1$ and, for $n \geq 2$, define

$$s_n = \frac{3}{2} \times \frac{5}{4} \times \cdots \times \frac{2n-1}{2n-2}.$$

- (a) Let $e_1 = 1$ and, for $n \geq 2$, let e_n denote the product of the first $2n-2$ factors of (2.47). (The letter e stands for even.) So,

$$e_1 = 1,$$

$$e_2 = \frac{4}{3} \quad \text{from} \quad \frac{2}{1} \times \frac{2}{3},$$

$$e_3 = \frac{64}{45} \quad \text{from} \quad \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5},$$

and so on. Write a similar definition for o_n (o for odd) and calculate o_1, o_2, o_3 .

- (b) Explain why

$$e_1 < e_2 < e_3 < \cdots < e_n < o_n < \cdots < o_3 < o_2 < o_1$$

and justify the identities

$$e_n = \frac{2n-1}{(s_n)^2} \quad \text{and} \quad o_n = \frac{2n}{(s_n)^2}.$$

- (c) From the results gathered so far, derive the inequality

$$\frac{2i-1}{e_n} \leq (s_i)^2 \leq \frac{2i}{o_n} \quad \text{for } 1 \leq i \leq n. \quad (2.48)$$

- (d) Now we appeal to geometry. Figure 2.5 shows the first quadrant of an xy -plane partitioned into rectangles $R_{i,j}$ with lower left corner (s_i, s_j) and upper right corner (s_{i+1}, s_{j+1}) . Let P_n denote the polygonal region consisting of rectangles $R_{i,j}$ with $i+j < n$. The corners of P_n farthest from the origin are the points (s_i, s_j) with $i+j = n+1$. Use the Pythagorean theorem and (2.48) to explain why the maximum distance from these corners to the origin is no more than

$$\sqrt{\frac{2(n+1)}{o_n}}. \quad (2.49)$$

Then explain why the maximum distance from the inner corners of P_n (where $i+j = n$) is no less than

$$\sqrt{\frac{2(n-1)}{e_n}}. \quad (2.50)$$

- (e) By the previous part, we know that P_n is contained in a quarter circle of radius (2.49) and that P_n contains a quarter circle of radius (2.50). Show how this sentence translates with the help of algebra to

$$\frac{(n-1)\pi}{2e_n} < \text{area of } P_n < \frac{(n+1)\pi}{2o_n}. \quad (2.51)$$

- (f) Now we will show that the area of polygonal region P_n simply equals n . The heights and widths of rectangles $R_{i,j}$ are differences in consecutive s_i values, so let's define

$$a_n = s_{n+1} - s_n.$$

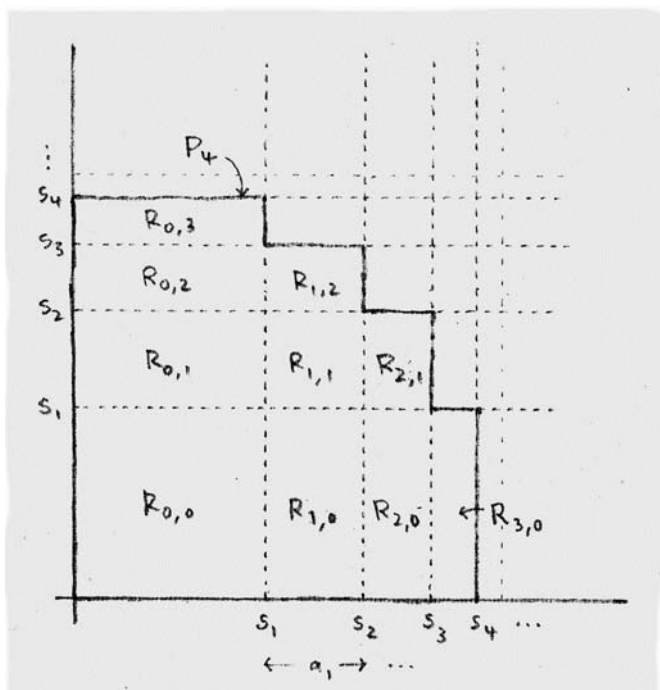


Figure 2.5

First, explain why

$$a_i a_j = \frac{j+1}{i+j+1} a_i a_{j+1} + \frac{i+1}{i+j+1} a_{i+1} a_j. \quad (2.52)$$

Then apply (2.52) to each term in

$$a_0 a_n + a_1 a_{n-1} + \cdots + a_n a_0$$

to convert this expression to

$$a_0 a_{n+1} + a_1 a_n + \cdots + a_{n+1} a_0.$$

Why does this equivalence imply that the area of P_n is n ? (Hint: start with $a_0 a_0 = 1$.)

- (g) So we may substitute n for the area of P_n in (2.51). Explain why it follows that

$$\frac{(n-1)\pi}{2n} < e_n < o_n < \frac{(n+1)\pi}{2n}. \quad (2.53)$$

In your own words, how does taking the limit as $n \rightarrow \infty$ in each expression in (2.53) yield Wallis's formula?

2.3 Converting an infinite product to a continued fraction. In section 2.6, we expressed a function involving sine as an infinite product. Here we discover that such products may be expressed as continued fractions. This in turn will unlock a lovely formula for π (in exercise 4).

An infinite product $a_1 a_2 a_3 a_4 \cdots$ resists conversion because continued fractions involve sums, so we introduce this operation by converting the equivalent infinite product

$$(1 + b_1)(1 + b_2)(1 + b_3)(1 + b_4) \cdots$$

to a continued fraction.

- (a) Skipping to the punchline would obscure the process by which one might discover the desired conversion. So we begin slowly, with the guess

$$(1 + b_1)(1 + b_2) = 1 + \frac{b_1}{1 + \frac{y}{z}}, \quad (2.54)$$

where y and z are to be determined. Show that from (2.54) we can derive

$$\frac{b_1 + b_2 + b_1 b_2}{b_1} = \frac{z}{z + y}.$$

Now if we choose $z = b_1 + b_2 + b_1 b_2$, explain why $y = -b_2(1 + b_1)$.

- (b) Substituting for y and z in (2.54) yields

$$(1 + b_1)(1 + b_2) = 1 + \frac{b_1}{1 - \frac{b_2(1 + b_1)}{b_1 + b_2 + b_1 b_2}}. \quad (2.55)$$

To extend the result, observe that

$$(1 + b_1)(1 + b_2)(1 + b_3) = (1 + b_1)(1 + (b_2 + b_3 + b_2 b_3))$$

rewrites the triple product on the left as a double product that compares to the left of (2.55), with $b_2 + b_3 + b_2 b_3$ in the role of

b_2 . Replacing b_2 with $b_2 + b_3 + b_2b_3$ throughout (2.55), we find that the bottom fraction becomes

$$\frac{(b_2 + b_3 + b_2b_3)(1 + b_1)}{b_1 + (b_2 + b_3 + b_2b_3) + b_1(b_2 + b_3 + b_2b_3)}.$$

Show that this expression is equivalent to

$$\frac{b_2(1 + b_1)}{b_1 + b_2 + b_1b_2 - \frac{b_1b_3(1 + b_2)}{b_2 + b_3 + b_2b_3}},$$

thus extending our result to the conversion of a triple product.

Note. The trick used in part (b) extends naturally to a proper induction proof of the full result that

$$(1 + b_1)(1 + b_2)(1 + b_3)(1 + b_4) \cdots \quad (2.56)$$

can be written as the continued fraction

$$1 + \frac{b_1}{1 - \frac{b_2(1 + b_1)}{b_1 + b_2 + b_1b_2 - \frac{b_1b_3(1 + b_2)}{b_2 + b_3 + b_2b_3 - \frac{b_2b_4(1 + b_3)}{b_3 + b_4 + b_3b_4 - \cdots}}}}.$$

Exercise 4 puts this conversion formula to good use.

2.4 A harmonic continued fraction for π . Here we may use the technique described in exercise 3 to transform

$$\frac{\sin \pi x}{\pi x} = (1 - x)(1 + x) \left(1 - \frac{x}{2}\right) \left(1 + \frac{x}{2}\right) \left(1 - \frac{x}{3}\right) \cdots \quad (2.57)$$

(which we obtain from (2.22) by replacing x with πx) into a continued fraction for π that features a simple and beautiful sequence.

From (2.57), we get

$$\frac{2}{\pi} = \left(1 - \frac{1}{2}\right) \left(1 + \frac{1}{2}\right) \left(1 - \frac{1}{4}\right) \left(1 + \frac{1}{4}\right) \left(1 - \frac{1}{6}\right) \cdots \quad (2.58)$$

by the substitution $x = 1/2$.

- (a) Comparing (2.58) with (2.56) from exercise 3, we see that

$$b_1 = -1/2, \quad b_2 = 1/2, \quad b_3 = -1/4, \quad \text{and so on.}$$

Substitute these values into the continued fraction version of (2.56) and reach

$$\frac{2}{\pi} = 1 - \frac{1}{2 + \frac{1 \cdot 2}{1 + \frac{2 \cdot 3}{1 + \frac{3 \cdot 4}{1 + \dots}}}} \quad (2.59)$$

via some appropriate algebra.

- (b) While the continued fraction (2.59) evidences a nice pattern, we can further tidy it up. Denote by Q the infinite tail end of (2.59), and observe that

$$\frac{2}{\pi} = 1 - \frac{1}{2 + Q} = \frac{1 + Q}{2 + Q}.$$

Inverting both sides, we have

$$\frac{\pi}{2} = \frac{2 + Q}{1 + Q} = 1 + \frac{1}{1 + Q}.$$

With the help of these observations and a bit of simplifying, discover the lovely continued fraction

$$\frac{\pi}{2} = 1 + \frac{1}{1 + \frac{1}{1/2 + \frac{1}{1/3 + \frac{1}{1/4 + \dots}}}} \quad (2.60)$$

with the *harmonic series* as its prominent feature.

- 2.5 Finding the n th digit of π .** As long as we are patient, we can produce any number of digits of π using formulas like

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots \quad \text{and}$$

$$\frac{\pi}{4} = 4 \arctan(1/5) - \arctan(1/239)$$

(Nilakantha's (2.8) and Machin's (2.10), respectively). Quite recently, a team of mathematicians discovered a way to leap over any number of digits of π and land where we wish. The only limitation is that we must work not in base ten but in a base (like binary or hexadecimal) that is a power of two.

- (a) We will get to π in a bit. First, by a similar but simpler process, we will calculate the n th binary digit of $\ln 2$. Integrate the series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

and substitute $x = 1/2$ into the result to prove that

$$\ln 2 = \sum_{k=1}^{\infty} \frac{1}{k \cdot 2^k} = \frac{1}{1 \cdot 2^1} + \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} + \dots \quad (2.61)$$

- (b) Expressed in binary,

$$\ln 2 = 0.101100010111001000010111111101 \dots \quad (2.62)$$

Let d_n represent the n th digit of $\ln 2$ in binary. Suppose that we wish to calculate d_{30} without calculating the preceding 29 digits. Note that multiplying (2.62) by 2^{29} yields

$$2^{29} \ln 2 = 10110001011100100001011111110.1 \dots ,$$

having the effect of shifting the decimal 29 places to the right, leaving d_{30} as the digit directly after the decimal. Thus, if we can calculate the fractional part of $2^{29} \ln 2$, then d_{30} will be its first digit.

So we multiply (2.61) by 2^{29} , simplify each term, and do a little grouping, to get

$$\begin{aligned} 2^{29} \ln 2 = & \left[\frac{2^{28}}{1} + \frac{2^{27}}{2} + \frac{2^{26}}{3} + \dots + \frac{2^2}{27} + \frac{2^1}{28} \right] \\ & + \left[\frac{1}{29} + \frac{1}{2 \cdot 30} + \frac{1}{2^2 \cdot 31} + \frac{1}{2^3 \cdot 32} + \dots \right]. \quad (2.63) \end{aligned}$$

We want to determine the first digit after the decimal of the expression on the right. The fractions in the second group are easy to translate to binary via long division, and they converge to 0

quite quickly, so we would stop when it is clear that they will no longer affect the value of d_{30} .

The fractions in the first group are trickier. The first two simplify to integers, so they have no bearing on the value of d_{30} (because it is the first digit after the decimal). But it is not immediately clear if $2^{26}/3$ is an integer. Further, if it has a fractional part, it will be $1/3$ or $2/3$, but which? And while 2^{26} has just eight digits, and is therefore accessible, what if we wanted the quadrillionth digit of $\ln 2$ instead of the 30th?

Modular arithmetic comes to our rescue. Two true equivalences multiply to yield a third:

$$\left. \begin{array}{l} a \equiv b \pmod{m} \\ c \equiv d \pmod{m} \end{array} \right\} \implies ac \equiv bd \pmod{m}.$$

To calculate $2^{26} \pmod{3}$, we find certain powers of 2 modulo 3:

$$2^1 \equiv 2 \pmod{3}$$

$$2^2 \equiv 1 \pmod{3} \leftarrow$$

$$2^4 \equiv 1 \pmod{3}$$

$$2^8 \equiv 1 \pmod{3} \leftarrow$$

$$2^{16} \equiv 1 \pmod{3} \leftarrow$$

by starting with the obvious top equivalence, then squaring each side (and reducing). Because $26 = 16 + 8 + 2$, we multiply the equivalences marked with arrows to reach

$$2^{26} \equiv 1 \pmod{3},$$

which tells us that the fractional part of $2^{26}/3$ is $1/3$. Because only the fractional part can affect d_{30} , we can ignore the integer part of $2^{26}/3$ (whatever it is). In this way, we can march through the terms that have powers of 2 in their numerators, reducing each one to a fraction.

Here are the partial results of applying this process to the fractions in (2.63). The $\rightarrow$ symbol indicates that we are keeping only

the fractional part.

$$2^{28}/1 \rightarrow 0$$

$$2^{27}/2 \rightarrow 0$$

$$2^{26}/3 \rightarrow 1/3$$

$$2^{25}/4 \rightarrow 0$$

$$2^{24}/5 \rightarrow 1/5$$

$$2^{23}/6 \rightarrow 2/6$$

$$2^{22}/7 \rightarrow 2/7$$

$$\vdots$$

$$2^2/27 \rightarrow 4/27$$

$$2^1/28 \rightarrow 2/28$$

Verify that each of these results is correct.

- (c) Adding together these fractions (as well as a few of the fractions in (2.63), starting with $1/29$) and converting to binary yields

$$1010.1111100011\dots,$$

and we can ignore everything except the first digit after the decimal, which establishes that $d_{30} = 1$.

Now that we know how to find the n th digit of $\ln 2$ by first expressing it as a sum of fractions, we can look for a way to express π as a similar sum. This task was completed in 1996 with the aid of a computer program, resulting in

$$\pi = \sum_{k=0}^{\infty} \frac{1}{2^{4k}} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right). \quad (2.64)$$

The terms in this sum differ a bit from those in (2.61), but only in that the factor k in the denominators has been replaced by polynomials in terms of k . This has no effect on the process described above.

We can verify (2.64) using techniques from elementary calculus. To begin, show that

$$\int_0^1 \left(\frac{4y}{y^2 - 2} - \frac{4y - 8}{(y - 1)^2 + 1} \right) dy = \pi$$

using (and these are merely suggestions) substitution and trigonometric substitution.

- (d) Combine the terms in the integrand and multiply the resulting numerator and denominator by

$$(y^2 + 2)(y^2 + 2y + 2)$$

to reach

$$\int_0^1 \frac{16(y^5 + y^4 + 2y^3 - 4)}{y^8 - 16} dy = \pi.$$

- (e) Substitute $y = \sqrt{2}x$ (and so $dy = \sqrt{2} dx$) to find that

$$\int_0^{1/\sqrt{2}} \frac{4\sqrt{2} - 8x^3 - 4\sqrt{2}x^4 - 8x^5}{1 - x^8} dx = \pi. \quad (2.65)$$

- (f) To link (2.65) to (2.64), we show that for $k \geq 1$ we have

$$\sqrt{2}^k \int_0^{1/\sqrt{2}} \frac{x^{k-1}}{1 - x^8} dx = \sum_{n=0}^{\infty} \frac{1}{2^{4n}(8n + k)}. \quad (2.66)$$

Establish this result using geometric series.

- (g) Finally, prove (2.64) by applying (2.66) to each of the four terms of (2.65).

Postscript. Using (2.64) to calculate digits of π is best done with computer help, so we will not do so here. But we pause to note, with satisfaction, that although the birth of elementary calculus is several hundred years in the past, and the study of the constant π is even more ancient, the discovery outlined in this exercise eluded us until not long ago. We now know, for example, the binary digit of π that sits in position 2×10^{15} .

CHAPTER 3

e

As a sociologist, my dad was familiar with studies that probe how the human mind works, and he occasionally performed similar experiments on my sister and me. For a Christmas gift, he gave each of us a two-pound chocolate kiss and a challenge: if either of us could eat the whole thing that day, he would reward us with another one. (This inverts the Stanford marshmallow experiment in which children are given a marshmallow and rewarded with another if they do *not* immediately eat the first.) I finally gave up gnawing on mine when it was a tooth-scarred ping-pong ball. My sister didn't try.

Dad also gave me an envelope containing a one dollar bill under the condition that I gift it to a total stranger with no explanation. (I still have it.) He once presented me a trade of all the money in his wallet for my next five weeks' allowance of five dollars. Conservative and suspicious, I opted not to play. (His wallet had over thirty dollars in it.)

3.1 The money puzzle

Many of my dad's challenges were puzzles, and one in particular takes us to our next constant. He offered me ten dollars or the option to split the ten and multiply the parts. Like five times five is twenty-five, I asked, and I get that much? Yup. So I went to work — two times eight is sixteen, three times seven is twenty-one — soon deciding that my original guess was best. But then inspiration struck: what about three times three times three times one is twenty-seven? Can I have that? Caught between the

wish to reward my initiative and the fear of the unknown, my dad decided to risk it, allowing more than two parts.

The wasted 1 in $3 \times 3 \times 3 \times 1 = 27$ bothered me until I overruled my desire to make the parts equal and lumped it in with one of the threes, obtaining $3 \times 3 \times 4 = 36$. My dad gulped.

It wasn't long before I hit on $2.5 \times 2.5 \times 2.5 \times 2.5 = 39.06$, prompting an emergency ruling. That's it, my dad said. I thought the answer was twenty-five. Stop. (But he needn't have panicked, because the maximum possible product is less than a dollar more.) Our search for that maximum product will lead us to our next constant.

A bit of experimentation would persuade anyone that the product of the parts of the ten dollars will be greatest if the parts are equal (and, in fact, the arithmetic geometric mean inequality (1.17) assures us^[31] of this). If x is the amount in each part, then $x^{10/x}$ is the product of the parts, so I was searching for the value of x that would maximize $x^{10/x}$. For example, when I split 10 into four equal parts of 2.5 each, the yield was $2.5^{10/2.5} = 2.5^4 = 39.06$. While as a child I would have only considered splitting the ten dollars into a whole number of parts, let's relax that stipulation and allow such products as

$$2.6^{10/2.6} \approx 2.6^{3.85} \approx 39.45,$$

which beats our current record.

A bit of exploring (as in Table 3.1) suggests that $x^{10/x}$ reaches a maximum at a value near $x = 2.7$. In the next few sections, we will capture

$2.5^{10/2.5}$	≈ 39.06
$2.6^{10/2.6}$	≈ 39.45
$2.7^{10/2.7}$	≈ 39.60
$2.8^{10/2.8}$	≈ 39.54
$2.9^{10/2.9}$	≈ 39.31

Table 3.1

the constant that would have maximized my dad's payout. Leonhard Euler generally receives credit for being the first person to incisively explore the properties and importance of the constant. It was he who chose the letter *e* to denote it. Although others, such as John Napier and Gottfried Leibniz, had knowledge of *e* before Euler, it was Euler's investigations that provided the deepest insights.

3.2 Euler's $e = \sum 1/k!$

Living before the advent of limits, Euler calculated with “infinitely small” numbers (or “infinitesimals”), that is, numbers that (metaphorically) are infinitely close to zero without equaling zero. His search for an infinite series involving e began, in his words, “Since $a^0 = 1$, when the exponent on a increases, the power itself increases, provided a is greater than 1. It follows that if the exponent is infinitely small and positive, then the power also exceeds 1 by an infinitely small number. Let ω be an infinitely small number, or a fraction so small that, although not equal to zero, still $a^\omega = 1 + \psi$, where ψ is also an infinitely small number.”

Examples may help make Euler's intent clear. Suppose that we choose $a = 2$ and $\omega = 0.001$. Then

$$2^{0.001} = 1.0006933874 \dots \quad \text{and so} \quad \psi = 0.0006933874 \dots$$

and we see that ψ is a bit less than ω and that both are close to zero. Neither is, of course, *infinitely* small, but together they illustrate Euler's point. If we change a to 3 and keep $\omega = 0.001$, we have

$$3^{0.001} = 1.0010992159 \dots \quad \text{and so} \quad \psi = 0.0010992159 \dots$$

instead, and we note that $\omega > \psi$ in the first case while $\omega < \psi$ in the second. Perhaps there is a number a between 2 and 3 for which $\omega = \psi$? Testing the special number 2.7 from section 3.1, we find that

$$2.7^{0.001} = 1.0009937452 \dots \quad \text{and so} \quad \psi = 0.0009937452 \dots$$

and we have our closest match between ω and ψ yet. It seems that pursuing the constant in Euler's context will also answer my dad's money puzzle.

So we are narrowing in on the particular number a that forces ω to equal ψ , so that

$$a^\omega = 1 + \omega.$$

To this end, let n be an infinitely large number (in the same poetic way that ω is infinitely small) such that $\omega = 1/n$, and rewrite the equation as

$$a^{1/n} = 1 + \frac{1}{n} \quad \text{or} \quad a = \left(1 + \frac{1}{n}\right)^n. \quad (3.1)$$

Now the generalized binomial theorem (2.23) allows us to expand the right side to obtain

$$\begin{aligned} a &= 1 + \frac{n}{1} \cdot \left(\frac{1}{n}\right)^1 + \frac{n}{1} \cdot \frac{n-1}{2} \cdot \left(\frac{1}{n}\right)^2 \\ &\quad + \frac{n}{1} \cdot \frac{n-1}{2} \cdot \frac{n-2}{3} \cdot \left(\frac{1}{n}\right)^3 + \cdots \\ &= 1 + \frac{1}{1!} + \frac{n-1}{n} \cdot \frac{1}{2!} + \frac{n-1}{n} \cdot \frac{n-2}{n} \cdot \frac{1}{3!} + \cdots \end{aligned}$$

as long as we accept that the theorem applies to exponents that are infinitely large. Dividing every term in each fraction (that contains n) by n , we obtain

$$a = 1 + \frac{1}{1!} + \frac{1 - (1/n)}{1} \cdot \frac{1}{2!} + \frac{1 - (1/n)}{1} \cdot \frac{1 - (2/n)}{1} \cdot \frac{1}{3!} + \cdots$$

Because n is infinitely large, all of the fractions within parentheses vanish. Thus we have

$$a = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \quad \text{or} \quad a = \sum_{k=0}^{\infty} \frac{1}{k!}. \quad (3.2)$$

As Euler did, we will let e denote this constant:

$$e \approx 2.71828.$$

Putting (3.2) together with (3.1), and using modern notation, we arrive at two beautiful truths about the constant e :

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad \text{and} \quad e = \sum_{k=0}^{\infty} \frac{1}{k!}. \quad (3.3)$$

Euler's argument fails at times to be rigorous in the modern sense. However, with the proper use of inequalities and limits, we can prove that these conclusions are correct. For example, we will see in section 3.4 that a judicious use of the arithmetic geometric mean inequality proves that the limit on the left of (3.3) exists.

Before we move on, let's investigate the expression

$$\left(1 + \frac{x}{n}\right)^n$$

where x is a constant. This expression is quite like the one on the right in (3.1), and in fact we will manipulate this new expression much as Euler did his. Using the generalized binomial theorem to expand, we find that

$$\left(1 + \frac{x}{n}\right)^n = 1 + \frac{n}{1} \cdot \frac{x}{n} + \frac{n}{1} \cdot \frac{n-1}{2} \cdot \frac{x^2}{n^2} + \frac{n}{1} \cdot \frac{n-1}{2} \cdot \frac{n-2}{3} \cdot \frac{x^3}{n^3} + \dots$$

and, as before, we can manipulate^[32] this equation to the form

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

If we set e^x equal to this limit, so that

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad (3.4)$$

then with $x = 1$ we recapture both formulas in (3.3). These sorts of observations prompted Euler to declare (3.4) as the definition of e^x .

A final note: because the sum on the right side of (3.4) appears to equal its own derivative, we may suspect that

$$\frac{d}{dx} e^x = e^x.$$

The endnotes^[33] contain a proper proof of this well-known fact.

3.3 The maximum of $x^{1/x}$

Armed with (3.4), we return to the problem (from section 3.1) of maximizing $x^{10/x}$ (and thus maximizing the amount of money that my dad would owe me). Writing $x^{10/x}$ as $(x^{1/x})^{10}$ allows us to ignore the 10 for now and concentrate on maximizing the function $x^{1/x}$. Now finding the maximum of a function typically involves its derivative. Here we take a different approach.

We need to establish two facts before we maximize $x^{1/x}$ directly. The first fact carries the name Bernoulli's inequality, after Jacob Bernoulli, who we met in section 2.9.

Claim. If m is a natural number and $y \geq -1$, then

$$(1 + y)^m \geq 1 + my. \quad (3.5)$$

For example, when $m = 1$ we have the obviously true statement

$$1 + y \geq 1 + y$$

and when $m = 2$ we observe that

$$(1 + y)^2 = 1 + 2y + y^2 \geq 1 + 2y \quad (3.6)$$

because $y^2 \geq 0$ (not just when $y \geq -1$, as the claim specifies, but for all values). So (3.6) says that

$$(1 + y)^2 \geq 1 + 2y$$

and if we multiply by $(1 + y)$ we get

$$(1 + y)^3 \geq (1 + 2y)(1 + y) \quad (3.7)$$

as long as $(1 + y) \geq 0$ (that is, as long as $y \geq -1$). The claim therefore contains this stipulation. So now from (3.7) we continue:

$$\begin{aligned} (1 + y)^3 &\geq 1 + 3y + 2y^2 \\ &\geq 1 + 3y \end{aligned}$$

and we see that the claim holds for $m = 3$. A proper induction proof of the claim is included^[34] in the endnotes.

Our second task (before we maximize $x^{1/x}$) begins with the limit

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$$

from (3.4). We wish to show that if we choose $x \geq -1$ then the expression (inside the limit on the right) is monotone increasing, which is to say that as we let n increase, every term is at least as large as the previous one. For example, letting $x = 1$ (in the left column) and $x = 2$ (in the right) and taking n to be the first few natural numbers, we generate the approximations in Table 3.2, and see that they each form an increasing sequence. We can formalize our goal by proving the following claim.

Claim. With $x \geq -1$ and n a natural number, we have

$$\left(1 + \frac{x}{n+1}\right)^{n+1} \geq \left(1 + \frac{x}{n}\right)^n.$$

The claim states that by bumping n up to $n + 1$, we get a new value that is at least as large as the previous value, just as Table 3.2 suggests. Note that the expression in the claim is equivalent to

$$\frac{\left(1 + \frac{x}{n+1}\right)^{n+1}}{\left(1 + \frac{x}{n}\right)^n} \geq 1, \quad (3.8)$$

$(1 + \frac{1}{1})^1 = 2$	$(1 + \frac{2}{1})^1 = 3$
$(1 + \frac{1}{2})^2 = 2.25$	$(1 + \frac{2}{2})^2 = 4$
$(1 + \frac{1}{3})^3 \approx 2.370$	$(1 + \frac{2}{3})^3 \approx 4.630$
$(1 + \frac{1}{4})^4 \approx 2.441$	$(1 + \frac{2}{4})^4 \approx 5.063$
$(1 + \frac{1}{5})^5 \approx 2.488$	$(1 + \frac{2}{5})^5 \approx 5.378$
$(1 + \frac{1}{6})^6 \approx 2.522$	$(1 + \frac{2}{6})^6 \approx 5.619$

Table 3.2: Some approximations that suggest that $(1 + x/n)^n$ is monotone increasing.

and we will prove this alternate inequality. We begin with the left side. The first few steps are merely algebraic manipulation:

$$\begin{aligned}
 \frac{\left(1 + \frac{x}{n+1}\right)^{n+1}}{\left(1 + \frac{x}{n}\right)^n} &= \frac{\left(1 + \frac{x}{n}\right) \left(1 + \frac{x}{n+1}\right)^{n+1}}{\left(1 + \frac{x}{n}\right) \left(1 + \frac{x}{n}\right)^n} \\
 &= \left(1 + \frac{x}{n}\right) \left(\frac{1 + \frac{x}{n+1}}{1 + \frac{x}{n}}\right)^{n+1} \\
 &= \left(1 + \frac{x}{n}\right) \left(\frac{n(n+1) + xn}{n(n+1) + x(n+1)}\right)^{n+1} \\
 &= \left(1 + \frac{x}{n}\right) \left(\frac{n^2 + (x+1)n + x - x}{(n+1)(n+x)}\right)^{n+1}.
 \end{aligned}$$

The quirky decision (in the last step above) to interject $+x - x$ into the numerator allows us to factor $n^2 + (x+1)n + x$ as $(n+1)(n+x)$ and therefore write

$$\begin{aligned}
 \frac{\left(1 + \frac{x}{n+1}\right)^{n+1}}{\left(1 + \frac{x}{n}\right)^n} &= \left(1 + \frac{x}{n}\right) \left(\frac{(n+1)(n+x) - x}{(n+1)(n+x)}\right)^{n+1} \\
 &= \left(1 + \frac{x}{n}\right) \left(1 - \frac{x}{(n+1)(n+x)}\right)^{n+1}. \quad (3.9)
 \end{aligned}$$

Now compare the second factor

$$\left(1 - \frac{x}{(n+1)(n+x)}\right)^{n+1}$$

with $(1+y)^m$ from the claim (3.5). If we put $n+1$ in the role of m and

$$-\frac{x}{(n+1)(n+x)}$$

in the role of y , then we can use claim (3.5) to proceed. Note, however, that (3.5) insists that $y \geq -1$, so we must check that

$$-\frac{x}{(n+1)(n+x)} \geq -1 \quad (3.10)$$

before we use it. Recall (from the current claim) that $x \geq -1$ and n is a natural number, so $n \geq 1$. Thus the denominator of (3.10) is positive, so we can rewrite it as

$$x \leq (n+1)(n+x) \quad \text{or} \quad 0 \leq n^2 + n(x+1),$$

which is clearly true. So we may use claim (3.5) to write (3.9) as

$$\frac{\left(1 + \frac{x}{n+1}\right)^{n+1}}{\left(1 + \frac{x}{n}\right)^n} \geq \left(1 + \frac{x}{n}\right) \left(1 - (n+1) \cdot \frac{x}{(n+1)(n+x)}\right).$$

It is easy to verify that the right side simplifies to 1 and thus that we have proved the claim in its alternate form (3.8).

Armed with these two claims, we may finally show how to maximize the curve $x^{1/x}$ without using its derivative. At (3.4) we showed that

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$$

and we have just proved that the expression inside the limit on the right is monotone increasing as long as $x \geq -1$ and n is a natural number. Thus,

$$e^x \geq \left(1 + \frac{x}{n}\right)^n \quad \text{when } x \geq -1 \text{ and } n \geq 1.$$

Now we use claim (3.5), with x/n in the role of y and n in the role of m to extend this inequality and say that

$$e^x \geq \left(1 + \frac{x}{n}\right)^n \geq 1 + n \cdot \frac{x}{n} = 1 + x \quad \text{when } x \geq -1.$$

So (changing the variable from x to y for a moment) we have established that

$$e^y \geq 1 + y \quad \text{when } y \geq -1.$$

Recall that our original goal was to maximize $x^{1/x}$ when $x \geq 0$. Observe^[35] that

$$x \geq 0 \quad \rightarrow \quad \frac{x - e}{e} \geq -1,$$

so we may let

$$y = \frac{x - e}{e}$$

in (3.3) and conclude that

$$e^{(x-e)/e} \geq 1 + \frac{x - e}{e}.$$

Because this inequality simplifies^[36] to

$$e^{1/e} \geq x^{1/x},$$

we learn that $x^{1/x}$ reaches its maximum when $x = e$. So, at worst, my dad would have had to pay me $(e^{1/e})^{10} \approx 39.60$ dollars.

3.4 The limit of $(1 + \frac{1}{n})^n$

As promised in section 3.2, we will now see an example of how we may rigorously prove one of the limits that Euler discovered (by ingenious but mathematically shaky methods). Specifically, we will prove that the limit

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \tag{3.11}$$

from (3.3) exists. Our primary prop will be the arithmetic geometric mean inequality. Let's start by defining a sequence $\{y_n\}$ (where n is a natural number) as

$$y_n = \left(1 + \frac{1}{n}\right)^n$$

and a companion sequence $\{z_n\}$ as

$$z_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

Table 3.3 shows each sequence for the first few n values. Our proof that

$(1 + \frac{1}{1})^1 = 2.0$	$(1 + \frac{1}{1})^2 = 4.0$
$(1 + \frac{1}{2})^2 = 2.25$	$(1 + \frac{1}{2})^3 = 3.375$
$(1 + \frac{1}{3})^3 \approx 2.37037$	$(1 + \frac{1}{3})^4 = 3.16049$
$(1 + \frac{1}{4})^4 \approx 2.44141$	$(1 + \frac{1}{4})^5 = 3.05176$
$(1 + \frac{1}{5})^5 \approx 2.48832$	$(1 + \frac{1}{5})^6 = 2.98598$
$(1 + \frac{1}{6})^6 \approx 2.52163$	$(1 + \frac{1}{6})^7 = 2.94190$

Table 3.3

limit (3.11) exists is in four acts:

- (1) Show that $y_n < z_n$ for every n .
- (2) Show that $\{y_n\}$ is monotone increasing.
- (3) Show that $\{z_n\}$ is monotone decreasing.
- (4) Show that $\{y_n\}$ and $\{z_n\}$ share a common limit.

These observations appear to be plausible when one reads Table 3.3. We can prove the first using simple algebra.

Claim. We have $y_n < z_n$ for every n .

Because n is a natural number, we know that

$$\left(1 + \frac{1}{n}\right) > 1.$$

This observation explains the inequality step in what follows:

$$z_n = \left(1 + \frac{1}{n}\right)^{n+1} = \left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n}\right) > \left(1 + \frac{1}{n}\right)^n = y_n.$$

And so $y_n < z_n$ as desired.

The second and third claims follow from the arithmetic geometric mean inequality. As discussed in section 1.7, it states that for any set of non-negative numbers $a_1, a_2, \dots, a_n$, their geometric mean

$$G_n = (a_1 a_2 \cdots a_n)^{1/n}$$

is less than or equal to their arithmetic mean

$$A_n = \frac{a_1 + a_2 + \cdots + a_n}{n},$$

so

$$(a_1 a_2 \cdots a_n)^{1/n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}, \quad (3.12)$$

with equality only when all of the a_j are equal. Now we will use this inequality to prove the second claim.

Claim. The sequence $\{y_n\}$ is monotone increasing.

Let's warm up to this task by checking a couple of examples. We can directly calculate that

$$y_1 = \left(1 + \frac{1}{1}\right)^1 = 2, \quad (3.13)$$

$$y_2 = \left(1 + \frac{1}{2}\right)^2 = \left(\frac{3}{2}\right)^2 = 2.25,$$

so $y_1 < y_2$. But consider this unusual argument as an alternative: let $n = 2$ in (3.12) and choose $a_1 = 1$ and $a_2 = 2$ as our two numbers. Substituting into (3.12), we get

$$(1 \cdot 2)^{1/2} < \frac{1+2}{2}$$

with strict inequality because $a_1 \neq a_2$. Squaring both sides yields

$$2 < \left(\frac{3}{2}\right)^2 \quad \text{or} \quad \left(1 + \frac{1}{1}\right)^1 < \left(1 + \frac{1}{2}\right)^2,$$

which matches the direct calculation in (3.13). So we have verified our direct calculation with a clever use of the arithmetic geometric mean inequality. We will generalize this clever trick after we verify (just for practice) that $y_2 < y_3$. Let $n = 3$ in (3.12) and choose $a_1 = 1$ and $a_2 = a_3 = 3/2$ as our three numbers. Substituting into (3.12), we get

$$(1 \cdot 3/2 \cdot 3/2)^{1/3} < \frac{1 + 3/2 + 3/2}{3} \quad \text{or} \quad \left(1 + \frac{1}{2}\right)^2 < \left(1 + \frac{1}{3}\right)^3,$$

so $y_2 < y_3$ as desired.

Now we generalize: let's choose the n numbers

$$a_1 = 1 \quad \text{and} \quad a_2 = a_3 = \cdots = a_n = \frac{n}{n-1}.$$

Substituting into (3.12) yields

$$\left(1 \cdot \frac{n}{n-1} \cdots \frac{n}{n-1}\right)^{1/n} < \frac{1 + n/(n-1) + \cdots + n/(n-1)}{n}.$$

There are $n - 1$ copies of $n/(n - 1)$ on each side, so we may simplify to

$$\left(\frac{n}{n-1}\right)^{(n-1)/n} < \frac{n+1}{n}.$$

Raising both sides (of the second inequality) to the power n and doing a little algebra, we find that

$$\left(1 + \frac{1}{n-1}\right)^{n-1} < \left(1 + \frac{1}{n}\right)^n,$$

so $y_{n-1} < y_n$. In other words, we now know that the entire sequence $\{y_n\}$ is monotone increasing.

It may come as no surprise that we can prove the third claim in much the same way.

Claim. The sequence $\{z_n\}$ is monotone decreasing.

As before, we will warm up with an example, say $z_3 < z_2$. Let $n = 4$ in (3.12) and choose $a_1 = 1$ and $a_2 = a_3 = a_4 = 2/3$ as our four numbers. Substituting into (3.12), we get

$$(1 \cdot 2/3 \cdot 2/3 \cdot 2/3)^{1/4} < \frac{1 + 2/3 + 2/3 + 2/3}{4} \quad \text{or} \quad \left(\frac{2}{3}\right)^{3/4} < \frac{3}{4}.$$

Raising both sides to the power 4 and then inverting each fraction (which changes the inequality sign), we can simplify this to

$$\left(1 + \frac{1}{3}\right)^4 < \left(1 + \frac{1}{2}\right)^3,$$

which says $z_3 < z_2$ as we wished. To generalize, choose the n numbers

$$a_1 = 1 \quad \text{and} \quad a_2 = a_3 = \cdots = a_n = \frac{n-2}{n-1}.$$

Substituting into (3.12) and simplifying, we obtain

$$\left(\frac{n-2}{n-1}\right)^{(n-1)/n} < \frac{n-1}{n}.$$

Raising both sides to the power n and then inverting each fraction, we reach

$$\left(1 + \frac{1}{n-1}\right)^n < \left(1 + \frac{1}{n-2}\right)^{n-1},$$

so $z_{n-1} < z_{n-2}$. So we now know that the entire sequence $\{z_n\}$ is monotone decreasing.

To establish the final claim, we simply note that $z_n = (1 + 1/n)y_n$ implies that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{n+1},$$

and of course we let e denote this common limit. Because the sequences in these limits are monotone increasing and monotone decreasing, respectively, we can conclude that

$$\left(1 + \frac{1}{n}\right)^n < e < \left(1 + \frac{1}{n}\right)^{n+1} \quad (3.14)$$

for any natural number n , a fact that we will return to in exercise 4 when we talk about Euler's constant.

3.5 A modern proof that $e = \sum 1/k!$

Here is a proof of the sum in (3.3) that uses integration by parts, a technique that relies on limits, not infinitesimals, for its reliability. Consider the family of integrals

$$a_n = \frac{1}{n!} \int_0^1 x^n e^{-x} dx \quad (3.15)$$

where n is a natural number. The definite integral equals the area over the interval $x \in (0, 1)$ under the curve

$$y_n = \frac{1}{n!} x^n e^{-x}.$$

The subscript in y_n indicates that the curve depends on our choice of natural number n . Plots of the first few members of this family of curves suggest^[37] that the area bounded by the curves approaches zero as n grows larger. That is, we think that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n!} \int_0^1 x^n e^{-x} dx = 0 \quad (3.16)$$

is a safe bet. We can prove (3.16) by noting a few characteristics of the curve y_n . At the left boundary $x = 0$, we have

$$y_n(0) = \frac{1}{n!} 0^n e^{-0} = 0,$$

while at the right boundary $x = 1$ we have

$$y_n(1) = \frac{1}{n!} 1^n e^{-1} = \frac{1}{n! e}, \quad \text{so} \quad \lim_{n \rightarrow \infty} y_n(1) = 0.$$

So every curve in the family passes through the origin $(0, 0)$ on the left and a point approaching $(1, 0)$ on the right. Now if every curve in the family lacks a critical point in the interval $(0, 1)$, then we know that each curve has no maximum points on the interval, so it must increase across the entire interval. To check for critical points, we set the derivative

$$\frac{d}{dx} x^n e^{-x} = x^{n-1} e^{-x} (n - x)$$

equal to zero, and conclude^[38] that the curve has no critical points in the interval $(0, 1)$. These observations together prove (3.16).

We now turn to the family of integrals (3.15). Starting with $n = 0$, we calculate

$$a_0 = \frac{1}{0!} \int_0^1 x^0 e^{-x} dx = \int_0^1 e^{-x} dx = [-e^{-x}]_0^1 = 1 - \frac{1}{e}.$$

When $n = 1$, we use integration by parts^[39] to find that

$$a_1 = -\frac{1}{e} + \int_0^1 e^{-x} dx \quad \text{or} \quad a_1 = -\frac{1}{e} + a_0.$$

Note that a_1 can be quickly found because we already calculated a_0 . We encounter this same shortcut when we integrate the general case. Using integration by parts with

$$\begin{aligned} u &= x^n & \text{so} & \quad du = nx^{n-1} dx, \\ dv &= e^{-x} dx & \text{so} & \quad v = -e^{-x}, \end{aligned}$$

we find that

$$\begin{aligned} a_n &= \frac{1}{n!} \int_0^1 x^n e^{-x} dx \\ &= \frac{1}{n!} [-x^n e^{-x}]_0^1 + \frac{n}{n!} \int_0^1 x^{n-1} e^{-x} dx \\ &= \frac{1}{n!} \left[-\frac{1}{e} \right] + \frac{1}{(n-1)!} \int_0^1 x^{n-1} e^{-x} dx. \end{aligned}$$

The second term is a_{n-1} , so we have established the recursion

$$a_n = -\frac{1}{n!e} + a_{n-1} = a_{n-1} - \frac{1}{n!e}.$$

Starting with a_0 and using it, we conclude that

$$a_0 = 1 - \frac{1}{e},$$

$$a_1 = a_0 - \frac{1}{1!e} = 1 - \frac{1}{e} - \frac{1}{1!e},$$

$$a_2 = a_1 - \frac{1}{2!e} = 1 - \frac{1}{e} - \frac{1}{1!e} - \frac{1}{2!e},$$

and so on. In general,

$$\begin{aligned} a_n &= 1 - \frac{1}{e} - \frac{1}{1!e} - \frac{1}{2!e} - \cdots - \frac{1}{n!e} \\ &= 1 - \frac{1}{e} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \right). \end{aligned}$$

Taking limits, we have

$$\lim_{n \rightarrow \infty} a_n = 1 - \frac{1}{e} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \right).$$

By (3.16), the left side equals zero, and then solving for e yields

$$\begin{aligned} e &= 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \\ &= \sum_{k=0}^{\infty} \frac{1}{k!}. \end{aligned} \tag{3.17}$$

Thus ends our modern proof of this sum, verifying that Euler's argument (in section 3.2) does indeed lead to a reliable conclusion.

3.6 e is irrational

It is certainly possible for an infinite series of rational numbers to equal a rational number; for example, the geometric series

$$1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots \tag{3.18}$$

sums via the formula

$$S = \frac{a}{1 - r},$$

where a is the initial term 1 and r is the common ratio $1/3$. So the series (3.18) sums to

$$\frac{1}{1 - 1/3} = \frac{3}{2},$$

a rational number.

This is not so for the series (3.17) for e , a fact that we will demonstrate now, using a proof by contradiction. Suppose that e is a rational number, so set $e = p/q$. We split^[40] the series (3.17) after the term $1/q!$:

$$e = \sum_{k=0}^q \frac{1}{k!} + \sum_{k=q+1}^{\infty} \frac{1}{k!}.$$

Replacing e with p/q , and subtracting the first sum from both sides, we have

$$\frac{p}{q} - \sum_{k=0}^q \frac{1}{k!} = \sum_{k=q+1}^{\infty} \frac{1}{k!}.$$

If we multiply the left side by $q!$, then every term simplifies to an integer, so the entire left side equals some integer. On the other hand, the right side's terms fall between 0 and 1 when multiplied by $q!$ because

$$\begin{aligned} \sum_{k=q+1}^{\infty} \frac{q!}{k!} &= \frac{1}{(q+1)} + \frac{1}{(q+1)(q+2)} + \frac{1}{(q+1)(q+2)(q+3)} + \cdots \\ &< \frac{1}{q+1} + \frac{1}{(q+1)^2} + \frac{1}{(q+1)^3} + \cdots, \end{aligned}$$

a geometric series with $a = 1/(q+1) = r$ that sums^[41] to $1/q$. Thus, the right side cannot equal an integer. This contradiction establishes that e cannot be rational.

3.7 Stirling's formula

Formula (2.19) gives a good approximation for a specific question (what is the probability of flipping exactly n heads and n tails in $2n$ coin flips?). Even a slight tweak to this question forces an entirely new approach to

such an approximation. For example, the probability of flipping 40 heads in 100 coin flips is

$$\frac{\binom{100}{40}}{2^{100}} = \frac{100!}{40! 60! 2^{100}}.$$

These factorials are murderous to find by hand.

James Stirling (Scotland, born 1692) found an approximation to $n!$ that, happily, involves both π and e . Our presentation uses a modern, calculus-based approach; Stirling's original discovery did not.

The addition property of logarithms makes it convenient to begin not with $n!$ itself but with $\ln n!$, because

$$\ln n! = \ln 1 + \ln 2 + \cdots + \ln n. \quad (3.19)$$

This sum is very nearly^[42] the area defined by

$$\int_1^n \ln x \, dx,$$

which via integration by parts with

$$\begin{aligned} u &= \ln x & \text{so} & \quad du = (1/x) \, dx, \\ dv &= dx & \text{so} & \quad v = x, \end{aligned}$$

gives us

$$\int_1^n \ln x \, dx = [x \ln x - x]_1^n = n \ln n - n + 1. \quad (3.20)$$

We can approximate this same area using trapezoids as in Figure 3.1. Let $R_1, R_2, \dots, R_{n-1}$ denote the areas of the “remainders” under the curve $y = \ln x$ that are not occupied by the trapezoids formed by the line segments shown. Then

$$\int_1^n \ln x \, dx = \sum \text{areas of trapezoids} + \sum_{k=1}^{n-1} R_k. \quad (3.21)$$

First we will calculate the areas of the trapezoids, and then we will show that the combined areas of the remainders is finite even if we let n grow

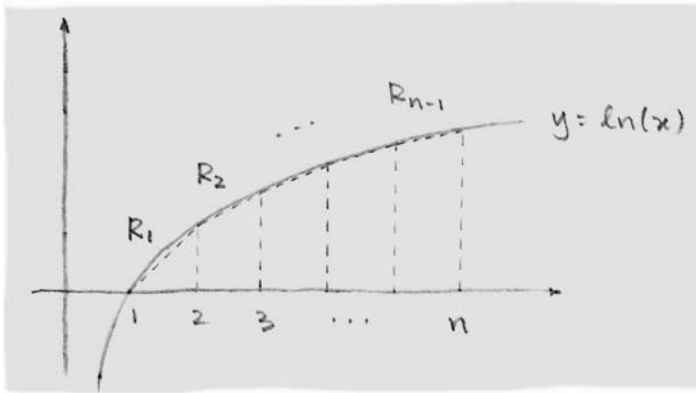


Figure 3.1

arbitrarily large. Recall that $\ln 1 = 0$, so

$$\begin{aligned} \sum \text{areas of trapezoids} &= \frac{\ln 1 + \ln 2}{2} + \frac{\ln 2 + \ln 3}{2} \\ &\quad + \cdots + \frac{\ln(n-1) + \ln n}{2} \\ &= \ln 2 + \ln 3 + \cdots + \ln(n-1) + \frac{1}{2} \ln n. \end{aligned}$$

Comparing to (3.19), we see that

$$\sum \text{areas of trapezoids} = \ln n! - \frac{1}{2} \ln n. \quad (3.22)$$

Using (3.20) to replace the left side of (3.21), and (3.22) to replace the first sum on the right of (3.21), we have

$$n \ln n - n + 1 = \ln n! - \frac{1}{2} \ln n + \sum_{k=1}^{n-1} R_k. \quad (3.23)$$

We must grapple with the sum of the remainders. First, we will establish that their combined area is limited even as $n \rightarrow \infty$, and then we will find the limit exactly.

Claim 1. The combined area of the remainders is limited, that is,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} R_k \text{ exists.}$$

The left drawing in Figure 3.2 shows the k th trapezoid $ABCD$ with the curve's tangent line DE drawn to create trapezoid $ABED$. The slope of DE is not only FE (because $DF = 1$), but also

$$\left[\frac{d}{dx} \ln x \right]_{x=k} = \frac{1}{k},$$

so

$$BE = BF + FE = AD + \frac{1}{k} = \ln k + \frac{1}{k}.$$

Thus,

$$\begin{aligned} \text{the area of trapezoid } ABED &= \frac{1}{2} \left[\ln k + \ln k + \frac{1}{k} \right] \\ &= \ln k + \frac{1}{2k}. \end{aligned}$$

In the right drawing of Figure 3.2, the tangent line at point C creates trapezoid $ABCG$. The slope of GC equals both $1/(k+1)$ and GH , so

$$\begin{aligned} \text{the area of trapezoid } ABCG &= \frac{1}{2} \left[\ln(k+1) + \ln(k+1) - \frac{1}{k+1} \right] \\ &= \ln(k+1) - \frac{1}{2(k+1)}. \end{aligned}$$

The arithmetic mean of the areas of trapezoids $ABED$ and $ABCG$ exceeds the combined areas of trapezoid $ABCD$ and the remainder area R_k , so

$$\frac{1}{2} \left[\ln k + \frac{1}{2k} + \ln(k+1) - \frac{1}{2(k+1)} \right] > \frac{1}{2} [\ln k + \ln(k+1)] + R_k.$$

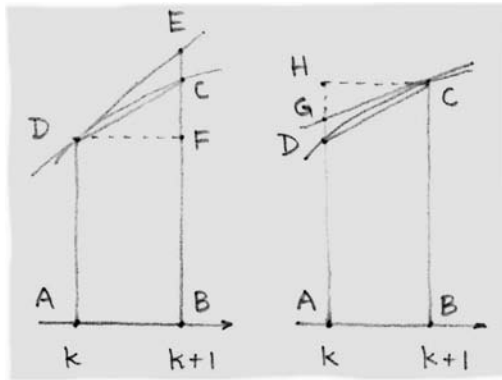


Figure 3.2

This simplifies nicely to

$$\frac{1}{4k} - \frac{1}{4(k+1)} > R_k.$$

The combined area of the remainders $R_1, R_2, \dots, R_{n-1}$ thus is bounded:

$$\sum_{k=1}^{n-1} R_k < \sum_{k=1}^{n-1} \left(\frac{1}{4k} - \frac{1}{4(k+1)} \right) = \frac{1}{4} - \frac{1}{4n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} R_k < \frac{1}{4}.$$

All R_k are positive, so their sum approaches a limit that we will call L . This establishes Claim 1.

Claim 2. The limit L of the combined area of the remainders equals

$$1 - \ln \sqrt{2\pi} \approx 0.08106.$$

Rearranging (3.23) yields

$$\sum_{k=1}^{n-1} R_k - 1 = n \ln n - n - \ln n! + (1/2) \ln n.$$

Taking the limit of both sides, we can write

$$L - 1 \asymp n \ln n - n - \ln n! + (1/2) \ln n.$$

Exponentiating and simplifying^[43], we reach

$$e^{L-1} \asymp \frac{n^n \sqrt{n}}{n! e^n}. \quad (3.24)$$

Now if we can evaluate the limit on the right, then we can solve for L as desired. Letting S_n denote the expression on the right, so that

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{n^n \sqrt{n}}{n! e^n},$$

note that

$$\lim_{n \rightarrow \infty} (S_n)^2 = \lim_{n \rightarrow \infty} \frac{n^{2n} n}{n! n! e^{2n}}$$

bears some resemblance to (2.16) and (2.19), which combined give

$$\frac{(2n)!}{n! n! 2^{2n}} \asymp \frac{1}{\sqrt{\pi(n+1/2)}}. \quad (3.25)$$

We can make the resemblance stronger by noticing that

$$S_{2n} = \frac{(2n)^{2n} \sqrt{2n}}{(2n)! e^{2n}}$$

and so the ratio of $(S_n)^2$ to S_{2n} is

$$\begin{aligned} \frac{(S_n)^2}{S_{2n}} &= \frac{n^{2n} n}{n! n! e^{2n}} \cdot \frac{(2n)! e^{2n}}{(2n)^{2n} \sqrt{2n}} \\ &= \frac{(2n)!}{n! n! 2^{2n}} \cdot \frac{n}{\sqrt{2n}}. \end{aligned}$$

Taking the limit of both sides and using (3.25), we find^[44] that

$$\lim_{n \rightarrow \infty} \frac{(S_n)^2}{S_{2n}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\pi(n+1/2)}} \cdot \frac{n}{\sqrt{2n}} = \frac{1}{\sqrt{2\pi}}.$$

The limit on the left side can be simplified^[45] another way:

$$\lim_{n \rightarrow \infty} \frac{(S_n)^2}{S_{2n}} = \frac{\lim_{n \rightarrow \infty} (S_n)^2}{\lim_{n \rightarrow \infty} S_{2n}} = \frac{\left(\lim_{n \rightarrow \infty} S_n \right)^2}{\lim_{n \rightarrow \infty} S_n} = \lim_{n \rightarrow \infty} S_n.$$

These last two results imply that

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{\sqrt{2\pi}}.$$

Now we can use (3.24) to calculate^[46] that, as claimed,

$$L = 1 - \ln \sqrt{2\pi}.$$

More importantly, however, we have from (3.24) that

$$\frac{1}{\sqrt{2\pi}} \asymp \frac{n^n \sqrt{n}}{n! e^n}$$

or

$$n! \asymp \sqrt{2\pi n} \left(\frac{n}{e} \right)^n. \quad (3.26)$$

This result is called *Stirling's formula*; its right side involves computations less cumbersome than those required to calculate $n!$ directly. For example, to approximate $40!$ to five significant digits without a calculator, we use

$$\sqrt{2\pi(40)} \left(\frac{40}{e}\right)^{40}.$$

Methods for computing square roots such as $\sqrt{2\pi(40)}$ by hand are as ancient as the invention of the wheel (not metaphorically!) and we can find the 40th power of $40/e$ using logarithms:

$$N = \left(\frac{40}{e}\right)^{40}$$

$$\ln N = 40 \ln \left(\frac{40}{e}\right)$$

$$\ln N = 40(\ln 40 - \ln e)$$

$$N = 5.135945 \times 10^{46}$$

The final approximation for $40!$ is 8.1592×10^{47} and required four multiplications, one division, one square root, and two uses of (in lieu of a calculator) a book of logarithms. These operations are not trivial, but they have the advantage of being possible in a way that a direct calculation of $40!$ is not.

3.8 Turning a series into a continued fraction

From (3.4) we have

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots,$$

and when $x = -1$,

$$\frac{1}{e} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots$$

or

$$1 - \frac{1}{e} = \frac{1}{1} - \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} - \frac{1}{1 \cdot 2 \cdot 3 \cdot 4} + \cdots. \quad (3.27)$$

The series on the right has the form

$$\frac{1}{a_1} - \frac{1}{a_1 a_2} + \frac{1}{a_1 a_2 a_3} - \frac{1}{a_1 a_2 a_3 a_4} + \dots \quad (3.28)$$

that we will now convert to a continued fraction.

The first two terms may be written

$$\frac{1}{a_1} - \frac{1}{a_1 a_2} = \frac{a_2 - 1}{a_1 a_2},$$

and the reciprocal of the right side may be written

$$\frac{a_1 a_2}{a_2 - 1} = \frac{a_1(a_2 - 1) + a_1}{a_2 - 1} = a_1 + \frac{a_1}{a_2 - 1}.$$

So we can express the first two terms of (3.28) as

$$\frac{1}{a_1} - \frac{1}{a_1 a_2} = \frac{1}{a_1 + \frac{a_1}{a_2 - 1}} \quad (3.29)$$

in continued fraction form.

Now we extend to the first three terms of (3.28), writing

$$\begin{aligned} \frac{1}{a_1} - \frac{1}{a_1 a_2} + \frac{1}{a_1 a_2 a_3} &= \frac{1}{a_1} - \frac{a_3 - 1}{a_1 a_2 a_3} \\ &= \frac{1}{a_1} - \frac{1}{a_1 \left(\frac{a_2 a_3}{a_3 - 1} \right)}. \end{aligned}$$

With the expression in parentheses taking the role of a_2 on the left of (3.29), we may conclude that the identity

$$\frac{1}{a_1} - \frac{1}{a_1 a_2} + \frac{1}{a_1 a_2 a_3} = \frac{1}{a_1 + \frac{a_1}{a_2 - 1 + \frac{a_2}{a_3 - 1}}}$$

properly converts the first three terms of series (3.28) to a continued frac-

tion. This method extends to an inductive proof of the general result

$$\begin{aligned} \frac{1}{a_1} - \frac{1}{a_1 a_2} + \frac{1}{a_1 a_2 a_3} - \frac{1}{a_1 a_2 a_3 a_4} + \cdots \\ = \frac{1}{a_1 + \frac{a_1}{a_2 - 1 + \frac{a_2}{a_3 - 1 + \frac{a_3}{a_4 - 1 + \cdots}}}} \end{aligned}$$

that we now apply specifically to (3.27). Setting $a_k = k$ yields

$$1 - \frac{1}{e} = \frac{1}{1 + \frac{1}{1 + \frac{2}{2 + \frac{3}{3 + \cdots}}}} \quad (3.30)$$

and, to create a continued fraction for e alone, we rewrite the left side

$$1 - \frac{1}{e} = \frac{1}{1} - \frac{1}{1 \cdot e}$$

and compare to (3.29) to obtain

$$\frac{1}{1} - \frac{1}{1 \cdot e} = \frac{1}{1 + \frac{1}{e - 1}}.$$

Now (3.30) reads

$$\frac{1}{1 + \frac{1}{e - 1}} = \frac{1}{1 + \frac{1}{1 + \frac{2}{2 + \frac{3}{3 + \cdots}}}}$$

from which^[47]

$$e = 2 + \frac{2}{3 + \frac{3}{4 + \frac{4}{5 + \frac{5}{6 + \cdots}}}}$$

follows. This continued fraction takes its place beside (1.16) and (2.60), continued fractions for φ and π respectively, in the museum of the most curious and graceful expressions in mathematics.

3.9 Further content

3.1 The probability of derangement. In 1708, **Pierre Renard de Montmort** (France, born 1678) posed a question about a card game named *Treize* (*thirteen* in French). In a simplified version of the game, a player shuffles a deck of thirteen cards that are numbered 1 through 13. The player flips a card face up from the deck and says “one”, then flips the next card and says “two”, and so on. If at any time the spoken number matches the number on the card, the player wins.

On the other hand, the player loses if no card sits in its numerical position in the shuffled deck. We call such an event a *derangement* of the cards. For example, the shuffled deck

10 7 13 9 11 8 5 2 4 3 5 1 12

leads to a loss because it is a derangement. Montmort’s question was the natural one: what is the probability of winning?

We win with each of the shuffles

9 13 5 12 4 8 1 2 10 7 11 3 6

and

7 9 3 11 5 12 2 8 6 13 10 4 1

because each underlined card sits in its numerical position. These examples illustrate the difficulty of Montmort’s question, in that we win when one card sits in its numerical position, but also when two

cards do, or three cards, and so on. Using a fact from the study of combinatorics called *the principle of inclusion-exclusion*, we can analyze this complex situation.

Here, however, we answer Montmort's question indirectly by calculating the probability that we lose the game. Now we need only count the derangements.

- (a) Let D_n denote the number of derangements of a deck of n cards numbered 1 through n . By writing all possible shufflings, check that $D_1 = 0$, $D_2 = 1$, $D_3 = 2$, and $D_4 = 9$.
- (b) Next we will argue that when the deck has $n > 2$ cards, then

$$D_n = (n - 1)(D_{n-1} + D_{n-2}). \quad (3.31)$$

But before we do, verify that (3.31) yields $D_3 = 2$, $D_4 = 9$, $D_5 = 44$, and $D_6 = 265$. (And if you enjoy it, continue until you reach $D_{13} = 2290792932$, the (huge) number of derangements of 13 cards.)

- (c) To prove (3.31), consider just the first flip in the game. We have a derangement only if we flip one of the $n - 1$ cards other than the 1 card. This explains the first factor on the right side of (3.31). Say that the first card shows the number K . One of two possibilities exist: either the K th card flipped shows the number 1, or it does not. The number of derangements of the rest of the deck is D_{n-1} in one case, and D_{n-2} in the other. Explain which is which, thus establishing the validity of (3.31).
- (d) For (3.31) to be useful when $n = 2$, we need to define D_0 , the number of derangements of a deck with no cards. Explain why (3.31) forces $D_0 = 1$. What common sense can you make of this answer?
- (e) Now rearrange (3.31) to obtain

$$D_n - nD_{n-1} = -[D_{n-1} - (n - 1)D_{n-2}]. \quad (3.32)$$

- (f) The expression in brackets on the right side of (3.32) equals the expression on the left but with every n replaced by $n - 1$. With

this in mind, check the next three cases after these two:

$$D_2 - 2D_1 = -[D_1 - 1D_0] = -[0 - 1] = 1,$$

$$D_3 - 3D_2 = -[D_2 - 2D_1] = -[1] = -1,$$

$$D_4 - 4D_3 = -[D_3 - 3D_2] = -[-1] = 1,$$

etc.

Argue that, in general,

$$D_n - nD_{n-1} = (-1)^n. \quad (3.33)$$

(g) Divide both sides of (3.33) by $n!$ and rearrange to obtain

$$\frac{D_n}{n!} = \frac{D_{n-1}}{(n-1)!} + \frac{(-1)^n}{n!}.$$

Explain why this equation implies that

$$\frac{D_n}{n!} = \sum_{k=0}^n \frac{(-1)^k}{k!}. \quad (3.34)$$

- (h) Because $n!$ counts the number of shufflings of a deck of n cards, then (3.34) with $n = 13$ calculates the probability of losing a game of Treize. Now compare the right side of (3.34) with (3.4) (evaluated at $x = -1$) to explain why

$$\lim_{n \rightarrow \infty} \frac{D_n}{n!} = \frac{1}{e}.$$

This result answers a stronger version of Montmort's question, namely, as the number of cards in the deck increases, the probability that the player wins approaches $1 - 1/e$, or about 63%. Are you surprised to learn that about twice as many games of Treize are won than lost?

3.2 Deranged matchings. This exercise appeals to the *principle of inclusion-exclusion*, and so might require some preliminary study.

On a school trip, kindergarten students are paired at random for breakfast seat assignments, and then again at lunch. In this exercise, we will find the probability that none of the breakfast pairings are repeated at lunch.

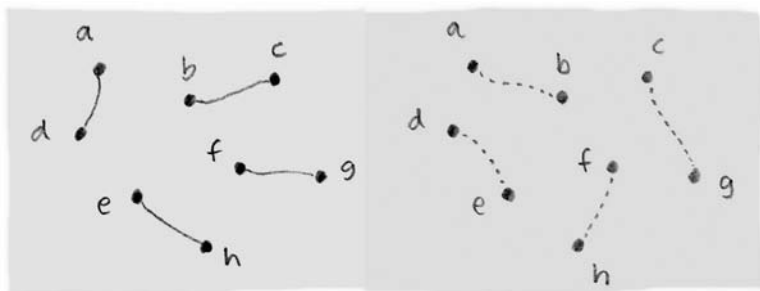


Figure 3.3

Using the language of graph theory, we can think of each child as a *vertex* (represented by dots labeled $a, b, c, \dots, h$ in Figure 3.3) and a pairing as an *edge* (the lines in the figure). There are two graphs in Figure 3.3, one for breakfast (the solid edges on the left) and one for lunch (the dashed edges on the right). This pair of graphs depicts a *deranged matching* because none of the breakfast pairs (ad, bc, eh, fg) are reunited at lunch.

- (a) Suppose that there are $2n$ children (we need an even number for the pairing to work). Prove that the total number of possible matchings at breakfast is

$$(2n - 1) \times (2n - 3) \times \cdots \times 5 \times 3 \times 1.$$

Because this expression resembles a factorial, we use the notation $(2n - 1)!!$ to abbreviate it. For example,

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040 \quad \text{but}$$

$$7!! = 7 \times 5 \times 3 \times 1 = 105.$$

Because $(-1)!!$ would equal the number of matchings when $n = 0$ (that is, when there are no children), we define $(-1)!! = 1$ to count the empty matching.

- (b) At breakfast, we choose pairs of children at random to be partners in one of these $(2n - 1)!!$ ways. Let's use the list

$$P_1 \quad P_2 \quad P_3 \quad \dots \quad P_{n-1} \quad P_n$$

to label the n pairs. At lunch, we match the children at random again. Define the event

$$A_k = \text{the breakfast pair } P_k \text{ is reunited at lunch}$$

so that, for example, the union $A_1 \cup A_2$ signals that pairs P_1 and P_2 ate both breakfast and lunch together. (This specific union does not preclude pairs other than P_1 and P_2 from eating both meals together as well.) Lastly, let

$$N_\ell = |\text{intersection of any } \ell \text{ of the sets } A_k|.$$

For example, to calculate N_3 we must intersect any three events ($A_1 \cap A_2 \cap A_4$, for example) and tally all possible pairings in which the corresponding breakfast pairs (P_1 , P_2 , and P_4 in this case) are also lunch partners.

Explain why, for any ℓ , the choice of which ℓ events to intersect is irrelevant.

- (c) Demonstrate that $N_\ell = (2n - 2\ell - 1)!!$.
- (d) Our strategy invokes the principle of inclusion-exclusion as a way to count the number of possible lunch matchings that repeat at least one pair from breakfast. Explain why the left side of

$$\left| \bigcup A_k \right| = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} N_k \quad (3.35)$$

counts this, and then use the principle to justify the right side.

- (e) To find the number of lunch matchings that do not contain any of the n breakfast pairs (that is, the number d_n of deranged matchings), we subtract (3.35) from the total number of possible matchings. Provide the missing details that bring us to

$$d_n = \sum_{k=0}^n (-1)^k \binom{n}{k} (2n - 2k - 1)!!. \quad (3.36)$$

- (f) The probability of a lunch matching being a deranged matching equals the left side of (3.36) divided by $(2n - 1)!!$, so we wish to divide the right side by the same expression. We introduce the *double falling factorial*

$$((2n - 1))_k = (2n - 1) \times (2n - 3) \times \cdots \times (2n - 2k + 1)$$

to streamline the notation. The subscript k on the left indicates the number of factors on the right. For example,

$$((9))_3 = 9 \times 7 \times 5 = 315.$$

With this in mind, explain how we may derive

$$\frac{d_n}{(2n-1)!!} = \sum_{k=0}^n (-1)^k \binom{n}{k} ((2n-1))_k \quad (3.37)$$

from (3.36).

- (g) Use (3.37) to find the probability of a deranged matching if there are eight children.
- (h) As in exercise 1, we reveal e when we let n become arbitrarily large. If we focus on a particular term of the sum in (3.37), so that k is fixed, we find that

$$\lim_{n \rightarrow \infty} \left[(-1)^k \binom{n}{k} ((2n-1))_k \right] = \frac{(-1)^k}{k! 2^k}.$$

Provide the necessary details. Hint: before taking the limit, divide every factor (that contains n) by n .

- (i) Finally, use (3.4) to show that

$$\lim_{n \rightarrow \infty} \frac{d_n}{(2n-1)!!} = \frac{1}{\sqrt{e}}.$$

As the number of children increases, then, the probability that none of the breakfast pairs are reunited at lunch is close to 61%.

3.3 Euler's constant γ . Here we detail the origins of a constant whose tale would easily expand to a full chapter of this book.

Figure 3.4 provides a visual inspiration for our upcoming work. To approximate the area under $y = 1/x$ between $x = 1$ and $x = n$ (where n is a whole number), we can draw the rectangles shown in the figure. Their total area exceeds the area beneath the curve by the areas of the shaded regions. Imagine sliding each shaded region, except the one above the interval $(1, 2)$, directly to its left until it is settled in that interval. None of the shaded regions would overlap, and together they would be contained in a unit square. Thus it appears intuitively clear that the combined area of the shaded regions converges to a value between 0 and 1 even as n approaches infinity. Euler's argument, which follows, allows us to draw this conclusion.

In what follows, let

$$H_{n,k} = \frac{1}{1^k} + \frac{1}{2^k} + \frac{1}{3^k} + \cdots + \frac{1}{n^k}.$$

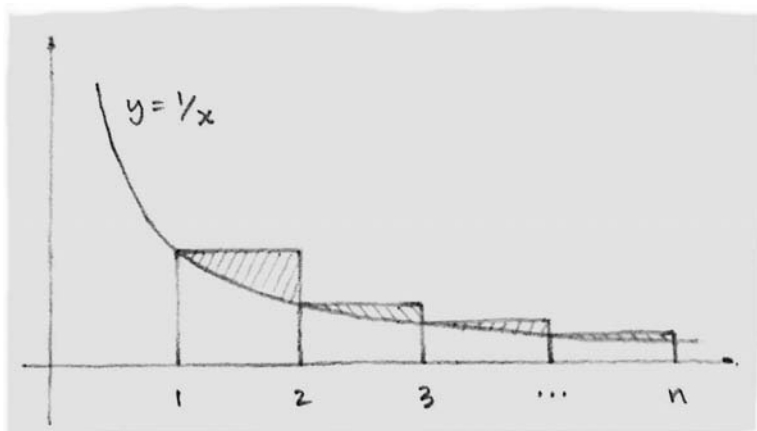


Figure 3.4

Note that $\lim_{n \rightarrow \infty} H_{n,1}$ gives the harmonic series (seen previously in exercise 4 where it appeared in a continued fraction for π) and that $\lim_{n \rightarrow \infty} H_{n,2}$ is the sum in the Basel problem (section 2.9).

(a) From elementary calculus, we know that the Maclaurin series

$$\ln(1+n) = n - \frac{n^2}{2} + \frac{n^3}{3} - \frac{n^4}{4} + \cdots \quad (3.38)$$

converges for $-1 < n \leq 1$ and that

$$\int_1^{n+1} \frac{1}{x} dx = \ln(1+n).$$

Now observe that

$$\begin{aligned} \ln(1+n) &= \ln\left(\frac{2}{1} \times \frac{3}{2} \times \frac{4}{3} \times \cdots \times \frac{n+1}{n}\right) \\ &= \ln(1+1) + \ln(1+1/2) + \cdots + \ln(1+1/n). \end{aligned}$$

Use (3.38) to expand each term in this last sum, and then obtain

$$\ln(1+n) = H_{n,1} - \frac{1}{2}H_{n,2} + \cdots \pm \frac{1}{n}H_{n,n} \quad (3.39)$$

by judicious factoring. (The sign on the final term depends on the parity of n .)

- (b) We may rearrange (3.39) to

$$H_{n,1} - \ln(1+n) = \frac{1}{2}H_{n,2} - \frac{1}{3}H_{n,3} + \cdots \pm \frac{1}{n}H_{n,n}. \quad (3.40)$$

Explain why the left side equals the desired area as n becomes infinite.

- (c) Now turn to the right side of (3.40). As n approaches infinity, why does the alternating sum converge?

Historical note. We use γ to denote the constant to which the right side of (3.40) converges as n approaches infinity. Euler's interest in $\gamma \approx 0.577218$ lasted for decades, during which he discovered other methods for calculating its digits. His primary motivation, however, was to link γ to other notable numbers such as e and π , something that γ has been somewhat resistant to in the years since. This discussion finds its home in this chapter because e^γ has made frequent appearances in the work of post-Eulerian mathematicians.

- 3.4 Euler's constant γ via e and π .** Here is the outline of a proof that Euler's constant γ (introduced in exercise 3) as defined by

$$\gamma = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \right) - \ln(n+1) \right] \quad (3.41)$$

exists. (This definition comes from the left side of (3.40).) The constants e and π each have a role in this argument.

- (a) We start with (3.14), which is

$$\left(1 + \frac{1}{n} \right)^n < e < \left(1 + \frac{1}{n} \right)^{n+1} \quad (3.42)$$

for any natural number n , and draw from it two conclusions. First, by taking the natural logarithm of each side of the inequality on the left, show that

$$\ln \left(1 + \frac{1}{n} \right) < \frac{1}{n}. \quad (3.43)$$

- (b) Again, from the left inequality of (3.42) we have

$$e > \left(1 + \frac{1}{n} \right)^n,$$

so define a_n as the multiplier that will raise the right side to equal the left side:

$$e = a_n \left(1 + \frac{1}{n}\right)^n. \quad (3.44)$$

From (3.44) and the inequality on the right of (3.42), derive the inequality

$$a_n < 1 + \frac{1}{n}. \quad (3.45)$$

- (c) If we take the natural logarithm of each side of (3.45) and use (3.43), we reach

$$\ln(a_n) < \frac{1}{n}.$$

Establish that

$$\frac{1}{n} = \ln(a_n^{1/n}) + \ln(n+1) - \ln(n) \quad (3.46)$$

by first taking the natural logarithm of each side of (3.44) and then simplifying appropriately.

- (d) We can create the following m equations from (3.46) by letting n equal $1, 2, \dots, m$:

$$1 = \ln(a_1^{1/1}) + \ln(2) - \ln(1)$$

$$\frac{1}{2} = \ln(a_2^{1/2}) + \ln(3) - \ln(2)$$

$$\frac{1}{3} = \ln(a_3^{1/3}) + \ln(4) - \ln(3)$$

...

$$\frac{1}{m} = \ln(a_m^{1/m}) + \ln(m+1) - \ln(m)$$

Add these equations and simplify to reach

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m} - \ln(m+1) = \sum_{k=1}^m \frac{1}{k} \ln a_k.$$

- (e) Finally, with the help of (3.45) and (2.35), conclude that

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m} - \ln(m+1) < \frac{\pi^2}{6}. \quad (3.47)$$

- (f) Thus, the left side of (3.47) is bounded above, so the limit in (3.41) exists as long as the sequence $\{\gamma_n\}$ defined by

$$\gamma_n = \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right) - \ln(n+1)$$

is monotone increasing. To show this, show that $\gamma_n - \gamma_{n-1} > 0$ with the help of (3.43).

- 3.5 The probability integral** $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. (This exercise requires knowledge of multivariate calculus and polar coordinates.) Certain probabilities may be estimated using the *normal distribution*, which involve a particular family of integrals. The integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx \quad (3.48)$$

is related to this family, and is of interest to us because, like Stirling's formula (3.26), this special *probability integral* links e to π . It cannot be computed directly by elementary means, but a foray into three dimensions sidesteps this difficulty.

- (a) Let I denote the integral (3.48) and observe that

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-x^2} dx.$$

As a cosmetic (but crucial) change, we replace x with another variable (we will use y) in the second integral to obtain

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy.$$

Explain why

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dy dx.$$

- (b) Convert this integral to polar coordinates to show that

$$I^2 = \int_0^{2\pi} \int_0^{\infty} r e^{-r^2} dr d\theta.$$

- (c) Use the substitution $u = r^2$ to evaluate it and conclude that $I^2 = \pi$, from which the desired conclusion follows by taking square roots.

CHAPTER 4

i

The area A of a circle with radius r can be calculated using the formula

$$A = \pi r^2,$$

so the area of a unit circle (where $r = 1$) is π . How much of this claim makes sense? If by “sense” we intend to involve our physical senses, then perhaps nothing about the claim makes sense. To settle the truth about the claim in a way that we can see and touch, someone would first need to produce a circle. But no one can.

Many natural and human-made objects are circular, to be sure, but circles exist only in the abstract. Yet they could hardly be simpler to define: On a flat surface, choose any point. Now choose a distance. A circle is determined when we locate all points on the surface that are equidistant from the first point.

The very symmetry and simplicity of this definition defeats any attempt to physically create a circle. If only the definition allowed some wiggle room, as in “all points on the surface that, *within a small error*, are equidistant from the first point,” then perhaps we could create a circle that we could see and touch. Then we would need to prepare a flat surface (none exist) and choose a point on it (whatever that means).

Perhaps “within a small error” can help us resolve the notions of *flat* and *point* in the same way that we used this phrase to relax the meaning of *equidistant*. Can we honestly claim to be defining a circle any more? This squirming would certainly cost us the elegant link between circles and the constant π .

Despite the lack of actual circles, many people find it fascinating to explore a claim like “the area of a unit circle is π ” as a purely imaginative task. Perhaps our willingness to imagine was sparked when we were young, while being taught to count with piles of buttons or pieces of chocolate, and our minds suddenly grasped that *there is no largest number*. So now, confronted with the number

56 873 403 284 337 447 984 278 897 484,

which quite likely has never been written or uttered by a human before this moment, you are perfectly willing to accept it as meaningful, and even feel confident that it could be the answer to some question. At noon last Tuesday, for example, perhaps a butterfly brought the combined total of wings flapped by creatures in the universe to this precise total.

4.1 Proportions

When you grew older, and there were three cookies to share with your one friend, you learned proportions. When ancient thinkers discussed proportion, they operated in the realm of geometry, interpreting proportion as, say, the ratio of the lengths of two line segments. For example, the concept of *one half* conjured in their minds a geometric analogue such as a bisected line segment or a square cut by its diagonal.

In fact, these thinkers discovered how from a unit segment and two given segments of lengths a and b (as in Figure 4.1) they could construct a segment of length a/b . Here is one such construction: arrange the segments of lengths a and b perpendicular to each other, in the manner of AB and BC in the figure. Now locate point D on BC so that $DC = 1$.

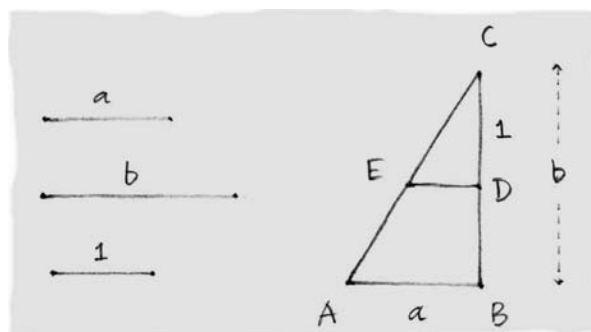


Figure 4.1

Draw DE perpendicular to BC . Because $\triangle ABC \sim \triangle EDC$, we have $a/b = DE/1$, so DE has the desired length.

So starting with a unit segment and using simple tools and logic, we can construct a segment that has, it would seem, any length that we wish. For example, we can produce a segment of length $3/7$ by laying 3 copies of the unit segment end to end (and assigning this segment of length 3 to the role of a in Figure 4.1), doing much the same with 7 copies to produce b , and then following the method described above to create a segment of length $3/7$. (If $a > b$, the process requires a tweak.) By this method, we can claim to have created all positive proportions.

Many ancient scholars believed that such proportions constituted all possible quantities, so they suspected that their method could create all numbers that exist. While they were rightly proud of having mastered proportion, they had only explored a tiny corner of what numbers have to offer.

Among these thinkers was one who produced a line segment in a perfectly ordinary way and then demonstrated that its length was not the ratio of two whole numbers. In other words, his line segment could easily be drawn, yet its length represented an entirely new kind of number. To create it, make square $ABCD$ from a unit segment. Bisect the square with a diagonal AC . By the Pythagorean theorem, $AC^2 = 1^2 + 1^2 = 2$. The irrationality of $AC = \sqrt{2}$ can be demonstrated in the same way that we proved $\sqrt{5}$ irrational in exercise 6. This segment does not follow the old rules; it opens a path to unexplored territory.

Such irrationals as $\sqrt{2}$ may be constructed much like the rationals we built in Figure 4.1. Given segments of lengths a and b , lay a and b end to end (as in Figure 4.2) to create segment AC . Locate midpoint M of AC

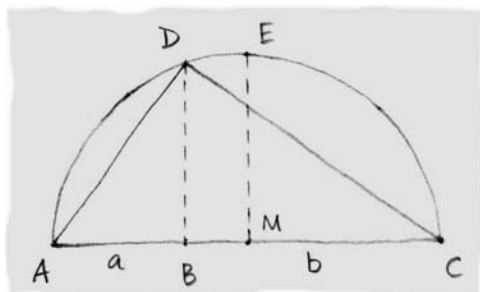


Figure 4.2

and draw the semicircle with center M and radius $MA = ME = MC$. Construct BD perpendicular to AC . Because $\angle ADC$ is inscribed in a semicircle, it must be a right angle. This implies that $\angle DAB + \angle DCB = 90^\circ$, but because $\angle BDC + \angle DCB = 90^\circ$ as well, we see that $\angle DAB = \angle BDC$. Thus $\triangle ABD \sim \triangle DBC$, so

$$\frac{AB}{BD} = \frac{BD}{BC} \quad \text{or} \quad BD = \sqrt{ab}.$$

To construct $\sqrt{6}$, for example, we can choose $a = 2$ and $b = 3$ and proceed as just described.

If we learned our lesson earlier, we will hesitate before claiming that now we can construct all possible lengths and therefore all numbers that exist. A moment of thought might spring negative numbers to mind. If all numbers must be tied to geometric constructs, such as line segments, then negative numbers appear to be either nonsense or off-limits. For centuries, in fact, mathematicians in some cultures considered negative numbers to carry no meaning whatsoever thanks to how tightly they linked numbers to geometry.

4.2 Negatives

Today, we have little difficulty understanding that if the temperature drops 20° from its current 15° , then the new temperature is $15 - 20$ or -5° . But there is a subtle change in the meaning of the symbol — as we process this statement; in $15 - 20$, it indicates an interaction between the two numbers, whereas in -5° , the symbol points us in a direction. Our ability to mentally navigate this subtlety is quite modern.

Yet this facility with negatives has ancient roots. As many as two thousand years ago, merchants in Asia employed counting rods, often colored black and red, to track payments and calculate bills. A customer who gave a merchant 10 coins, for example, but selected goods worth 14, would prompt the merchant to show the customer 10 black rods and 14 red, removing them in black-red pairs until 4 reds remain. Neither person has written “ $10 - 14 = -4$ ” but certainly the customer understands that she needs to fork over four more coins.

Despite this common and natural interaction with negatives, the mathematical community grappled with the idea of numbers that could be “less than nothing” for centuries before coming to a consensus about how they should be understood and used in calculations. Among the puzzling

questions they considered were: Must we reject wholesale the link between numbers and geometry to accommodate negatives? What happens when negatives are multiplied or divided? Does a negative solution to an algebra problem carry any meaning?

By way of illustrating the complexity of this long discussion, consider this argument proposed by John Wallis (who we met in Chapter 2). The values of the fractions

$$\cdots \quad \frac{1}{5} \quad \frac{1}{4} \quad \frac{1}{3} \quad \frac{1}{2} \quad \frac{1}{1}$$

increase at each step because the denominators are decreasing. The sequence

$$\frac{1}{1/2} \quad \frac{1}{1/3} \quad \frac{1}{1/4} \quad \frac{1}{1/5} \quad \frac{1}{1/6} \quad \cdots$$

exhibits the same behavior. As the denominators approach zero, the value of each successive fraction grows without bound, prompting Wallis to hypothesize that

$$\frac{1}{0} = \infty$$

might mean something. But if negatives are the natural continuation of decreasing positives as we pass through zero, then isn't it plausible to say that dividing 1 by a negative number yields a quantity greater than infinity? Can we write

$$\frac{1}{-1} > \infty$$

and possibly be making sense?

This musing by Wallis represents just one voice in the chorus of ideas prompted by negative numbers. Arguments were passionate; minds were changed; definitions were proposed, confronted, discarded and rebooted. And if this flow of ideas was turbulent, imagine how vibrant the discussion became when someone proposed the idea of subjecting a negative number to a square root.

4.3 Chimeras

A square root of a number N is a number that, when squared, equals N . Because $3 \times 3 = 9$, for example, we say that 3 is a square root of 9. And because the product of negative numbers is positive, a square root of a negative number cannot exist. An expression such as $\sqrt{-9}$, then,

is buried beneath two layers of obscurity; if -9 itself triggers cognitive chaos, because it is less than nothing, then surely its square root must be banished to the universe of pure imagination.

As late as the 1800s, in fact, a number such as $\sqrt{-9}$ was decried as “chimerical” (that is, monstrous, in reference to a creature from Greek myth that was built from parts of a snake, a goat, and a lion). A modern definition of *chimerical* good-naturedly flips its meaning by defining it as a desirable quality that is, sadly, impossible to attain. This transformation in the meaning of the word mirrors the shift in tone with which thinkers, over time, spoke of chimeras like $\sqrt{-9}$.

John Wallis thought deeply about the connections between geometry, its younger sibling algebra, and about what each might explain about negative numbers and chimeras. Wallis suggested that we extend the metaphor that connects a number to the length of a line segment by caring as well about the *direction* of the segment. To illustrate, we revisit Figure 4.2 and treat point B as a sort of origin, like a zero, and stipulate that segments lying to the right of B have positive length (so BC has length b) while segments lying to the left of B have negative length (so BA has length $-a$). Now if we follow the same argument as presented with that figure, we conclude that

$$BD = \sqrt{-ab}.$$

This interpretation of the figure potentially offers us a snapshot of a chimera caught in the wild.

While Wallis’s argument by no means settles any matters, his inspiration helped to loosen centuries-old biases against negatives and their square roots. There are, after all, scientific reasons to assign a direction to a number, whether we are describing physical forces such as gravity or trends such as rising temperatures. In fact, as we will soon see, our modern understanding of chimerical numbers relies heavily on the concept of direction. Yet it should be noted how carefully and patiently the conversation progressed: about 150 years after Wallis’s thought experiment, the label “chimerical” appeared in print. Clearly a consensus was difficult to attain.

4.4 Cubics

The cubic equation $Ax^3 + Bx^2 + Cx + D = 0$ played a crucial role in the acceptance of chimerical numbers as meaningful. In exercise 1, we will

witness the geometric approach of **Omar Khayyám** (Persia, born 1048), whose methods drew inspiration from centuries of effort by other Persian mathematicians. **Qin Jiushao** (born c. 1200) spearheaded a millennium of Chinese attacks on cubics, following algorithmic lines in the tradition of his predecessors.

Working with cubics is so difficult that **Luca Pacioli** (Italy, born c. 1445) suggested that a full understanding was impossible. Nevertheless, several other Italian mathematicians made progress during the ensuing decades, despite the additional burden of the lack of a symbolic algebra. They described the expression $x^3 = 6x + 10$, for example, as “a cube equals six things and ten”. Bear this in mind during what follows, where we use the notation of algebra in a decidedly non-1500s way.

Here we consider just two of the mathematicians involved with the story of cubics, **Scipione del Ferro** (Italy, born 1465) and **Girolamo Cardano** (Italy, born 1501). It is said that on his deathbed, del Ferro told one of his students his solution to cubics of the form taken by $x^3 = 6x + 10$. Such cubics are called *depressed* thanks to the missing x^2 term.

The trick that del Ferro discovered amounted to the change of variables $x = u + v$. Remember that he would not use these symbols, but would instead consider x as a length and thus x^3 as the volume of a cube, so our modern statement $x = u + v$ would for del Ferro have signified the chopping of a cube into smaller bricks.

Applying $x = u + v$ to $x^3 = 6x + 10$ yields

$$(u + v)^3 = 6(u + v) + 10,$$

which can be tinkered with^[48] to reach

$$u^3 + v^3 + (3uv - 6)(u + v) = 10. \quad (4.1)$$

Now we are welcome to place a stipulation on u and v in any way that we choose because we set $x = u + v$ and it is x for which we are solving. We could require, for example, that u be twice v , or that the product uv must be 10. Here, it is convenient to force $3uv - 6 = 0$, so that (4.1) becomes

$$u^3 + v^3 = 10 \quad \text{with} \quad 3uv - 6 = 0.$$

The second equation simplifies to $v = 2/u$, which substituted into the first equation gives us

$$u^3 + \left(\frac{2}{u}\right)^3 = 10, \quad \text{or} \quad u^6 - 10u^3 + 8 = 0.$$

Letting $z = u^3$ allows us to apply the quadratic formula to

$$z^2 - 10z + 8 = 0$$

to obtain the solutions

$$u = (5 + \sqrt{17})^{1/3} \text{ and } u = (5 - \sqrt{17})^{1/3}.$$

Combining the first of these solutions with $u^3 + v^3 = 10$ reveals^[49] that

$$v = (5 - \sqrt{17})^{1/3},$$

which is the second solution for u above. Thus, because $x = u + v$,

$$x = (5 + \sqrt{17})^{1/3} + (5 - \sqrt{17})^{1/3} \quad (4.2)$$

solves the equation $x^3 = 6x + 10$.

This clever tactic was like a treasure to del Ferro, because in his day it was the custom that teaching positions could be lost to aspiring challengers. Decades later, **Niccolò Fontana** (Italy, born c. 1500), also known as Tartaglia, rediscovered del Ferro's manipulation, and he shared it with Cardano after much cajoling. The widely available details of this story could pass for overwrought fiction.

Cardano's book *Ars Magna* (The Great Art) contains not only del Ferro's technique but also a method for "depressing" the equation

$$Ax^3 + Bx^2 + Cx + D = 0$$

by eliminating the x^2 term. Divide by the leading coefficient A to obtain

$$x^3 + bx^2 + cx + d = 0, \quad (4.3)$$

where $b = B/A$ and so on. We briefly foray into calculus to motivate the forthcoming change of variables that will depress this cubic. Let $f(x)$ denote the left side of (4.3), so $f''(x) = 6x + 2b$. Thus the curve f will have its inflection point on the y axis when $b = 0$. So all that we must do to drop the x^2 term from (4.3) is shift the curve left or right to place the curve's inflection point on the y axis. This horizontal shift requires the substitution of $x - \alpha$ for x , so (4.3) becomes

$$(x - \alpha)^3 + b(x - \alpha)^2 + c(x - \alpha) + d = 0. \quad (4.4)$$

Now $(b - 3\alpha)x^2$ is the x^2 term in the expansion of (4.4), so to drop this term, we require that $\alpha = b/3$. We therefore substitute^[50] $x - b/3$ for x . The shifted curve has different roots than the original (4.3), but reversing the shift poses no difficulty.

4.5 A truly curious thing

Let's revisit del Ferro's technique, this time watching it work on the depressed cubic

$$x^3 = 3px + 2q$$

with p and q positive numbers (the 3 and 2 are choices that turn out convenient in a bit). Now because p and q are positive, then the terms on the right would have had negative coefficients in their original positions on the left. Many mathematicians of Cardano's day still disallowed negative numbers, so for them our moves would be nonsense.

Let $x = u + v$ as before. Then

$$(u + v)^3 = 3p(u + v) + 2q$$

can be rewritten

$$u^3 + v^3 + (3uv - 3p)(u + v) = 2q,$$

whereupon we choose to set $3uv - 3p = 0$, or $v = p/u$. Replacing v in $u^3 + v^3 = 2q$, we obtain

$$u^3 + \left(\frac{p}{u}\right)^3 = 2q.$$

We multiply by u^3 and let $z = u^3$. The result

$$z^2 - 2qz + p^3 = 0$$

leads via the quadratic formula to the solutions

$$u = \left(q \pm \sqrt{q^2 - p^3}\right)^{1/3}.$$

As in the previous section, we conclude that

$$x = \left(q + \sqrt{q^2 - p^3}\right)^{1/3} + \left(q - \sqrt{q^2 - p^3}\right)^{1/3} \quad (4.5)$$

is a solution to the original depressed cubic. We can check (4.5) against (4.2) if we note that (4.2) is a solution to

$$x^3 = 6x + 10$$

and so $p = 2$ and $q = 5$ in

$$x^3 = 3px + 2q. \quad (4.6)$$

Now if we think about the depressed cubic (4.6) as the intersection of the curves x^3 and $3px + 2q$ and do a little calculus, we discover a curious thing. Consider the situation in Figure 4.3, where both p and q are positive, and the straight line $y = 3px + 2q$ is drawn tangent to $y = x^3$ at some point (a, a^3) . The y intercept of the straight line is $(0, 2q)$ and its slope is $3p$. We also know that its slope is $3a^2$ from the derivative of x^3 , so $p = a^2$.

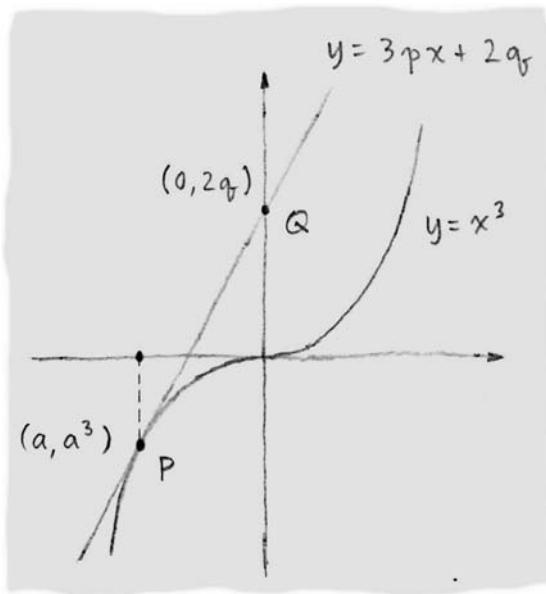


Figure 4.3

If we adjust the slope of the straight line so that its slope $3p$ is greater than the slope of the tangent line PQ , then the straight line will intersect the curve in the first quadrant and twice more in the third quadrant. In other words, when

$$3p > \frac{a^3 - 2q}{a},$$

then (4.6) will have three real solutions. Knowing that $p = a^2$ from above (and bearing in mind that $a < 0$ and $p > 0$), we may write^[51] the inequality as

$$q^2 - p^3 < 0. \quad (4.7)$$

Thus we have shown that $x^3 = 3px + 2q$ has three real solutions precisely when $q^2 - p^3$ in (4.5) is negative; in other words, the equation's solutions are all non-chimerical only when the formula generates chimerical numbers.

Take $x^3 = 6x + 4$ as an example. Here $p = 2$ and $q = 2$, so (4.5) gives the solution

$$x = (2 + \sqrt{-4})^{1/3} + (2 - \sqrt{-4})^{1/3}. \quad (4.8)$$

But because (4.7) holds for $p = 2$ and $q = 2$, we know that the curve x^3 intersects the line $6x + 4$ three times. In fact, we can check that the intersections occur at $x = -2$ and $x = 1 + \sqrt{3}$ and $x = 1 - \sqrt{3}$.

Yet not only does (4.8) contain the chimera $\sqrt{-4}$, but also it appears to give us just one solution, not three. To probe these mysteries, we must learn how expressions like $2 + \sqrt{-4}$ can be visualized.

4.6 The complex plane

A few decades after Cardano published his results, **Rafael Bombelli** (Italy, born c. 1526) coaxed them into revealing the real numbers hiding within. Provoked by the curious fact that (4.5) produces three real solutions precisely when its square roots contain negative numbers, Bombelli extended the rules that govern the behavior of real numbers to apply to chimerical numbers as well.

To illustrate, we will simplify (4.8) to $x = -2$ as Bombelli would have done it (in fact, Bombelli used our $x^3 = 6x + 4$ as an example himself). Let's assume that the cube root of $2 + \sqrt{-4}$ has the same structure $p + \sqrt{-q}$. Then

$$(p + \sqrt{-q})^3 = 2 + \sqrt{-4}, \quad (4.9)$$

$$(p - \sqrt{-q})^3 = 2 - \sqrt{-4}.$$

Multiply the corresponding sides together to obtain

$$p^2 + q = 2. \quad (4.10)$$

This maneuver requires trust in the rule

$$\sqrt{-q} \times \sqrt{-q} = -q. \quad (4.11)$$

It mimics the way that square roots act on positive numbers, but it need not necessarily be the way that square roots act on negatives. Bombelli gamely took (4.11) as a hypothesis and continued.

We expand the left side of the top equation in (4.9) to see that

$$(p^3 - 3pq) + (3p^2 - q)\sqrt{-q} = 2 + \sqrt{-4}$$

and, matching the non-chimerical terms left and right, we have

$$p^3 - 3pq = 2.$$

Together with (4.10), this yields

$$2p^3 = 3p + 1, \quad (4.12)$$

a cubic equation with one accessible solution $p = -1$, which then gives us $q = 1$. Thus $-1 + \sqrt{-1}$ is a cube root of $2 + \sqrt{-4}$. Let's check:

$$\begin{aligned} (-1 + \sqrt{-1})^3 &= (-1)^3 + 3(-1)^2\sqrt{-1} + 3(-1)(\sqrt{-1})^2 + (\sqrt{-1})^3 \\ &= -1 + 3\sqrt{-1} + 3 + (-1)\sqrt{-1} \\ &= 2 + 2\sqrt{-1} \\ &= 2 + \sqrt{-4}. \end{aligned} \quad (4.13)$$

Similarly, we could calculate that $(-1 - \sqrt{-1})^3 = 2 - \sqrt{-4}$ and therefore that at (4.8) we have $x = -2$ as desired. All of this presupposes the good behavior of chimerical numbers, such as in the last step, where we used the rule

$$\sqrt{m} \times \sqrt{n} = \sqrt{mn}$$

as if it applies to negative m and n . But the fact that Bombelli's approach appears to work certainly solidifies our confidence in these extensions of the rules of algebra. (See the endnotes^[52] for another approach.)

Bombelli's efforts, while admirable and evidently successful, leave us with another cubic to solve at (4.12). In this case, the solution $p = -1$ can be guessed. But what if guessing fails? And are there other solutions to (4.12)?

It took more than two centuries before an ingenious insight answered these questions almost effortlessly. This innovation, a way of visually capturing the behavior of chimeras like $2 + \sqrt{-4}$, narrowly escaped John Wallis in the late 1600s. Recall that Wallis suggested that a chimerical

number might carry a geometric meaning if we allow numbers to somehow point in a direction. In 1799, **Caspar Wessel** (Norway, born 1745) argued that we might benefit from treating numbers like $2 + \sqrt{-4}$ as a pair of numbers, first by factoring $\sqrt{-4}$ into $2\sqrt{-1}$ and then by lifting the coefficients of $2 + 2\sqrt{-1}$ (as if $\sqrt{-1}$ were a variable) into an ordered pair $(2, 2)$. Now this pair can be treated like the tip of a vector in the plane, forcing the vertical axis to represent a $\sqrt{-1}$ number line while the horizontal axis continues to represent the usual number line.

Not long after, **Carl Gauss** (Germany, born 1777) independently conceived of such an approach. Gauss used the symbol i to represent $\sqrt{-1}$ and the term *complex* for numbers of the form $a + bi$. The constant a is the *real part* of the complex number, and the coefficient b is the *imaginary part*. From here, we will follow these conventions.

Although Wessel and Gauss took different approaches to visualizing complex numbers, their efforts were at heart quite similar to our modern approach. We associate the complex number $a + bi$ with the point (a, b) and the vector $\langle a, b \rangle$ shown in Figure 4.4. We call this plane the *complex plane*. The *magnitude* of a complex number $a + bi$ is denoted $|a + bi|$ and refers to the length r of the vector $\langle a, b \rangle$, which is found by the Pythagorean theorem to be

$$|a + bi| = \sqrt{a^2 + b^2}.$$

The angle θ created by the vector $\langle a, b \rangle$, as measured (in the usual way) counterclockwise from the positive horizontal axis, is denoted $\arg(a + bi)$ (for *argument*) in most venues, but here, in this less intense setting, we

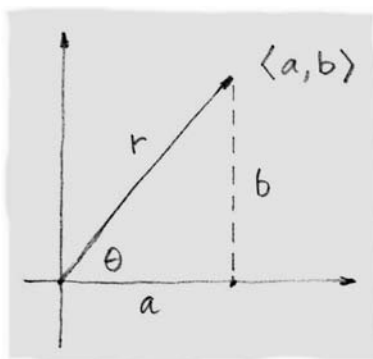


Figure 4.4

will use the usual angle symbol; for example,

$$\angle(1 + i) = \frac{\pi}{4} \quad \text{and} \quad \angle(-2) = \pi.$$

A complex number $a + bi$ also has a polar form

$$a + bi = r(\cos \theta + i \sin \theta) \quad (4.14)$$

as indicated^[53] by the figure.

Earlier in this section, we followed Bombelli's approach in turning $(2 + 2i)^{1/3}$ into $-1 + i$. His method was inspired but technical. We turn our attention now toward finding a way to raise each side of (4.14) to powers. This goal requires an understanding of how complex numbers multiply. In short, to multiply two complex numbers with the help of the complex plane, we multiply their magnitudes and add their angles; that is, to calculate

$$p + qi = (a + bi)(c + di)$$

geometrically, we can use

$$(1) \quad |p + qi| = |a + bi| \cdot |c + di|,$$

$$(2) \quad \angle(p + qi) = \angle(a + bi) + \angle(c + di).$$

Anyone unfamiliar with how or why these observations are true can find help^[54] in the endnotes. Armed with this technique for multiplying numbers in the complex plane, we can recalculate Bombelli's $(2 + 2i)^{1/3}$ in a new way. Because the angle created by $2 + 2i$ is $\pi/4$, we know that its cube root must create the angle $\pi/12$. And because the magnitude of $2 + 2i$ is $\sqrt{8}$ or $(\sqrt{2})^3$, we know that its cube root must have magnitude $\sqrt{2}$. These observations lead to Figure 4.5, where we see^[55] that

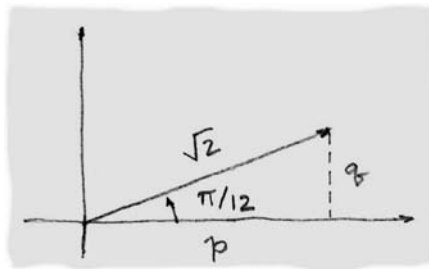


Figure 4.5

$$p = \sqrt{2} \cos(\pi/12) = \frac{1 + \sqrt{3}}{2},$$

$$q = \sqrt{2} \sin(\pi/12) = \frac{\sqrt{3} - 1}{2}.$$

So

$$\frac{1 + \sqrt{3}}{2} + \frac{\sqrt{3} - 1}{2}i = (2 + 2i)^{1/3},$$

which we can check by cubing both sides.

But where is Bombelli's cube root $-1 + i$ from (4.13) earlier in this section? This solution hides in the fact that there are two other angles besides $\pi/12$ that triple to $\pi/4$, namely $3\pi/4$ and $17\pi/12$. (In fact, there are infinitely many other angles that triple to $\pi/4$, but these are the three that lie between 0 and 2π .) The first of these alternative angles leads us to calculate

$$\begin{aligned} \sqrt{2} (\cos(3\pi/4) + i \sin(3\pi/4)) &= \sqrt{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \\ &= -1 + i, \end{aligned}$$

thus unearthing Bombelli's solution. The third angle yields^[56] a third cube root

$$\sqrt{2} (\cos(17\pi/12) + i \sin(17\pi/12)) = \frac{\sqrt{2}}{2} \left(\sqrt{2 - \sqrt{3}} + i \sqrt{2 + \sqrt{3}} \right),$$

which we can check by cubing the right side to get $2 + 2i$. Once we see how the multiplication of complex numbers connects to the addition of the angles of their corresponding vectors, it becomes clear that there are n versions of the n th root of a complex number. This appealing symmetry helped make the use of complex numbers as routine as it is today.

4.7 $\ln(i)$

As we have seen, cubic equations revealed a surprising and undeniable link between complex numbers and real numbers, and innovations in visualizing complex numbers simplified our manipulation of them. These investigations helped to bring an end to widespread suspicion of complex numbers during the centuries spanning the 1500s to the 1800s.

But even while suspicion faded, puzzles continued to vex the mathematical community. In the early 1700s, for example, **Johann Bernoulli** (Switzerland, born 1667) compared the integral

$$\int \frac{1}{1+x^2} dx = \arctan x \quad (4.15)$$

to the very similar

$$\int \frac{1}{1-x^2} dx = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right).$$

We typically use trigonometric substitution to calculate the first integral, but partial fractions to find the second, as follows: because

$$\frac{1}{1-x^2} = \frac{1}{(1+x)(1-x)} = \frac{1/2}{1+x} + \frac{1/2}{1-x},$$

we can say that

$$\int \frac{1}{1-x^2} dx = \frac{1}{2} \ln(1+x) - \frac{1}{2} \ln(1-x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right).$$

Now Bernoulli wondered what might happen if he applied the method of partial fractions to the first integral. Factoring

$$\frac{1}{1+x^2} = \frac{1}{(1+ix)(1-ix)}$$

requires complex coefficients, so we might have complex numerators when we proceed to the partial fractions decomposition

$$\frac{1}{(1+ix)(1-ix)} = \frac{A+Bi}{1+ix} + \frac{C+Di}{1-ix}.$$

It follows^[57] that $A = C = 1/2$ and $B = D = 0$, so

$$\begin{aligned} \int \frac{1}{1+x^2} dx &= \frac{1}{2} \int \frac{1}{1+ix} dx + \frac{1}{2} \int \frac{1}{1-ix} dx \\ &= \frac{1}{2i} \ln(1+ix) - \frac{1}{2i} \ln(1-ix) \\ &= \frac{1}{2i} \ln \left(\frac{1+ix}{1-ix} \right). \end{aligned}$$

Using (4.15) to replace the left side, we get

$$\arctan x = \frac{1}{2i} \ln \left(\frac{1+ix}{1-ix} \right),$$

a perplexing statement that links a function from trigonometry to the logarithm of a complex expression. When $x = 1$, the left side equals

$$\arctan(1) = \frac{\pi}{4}$$

while on the right we get

$$\frac{1}{2i} \ln \left(\frac{1+i}{1-i} \right) = \frac{1}{2i} \ln(i).$$

Equating the last two results leads to

$$\ln(i) = \frac{i\pi}{2},$$

which appears to be extending the domain of the logarithm function beyond what we may be used to. Quirky results like this sparked a debate between Bernoulli and his mentor Gottfried Leibniz over the meaning of logarithms of negative or complex numbers. Both men wished to honor the rules of algebra and logarithms as they pertain to real numbers, but their conclusions differed nevertheless. For example, Bernoulli argued that because

$$\frac{d}{dx} \ln(x) = \frac{1}{x}, \quad \text{and} \quad \frac{d}{dx} \ln(-x) = \frac{-1}{-x} = \frac{1}{x},$$

then $\ln(x)$ must equal $\ln(-x)$, as they share the same derivative. Leibniz countered that one of Bernoulli's two derivatives must be meaningless because one cannot find the logarithm of a negative number. Their dispute occupied them over a lengthy correspondence, with no agreement reached. It fell to Bernoulli's own student Euler to sort this out.

4.8 $i\theta = \ln(\cos \theta + i \sin \theta)$

Euler objected to Bernoulli's conclusion that functions with equal derivatives must themselves be equal, using

$$\frac{d}{dx} \ln(2x) = \frac{2}{2x} = \frac{1}{x} = \frac{d}{dx} \ln(x)$$

as a counterexample. But Euler also pushed back against limiting logarithmic differentiation to positive numbers, writing that Leibniz's suggestion, "if it were right, would shake the foundation of all analysis, which consists principally in the generality of the rules and operations which are judged to be true, whatever the nature we suppose the quantities to be that we apply them to."

After reacting further to both Bernoulli and Leibniz, Euler detailed his own mental travails as he grappled with what meaning the logarithm of a negative number or the logarithm of a complex number could carry. In particular, he explained that his original hypothesis, that such a logarithm could only have one value, evolved to include the possibility that there may be more ways, even infinitely many, to evaluate such a logarithm. Ultimately, he struck upon the following argument and its almost preternatural conclusion.

Start with $x = \cos \theta$ and $y = \sin \theta$ as labeled on the unit circle in Figure 4.6. By the Pythagorean theorem, we know that $x = \sqrt{1 - y^2}$, and from $y = \sin \theta$ we find $dy = \cos \theta d\theta$ as the differential. Thus

$$d\theta = \frac{dy}{\cos \theta} = \frac{dy}{x} = \frac{dy}{\sqrt{1 - y^2}}.$$

Substitute iz for y (so $dy = i dz$ and $1 - y^2 = 1 + z^2$) and then integrate to obtain

$$\int d\theta = \int \frac{i dz}{\sqrt{1 + z^2}} \quad \text{or} \quad \theta = i \int \frac{dz}{\sqrt{1 + z^2}}.$$

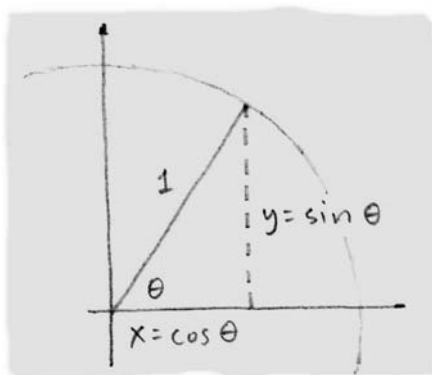


Figure 4.6

This integral succumbs to the trigonometric substitution

$$z = \tan \alpha, \quad \text{so} \quad dz = \sec^2 \alpha \, d\alpha,$$

leading^[58] to

$$\theta = i \int \sec \alpha \, d\alpha.$$

From elementary calculus we know that we can write

$$\theta = i \ln (\sec \alpha + \tan \alpha) + C$$

and so, because $z^2 = \tan^2 \alpha = \sec^2 \alpha - 1$, we get

$$\begin{aligned} \theta &= i \ln (\sqrt{1+z^2} + z) + C \\ &= i \ln (\sqrt{1-y^2} + y/i) + C \\ &= i \ln (\sqrt{1-y^2} + iy/i^2) + C \\ &= i \ln (\sqrt{1-y^2} - iy) + C. \end{aligned}$$

As Figure 4.6 makes clear, when $\theta = 0$ we have $y = 0$, so $C = 0$ as well. We may tweak^[59] the last expression to obtain

$$\theta = -i \ln (\sqrt{1-y^2} + iy) \quad (4.16)$$

and multiplying both sides by i we get

$$i\theta = \ln (\cos \theta + i \sin \theta). \quad (4.17)$$

Here, Euler noted that the periodic natures of the trigonometric functions cosine and sine allow for infinitely many solutions, so (with n as an integer) he wrote (4.17) as

$$i(\theta + 2\pi n) = \ln (\cos \theta + i \sin \theta) \quad (4.18)$$

to reflect this.

Euler checked the plausibility of (4.18), with judicious choices of θ , against the existing properties of logarithms. Letting $\theta = 0$ we get

$$i(2\pi n) = \ln (1),$$

and $n = 0$ yields the familiar $\ln (1) = 0$. With n ranging throughout the integers, we see that

$$\ln (1) = 0, \quad \pm 2\pi i, \quad \pm 4\pi i, \quad \pm 6\pi i, \quad \text{etc.} \quad (4.19)$$

Setting $\theta = \pi$, on the other hand, produces

$$i(\pi + 2\pi n) = \ln(-1),$$

which as a list is

$$\ln(-1) = \pm\pi i, \pm3\pi i, \pm5\pi i, \pm7\pi i, \text{ etc.}$$

One property of logarithms is $\ln(x^m) = m \ln(x)$, so Euler checked it with his results. Because $(-1)^2 = 1$, we can write

$$\ln(1) = \ln((-1)^2) = 2 \ln(-1) = \pm2\pi i, \pm6\pi i, \pm10\pi i, \text{ etc.}, \quad (4.20)$$

all solutions that appear in the list (4.19). (We do not generate the entire list because -1 is just one of the two square roots of 1 , so we could have written $\ln(1^2)$ in the second spot of (4.20), which would have generated the missing entries.)

Euler carefully checked other cases until he was satisfied that his argument had produced a conclusion that did not contradict existing knowledge. With his results, he settled the dispute between his mentor Bernoulli and his mentor Leibniz. But what he discovered was much more, like a path leading to even more curious places.

4.9 $e^{i\theta} = \cos \theta + i \sin \theta$

A few decades before Euler set the argument in the previous section on paper, **Roger Cotes** (England, born 1682) drew conclusion (4.17) from his study of the lengths of certain curves. Exponentiating (4.17), we reach

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (4.21)$$

and though Cotes did not take this final step, we commonly call this the Euler-Cotes formula.

The formula's beauty originates not only in the bond that it forges between e and i and thus between the real and the complex, but also in the intriguing results that it implies. Both of these points can be made at once by obtaining

$$e^{i\pi} = -1 \quad (4.22)$$

from (4.21) when $\theta = \pi$. In this equation, a real irrational number raised to a complex power equals an integer, and three of this book's four constants coalesce in a six symbol truth. In a moment, we will learn to

view this arcane equation as something accessible and plausible, but that knowledge need not stifle our admiration of the thinkers who alchemized this equation.

If we choose to set $\theta = \pi/2$ in (4.21), we can then manipulate^[60] the result to reach either

$$\pi = \frac{2 \ln i}{i} \quad \text{or} \quad i^i = e^{-\pi/2}, \quad (4.23)$$

depending on the choices that we make as we simplify. The first expresses π as an unexpected ratio of complex quantities, and the second reveals that a complex number raised to a complex power can simplify to a real number. In this case,

$$i^i = 0.2078795763507619 \dots$$

These peculiar results are merely instances of the Euler-Cotes formula

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

and a visual interpretation of it can help us assess its odd offspring. Recall from section 4.6 that the right side is a complex number that can be identified with a unit vector in the complex plane that makes the angle θ with the positive horizontal axis. Thus, the left side $e^{i\theta}$ can itself be linked to this vector, as in Figure 4.7. So when $\theta = \pi$, the vector associated with $e^{i\pi}$ points to the location -1 on the horizontal axis. So now we can *see* that

$$e^{i\pi} = -1.$$

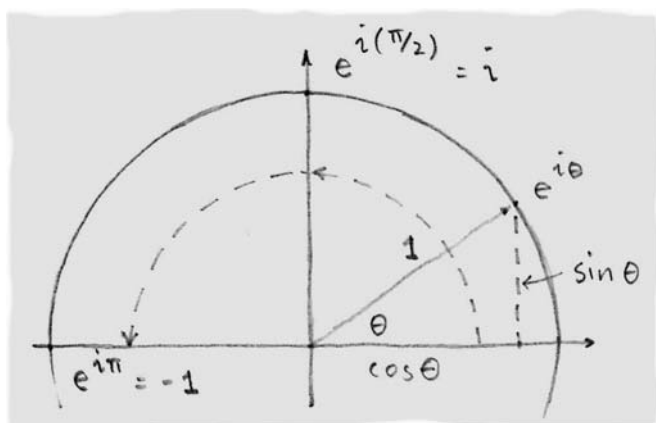


Figure 4.7

Using the same technique, we can set $\theta = \pi/2$ and *see* that

$$e^{i(\pi/2)} = i,$$

from which we generated the mysteries in (4.23).

4.10 The shortest path

Much of the final resistance to complex numbers faded as it became clear that their behavior posed no threat to the rules and operations of algebra. On the contrary, quite often the complex realm opened paths that made already existing results easier to prove. A useful tool for illustrating this effect is named after **Abraham de Moivre** (France, born 1667), and follows easily from the Euler-Cotes formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

by replacing θ with $n\theta$. On the left,

$$e^{i(n\theta)} = (e^{i\theta})^n = (\cos \theta + i \sin \theta)^n,$$

so

$$(\cos \theta + i \sin \theta)^n = \cos (n\theta) + i \sin (n\theta). \quad (4.24)$$

From this formula we can derive trigonometric identities using just a smidgen of algebra. For example, let $n = 2$ in (4.24) and expand the left side to get

$$\cos^2 \theta - \sin^2 \theta + 2i \cos \theta \sin \theta = \cos (2\theta) + i \sin (2\theta).$$

Comparing the real parts, we see that

$$\cos (2\theta) = \cos^2 \theta - \sin^2 \theta,$$

and doing the same with the coefficients of i yields

$$\sin (2\theta) = 2 \cos \theta \sin \theta.$$

Both of these identities are common and useful, and this derivation of them could hardly be more straightforward.

If we let $n = 3$ in (4.24) and carry out the same procedure, we can show that

$$\cos (3\theta) = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

and, using $\sin^2 \theta = 1 - \cos^2 \theta$, we can simplify to

$$\cos(3\theta) = 4\cos^3 \theta - 3\cos \theta. \quad (4.25)$$

Again the derivation is simple, but notice that in this case, with $x = \cos \theta$, the right side of (4.25) becomes $4x^3 - 3x$, which mimics the structure of a depressed cubic equation. Exercise 2 explains how **François Viète** (France, born 1540) used (4.25) to solve such equations.

Further, de Moivre's formula (4.24) simplifies calculations that involve powers of complex numbers. As an example, we will find a fifth root of the number 1 (specifically because we will be reminded of our study of the golden ratio φ). As we saw in section 4.6, the multiplication of complex numbers is linked to the addition of the angles that they make with the positive horizontal axis. This fact distributes the vectors that correspond to the fifth roots of 1 in the pentagonal fashion depicted in Figure 4.8. We will not be surprised, then, when φ appears in our work, knowing the pentagon as its birthplace.

Focus on the vector pointing at P in Figure 4.8. Its angle equals $2\pi/5$ (or 72°), so we choose $\theta = 2\pi$ and $n = 1/5$ in (4.24) to get

$$(\cos(2\pi) + i \sin(2\pi))^{1/5} = \cos(2\pi/5) + i \sin(2\pi/5).$$

The left simplifies to $1^{1/5}$ as desired. For the right, recall that $\tan(2\pi/5) = \sqrt{4\varphi + 3}$ from section 1.3. Thus,

$$\tan(2\pi/5) = \frac{y}{x} = \sqrt{4\varphi + 3} \quad \text{and} \quad x^2 + y^2 = 1$$

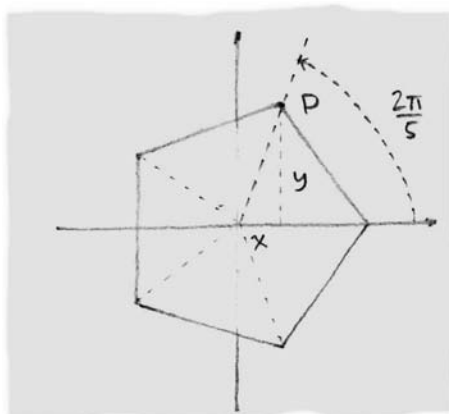


Figure 4.8

(this second fact by the Pythagorean theorem). Solving^[61] for x yields

$$x = \cos(2\pi/5) = \frac{1}{2\varphi}, \quad \text{and so} \quad y = \sin(2\pi/5) = \frac{\sqrt{\varphi+2}}{2}. \quad (4.26)$$

So we have our fifth root of 1:

$$1^{1/5} = \frac{1}{2\varphi} + \frac{i\sqrt{\varphi+2}}{2}. \quad (4.27)$$

Raising the complex right side to the power 5 gives the simple result 1. We could find the three remaining fifth roots at the other vertices of the pentagon by the same technique. Alternatively, because they are

$$1^{2/5}, 1^{3/5}, \text{ and } 1^{4/5},$$

we could instead take powers of (4.27). Either way, the symmetry in Figure 4.8 plays to our advantage.

4.11 $\varphi = e^{i\pi/5} + e^{-i\pi/5}$

What better punchline for this book than the formula in the heading of this section, which binds all four of our constants into one elegant expression? We have just the results in our toolkit to reach it quickly.

Replace θ with $-\theta$ in the Euler-Cotes formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

and use the negative angle trigonometric identities

$$\cos(-\theta) = \cos \theta \quad \text{and} \quad \sin(-\theta) = -\sin \theta$$

to reach

$$e^{-i\theta} = \cos \theta - i \sin \theta.$$

Adding this to the original formula gives us

$$2 \cos \theta = e^{i\theta} + e^{-i\theta}. \quad (4.28)$$

From (4.26) we have $\cos(2\pi/5) = 1/(2\varphi)$, and with the help^[62] of the half-angle identity for cosine we can convert this to $\cos(\pi/5) = \varphi/2$. And so

$$\varphi = e^{i\pi/5} + e^{-i\pi/5}$$

pulls everything together into a fitting last word.

4.12 Further content

- 4.1 **Khayyám's geometric solution to a cubic.** Many centuries before Cardano and Bombelli studied cubic equations, **Omar Khayyám** (Persia, born 1048) published a treatise on mathematics that included a thorough treatment of such equations. True to his time, his approach was geometric, not heavily symbolic, and averse to negative numbers. For example, to describe the cubic

$$x^3 + ax^2 + bx = c, \quad (4.29)$$

Khayyám wrote,

A cube plus squares plus edges equals a number,

and he assumed that all necessary numbers were positive.

Limiting his tools to compass, straightedge, and hyperbola, Khayyám showed how to start with segments of lengths a, b, c (the coefficients in the cubic equation (4.29)) and construct a segment with length equal to the solution to the equation. This exercise outlines his procedure.

We begin with a hyperbola. In Figure 4.9, focus on segments AB (think of it as a horizontal axis) and AC (think of it as a vertical axis) and the curve TE , which depicts a hyperbola with AB and AC as its asymptotes. Using modern notation, we could describe this curve with the equation $y = \alpha/x$ for some $\alpha > 0$ and say that we are only showing a piece of the curve in the first quadrant. Khayyám had none of this modern apparatus. For him, the relationship between a hyperbola and its asymptotes kept a particular product fixed: at any point on the hyperbola, the product of the distances to each asymptote from that point equaled a constant. Thus, for example,

$$GC \cdot GJ = ED \cdot EB. \quad (4.30)$$

As Khayyám would phrase this, the areas of rectangles $GCAJ$ and $EDAB$ are equal.

Now let's build Figure 4.9 from a blank page using our simple tools. Draw a segment AB with length c/b and then construct AD perpendicular to AB so that AD has length $\sqrt{b}$. (We learned how to

- (b) When a perpendicular connects a point on a semicircle to the diameter, as is the case with GH and FE , then the square of the perpendicular's length equals the product of the lengths of the two pieces of the diameter. (Using Figure 4.2, we saw this earlier.) Thus,

$$GH^2 = FH \cdot HE. \quad (4.32)$$

From (4.31) and (4.32), derive

$$AD^2 \cdot HE = DH^2 \cdot FH. \quad (4.33)$$

- (c) Replace FH with $DH + FD$ in (4.33), and then add $DH \cdot AD^2$ to both sides. Simplify the result to obtain

$$DH^3 + FD \cdot DH^2 + AD^2 \cdot DH = AD^2 \cdot ED. \quad (4.34)$$

- (d) Explain why (4.34) provides the answer that Khayyám sought.

Historical note. Khayyám, bound by his rejection of negative numbers, provided similar constructions for various versions of the cubic equation (4.29).

4.2 Viète's trigonometric approach to cubics. In section 4.10, we used the Euler-Cotes formula to generate the trigonometric identity

$$\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta,$$

which may be written

$$\cos^3 \theta = \frac{3}{4} \cos \theta + \frac{1}{4} \cos(3\theta). \quad (4.35)$$

This formula's resemblance to the depressed cubic equation

$$x^3 = 3px + 2q \quad (4.36)$$

(from section 4.5) prompted Viète to discover a way to solve the depressed cubic equation with the help of (4.35).

- (a) The conspicuous choice $x = \cos \theta$ fails because x represents a solution to the cubic equation, so can potentially take on any value, while $\cos \theta$ is trapped between -1 and 1 . To free $\cos \theta$ from its bounds, we multiply it by an arbitrary constant a and let $x = a \cos \theta$. Use this substitution to write (4.35) as

$$x^3 = 3 \left(\frac{a^2}{4} \right) x + \frac{a^3}{4} \cos(3\theta). \quad (4.37)$$

- (b) Comparing (4.37) to (4.36), it seems prudent to let

$$p = a^2/4, \quad \text{which is equivalent to} \quad a = 2\sqrt{p}, \quad (4.38)$$

and

$$2q = \frac{a^3}{4} \cos(3\theta).$$

Show how from these results we can reach

$$\theta = \frac{1}{3} \arccos\left(\frac{q}{p^{3/2}}\right). \quad (4.39)$$

- (c) We initially let $x = a \cos \theta$, so (4.38) and (4.39) together yield

$$x = 2\sqrt{p} \cos\left[\frac{1}{3} \arccos\left(\frac{q}{p^{3/2}}\right)\right]. \quad (4.40)$$

This is Viète's formula. In section 4.5, we found the solutions of $x^3 = 6x + 4$ to be $x = -2$ and $x = 1 \pm \sqrt{3}$. Use (4.40) to find one of them.

- (d) The expression $q/p^{3/2}$ in (4.40) must be contained between -1 and 1 for the inverse cosine function to give a real number. Explain the connection between this observation and (4.7) from section 4.5.
- (e) If the expression $q/p^{3/2}$ in (4.40) is not contained between -1 and 1 , then the inverse cosine function outputs a complex number. Describe how (4.16) and Figure 4.6 may be used to calculate

$$\arccos(2) = i \ln(2 + \sqrt{3})$$

as an example of this phenomenon. Note that the cosine of each side results in

$$\cos(i \ln(2 + \sqrt{3})) = 2,$$

and because we evaluate trigonometric functions at angles, here we have found the cosine of a complex angle.

4.3 A complex approach to the Basel problem. Section 2.9 gave one of Euler's proofs that

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots \quad (4.41)$$

(the solution to what is known as the Basel problem). Here we outline a clever alternative that makes use of our knowledge of complex numbers.

- (a) From elementary calculus, we know that the series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad (4.42)$$

converges for $-1 < x \leq 1$. May we safely replace x with a complex expression? The answer is yes, so long as certain conditions are met. The note at the end of this exercise elaborates. For now, we will trust that replacing x with e^{-2ix} poses no threat.

Make this substitution in (4.42) and then integrate both sides from $x = 0$ to $x = \pi/2$ to show that

$$\int_0^{\pi/2} \ln(1 + e^{-2ix}) dx$$

is equal to

$$\frac{-1}{2i} \left(e^{-i\pi} - 1 - \frac{e^{-2i\pi} - 1}{2^2} + \frac{e^{-3i\pi} - 1}{3^2} - \cdots \right).$$

- (b) Explain how $e^{i\pi} = -1$ (from (4.22)) helps us simplify this to

$$\frac{1}{i} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots \right).$$

As we saw in section 2.9, the sum in the parentheses is $3/4$ of the sum on the right side of (4.41).

- (c) By factoring e^{-ix} from both terms inside $\ln(1 + e^{-2ix})$, and then using properties of logarithms along with (4.28), show that

$$\ln(1 + e^{-2ix}) = -ix + \ln(2 \cos x).$$

- (d) Use all results that we have thus far established to obtain

$$\int_0^{\pi/2} \ln(2 \cos x) dx = i \left[\frac{\pi^2}{8} - \frac{3}{4} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \right) \right]. \quad (4.43)$$

- (e) The integral on the left of (4.43) is improper (at $\pi/2$), and the antiderivative of $\ln(2 \cos x)$ is not elementary, but it can be shown (using theory that is beyond the scope of this book) that the integral converges to a real number. The right side of (4.43), by contrast, is merely a real coefficient times i and is thus purely imaginary. Therefore, to be equal, each side must equal 0. Set the right side equal to 0 to establish (4.41).

Note. As hinted at in part (a), we may only replace x with e^{-2ix} under the condition that $|e^{-2ix}| < 1$. But

$$\begin{aligned} |e^{-2ix}| &= |\cos(2x) + i \sin(2x)| \\ &= (\cos^2(2x) + \sin^2(2x))^{1/2} \\ &= 1, \end{aligned}$$

which then (it turns out) requires that we carefully take limits. The details do not lie within the purview of this book, although they lie right on the boundary. Investigations into matters like this occupied mathematicians long after complex numbers were established as meaningful and useful.

4.4 Hamilton discovers the quaternions. The complex number

$$a + bi \quad \text{where } i^2 = -1$$

can be associated with the ordered pair (a, b) and visualized on the complex plane as a vector that extends to that point. Further, the multiplication of two complex numbers in the complex plane involves the addition of angles, and therefore rotation. Like us, **William Hamilton** (England, born 1805) knew these facts, and he asked the natural questions: might there be *triplet* numbers with two imaginary parts, of the form

$$a + bi + cj \quad \text{where } i^2 = -1 \text{ and } j^2 = -1 \text{ but } i \neq j$$

as well? And might the product of two such numbers involve rotation in a *three* dimensional space?

Hamilton wished to preserve the behaviors of complex numbers while he pushed into new territory, so he stipulated that

The magnitude of the product of two triplets should equal the product of their individual magnitudes.

Complex numbers obey this rule (see (13) in Extra Help if need be).

- (a) To find the magnitude of a triplet $a + bi + cj$, we need a three dimensional extension of the Pythagorean theorem. Sketch a vector that extends to the point (a, b, c) and use the Pythagorean

theorem twice to show that

$$|a + bi + cj| = (a^2 + b^2 + c^2)^{1/2}.$$

Recall that the notation on the left signifies the magnitude (or length) of the triplet.

- (b) Hamilton checked if the multiplication rule held when a triplet was multiplied by itself. He computed

$$(a + bi + cj)^2 = (a^2 - b^2 - c^2) + i(2ab) + j(2ac) + ij(2bc). \quad (4.44)$$

Verify his result. Note that to reach his conclusion, Hamilton assumed that $ij = ji$, a reasonable assumption given that multiplication is commutative for real numbers.

- (c) Now show that when we find the magnitude of each side of (4.44), the right side's magnitude exceeds the left side's by the term $(2bc)^2$.
- (d) This mismatch of the two magnitudes could be corrected if we set $ij = 0$, but this would contradict the multiplication rule. Explain both the correction and the contradiction.
- (e) Hamilton realized that the mismatch could also be repaired if he abandoned his assumption that $ij = ji$, replacing it with $ij = -ji$. How does this do the trick?
- (f) Updating his assumptions, Hamilton asserted that

$$i^2 = -1 \quad \text{and} \quad j^2 = -1 \quad \text{but} \quad ij = -ji.$$

This last hypothesis fixes (4.44), as we have just seen. So Hamilton tried multiplying two different triplets $a + bi + cj$ and $x + yi + zj$ using his new assumptions. Check that the product $(a + bi + cj)(x + yi + zj)$ equals

$$(ax - by - cz) + i(ay + bx) + j(az + cx) + ij(bz - cy)$$

and then explain why the magnitudes differ by a factor of $4bcyz$.

Historical note. After further struggle to repair this new error (his son routinely asked him at breakfast if Papa had solved his puzzle yet), Hamilton was struck by inspiration while on a walk with his wife. He realized that he could allow a third imaginary number k to join i and j , forming the *quaternion* $a + bi + cj +$

dk , and let the troublesome product ij equal k . He once again tweaked his assumptions to read

$$i^2 = -1 \quad j^2 = -1 \quad k^2 = -1$$

$$ij = k \quad jk = i \quad ki = j$$

and then carefully checked the magnitudes of multiplied quaternions in the way that we have done throughout this exercise. This time, everything fell into place. Hamilton had discovered a new set of numbers that behaved in many ways like complex numbers, but with a remarkable new structure that had never before been thoroughly studied.

Hamilton's results conclude this chapter on i for three reasons. His story reminds us that in mathematics, the door just opened is rarely the last; almost every discovery results from persistence and risk taking; and one need not be a lifelong expert to be a pioneer. These takeaways underlie everything contained in this book.

APPENDIX **A**

Wallis's original derivation of his formula for π

At (2.11), we saw John Wallis's formula

$$\frac{\pi}{2} = \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \frac{6}{7} \times \cdots$$

but did not pursue his original derivation due to the extensive work required. His clever argument, however, belongs in this book, as it testifies to the ingenuity of mathematicians who lacked the tools of calculus and were forced to make do with basic algebra and geometry. Here is how Wallis proceeded.

On the left in Figure A.1, we see a quarter circle inside a unit square. The quarter circle can be described by the equation $y = (1 - x^2)^{1/2}$, and we will use this description despite the fact that Wallis did not have this convenient notation available to him. (Wallis did draw figures like those in Figure A.1, but described the proportions of each curve in words.) The area of the quarter circle is $\pi/4$ and, wishing to derive a formula for π , Wallis guessed that the study of curves that fit the pattern $y = (1 - x^m)^n$ would prove fruitful.

Let's begin to follow Wallis's path by considering the curve $y = x^3$, inscribed in a unit square on the right in Figure A.1. What proportion of the square is shaded (so, below the curve)? Wallis argued that we could find this proportion by this strategy:

1. Slice the square vertically (like from A to C) at equal intervals (as indicated by the marks along the bottom edge),

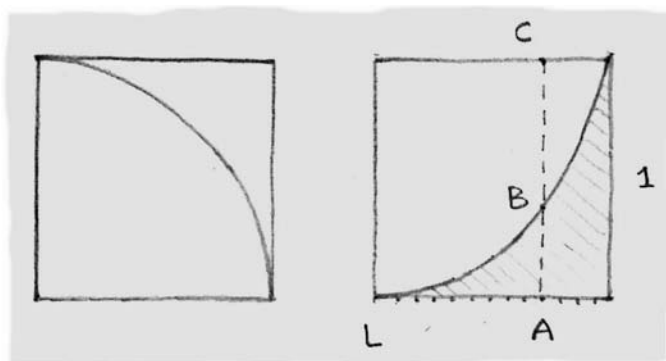


Figure A.1

2. sum the lengths of each part of each slice that land below the curve (like AB),
3. put this sum in ratio to the sum of the entire length (AC) of each slice, and
4. let the number of slices become infinite.

This slicing method has ancient roots. **Democritus** (Greece, born c. 460 BCE), for example, suggested that if cross sectional areas of two volumes were equal throughout their equal heights, then the volumes must be equal.

If there are n equidistant vertical slices, then their distances from the lower left corner L (see Figure A.1 again) can be marked

$$0/n, 1/n, 2/n, \dots, n/n.$$

Thus, the ratio sought in step 3 above is

$$\frac{(0/n)^3 + (1/n)^3 + (2/n)^3 + \dots + (n/n)^3}{1 + 1 + 1 + \dots + 1}$$

or, multiplying every term by n^3 ,

$$\frac{0^3 + 1^3 + 2^3 + \dots + n^3}{n^3 + n^3 + n^3 + \dots + n^3}.$$

Wallis experimented with this last expression in hopes of pinning down its behavior as n becomes infinitely large (part 4 of his strategy). The first

few cases are

$$\begin{aligned}\frac{0+1}{1+1} &= \frac{1}{2} = \frac{1}{4} + \frac{1}{4} \\ \frac{0+1+8}{8+8+8} &= \frac{9}{24} = \frac{1}{4} + \frac{1}{8} \\ \frac{0+1+8+27}{27+27+27+27} &= \frac{36}{108} = \frac{1}{4} + \frac{1}{12} .\end{aligned}$$

Wallis concluded that "If an infinite series of quantities which are the cubes of a series of continuously increasing numbers in arithmetic progression, beginning, say, with 0, is divided by the sum of numbers all equal to the highest and equal in number, then we obtain $1/4$." So the shaded region on the right in Figure A.1 occupies $1/4$ of the unit square.

In a similar way, Wallis concluded that

The region below the curve $y = x^n$ (where n is any positive rational number) occupies $1/(n+1)$ part of the unit square (W1) that encloses it.

In modern notation,

$$\int_0^1 x^n dx = \frac{1}{n+1} .$$

Now recall that Wallis was specifically interested in the area defined by curves of the form $y = (1 - x^m)^n$ so that he can investigate the quarter circle $y = (1 - x^2)^{1/2}$. Take $y = (1 - x)^2$ as an example, and remember that all areas are being found inside a unit square. Expanding and applying (W1), we have

$$\begin{aligned}\text{area under } (1 - x)^2 &= \text{area under } (1 - 2x + x^2) \\ &= 1 - 2 \cdot \frac{1}{2} + \frac{1}{3} \\ &= \frac{1}{3} .\end{aligned}\tag{A.1}$$

Wallis created Table A.1 to store results like this one. Each cell holds the number N so that the region under the curve $y = (1 - x^{1/p})^k$ occupies $1/N$ of the unit square. (Wallis put reciprocals in his table for clarity. Every fractional area has the numerator 1, so by using reciprocals, Wallis was simply ignoring their clutter.) So 3 appears in the cell for $p = 1$ and $k = 2$ thanks to (A.1). The cell containing $\square$ marks the key quantity that

Table A.1: The reciprocal of each entry is the portion of a unit square (as in Figure A.1) beneath $y = (1 - x^{1/p})^k$.

		k						
		0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3
p	0	1		1		1		1
	$\frac{1}{2}$		$\square$					
	1	1		2		3		4
	$\frac{3}{2}$							
	2	1		3		6		10
	$\frac{5}{2}$							
	3	1		4		10		20

Wallis was pursuing. The extra help^[63] has further examples of how the entries in Table A.1 are calculated.

We might also have filled in the rows for fractional p and whole numbered k using an approach like in calculation (A.1), but delayed in order to highlight the sequences that appear in rows $p = 2$ and $p = 3$. The sequence 1, 3, 6, 10, . . . consists of the *triangular numbers* and the sequence 1, 4, 10, 20, . . . lists the *tetrahedral numbers*. To fill in the entries between those in these sequences, in the columns that correspond to $k = 1/2$ and $k = 3/2$ and so on, Wallis would have needed the ability to expand expressions with fractional exponents like $(1 - x^{1/3})^{1/2}$ and $(1 - x^{1/2})^{3/2}$. He lacked this knowledge, but it would not be long until Isaac Newton discovered a way. (Section 2.7 elaborates.)

Wallis instead appealed to a process called *interpolation*, which in this case amounts to making smart guesses as to what numbers come between those in a given sequence. The sequence 1, 2, 3, 4, . . . in row $p = 1$, for example, interpolates as

$$1 \qquad 3/2 \qquad 2 \qquad 5/2 \qquad 3 \qquad 7/2 \qquad 4 \qquad \dots$$

by common sense. But what numbers should occupy the blanks in the sequence

$$1 \qquad \rule{1cm}{0.4pt} \qquad 3 \qquad \rule{1cm}{0.4pt} \qquad 6 \qquad \rule{1cm}{0.4pt} \qquad 10 \qquad \dots$$

from row $p = 2$ in order to properly interpolate the triangular numbers? Wallis appealed to geometry. The modifier "triangular" is prompted by

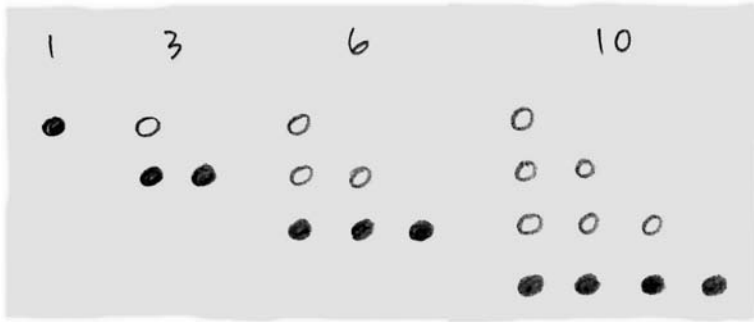


Figure A.2

the visual representation in Figure A.2. This chart lists the number of dots in each figure and the number of dots along the base:

dots total: 1 3 6 10 ...

dots along base: 1 2 3 4 ...

Wallis guessed that the second row should be interpolated in the obvious way, as in

dots total: 1 ____ 3 ____ 6 ____ 10 ...

dots along base: 1 3/2 2 5/2 3 7/2 4 ...

No matter that a triangle like those in Figure A.2 can't have a fractional number of dots along its base. Audacious leaps are hallmarks of noteworthy mathematicians like John Wallis.

To interpolate the top row, Wallis determined the ratio of each entry in the top row to its counterpart in the bottom row. The n th triangular number has $1 + 2 + 3 + \cdots + n$ dots, and the well known formula

$$1 + 2 + 3 + \cdots + n = \frac{n}{2}(n + 1)$$

enabled Wallis to find that

$$\frac{\text{\# dots total}}{\text{\# dots along base}} = \frac{\frac{n}{2}(n + 1)}{n} = \frac{1}{2}(n + 1).$$

Substituting $3/2$ for the number of dots on a side yields

$$\frac{\text{\# dots total}}{3/2} = \frac{1}{2} \left(\frac{3}{2} + 1 \right) \quad \text{and so} \quad \text{\# dots total} = \frac{15}{8}.$$

Wallis therefore concluded that there are $15/8$ dots in a triangle that has $3/2$ dots along its base. (Again, this imaginative result is not meant to be drawn.) The same process^[64] applied to $5/2$ and $7/2$ fills in the top row:

# dots total:	1	15/8	3	35/8	6	63/8	10	...
# dots along base:	1	3/2	2	5/2	3	7/2	4	...

These results fill in row $p = 2$ of Table A.2. Note the symmetry that row $p = 2$ shares with column $k = 2$, which contains entries that can be calculated directly with (W1) without interpolation. Row $p = 1$ of Table A.2 has been filled in using interpolation and column $k = 1$ by direct calculation. Wallis interpolated the tetrahedral numbers $1, 4, 10, 20, \dots$ by the same technique that he applied to the triangular numbers (but the table only includes the results for $p = 3, k = 1/2$ and $p = 1/2, k = 3$).

Wallis was now ready to move on $\square$. Focus on row $p = 1/2$ of Table A.2, which has

$$1 \qquad 3/2 \qquad 15/8 \qquad 105/48 \qquad \dots$$

as its numerical entries. Lacking a geometric metaphor (such as he found when he interpolated the triangular numbers) for these fractions, Wallis sought a purely arithmetic pattern within their structures. He found it: the sequence is generated by the partial products of

$$1 \times \frac{3}{2} \times \frac{5}{4} \times \frac{7}{6} \times \dots .$$

The numerators being all odds and the denominators all evens, Wallis reasoned that he could calculate the missing entries in row $p = 1/2$

Table A.2: The reciprocal of each entry is the portion of a unit square beneath $y = (1 - x^{1/p})^k$. (Here, a few more entries of Table A.1 have been filled in.)

		k						
		0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3
p	0	1		1		1		1
	$\frac{1}{2}$	1	$\square$	$\frac{3}{2}$		$\frac{15}{8}$		$\frac{105}{48}$
	1	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
	$\frac{3}{2}$			$\frac{5}{2}$		$\frac{35}{8}$		
	2	1	$\frac{15}{8}$	3	$\frac{35}{8}$	6	$\frac{63}{8}$	10
	$\frac{5}{2}$			$\frac{7}{2}$		$\frac{63}{8}$		
	3	1	$\frac{105}{48}$	4		10		20

using the partial products of

$$\square \times \frac{4}{3} \times \frac{6}{5} \times \frac{8}{7} \times \cdots ,$$

yielding the interpolated row

$$1 \quad \square \quad 1 \times \frac{3}{2} \quad \square \times \frac{4}{3} \quad 1 \times \frac{3}{2} \times \frac{5}{4} \quad \square \times \frac{4}{3} \times \frac{6}{5} \quad \cdots .$$

These quantities grow at a slowing rate that Wallis analyzed in the following way. Label the quantities in the sequence $a_1, a_2, a_3, \dots$, and note that

$$\frac{a_3}{a_1} > \frac{a_5}{a_3} > \frac{a_7}{a_5} > \cdots \quad \text{and} \quad \frac{a_4}{a_2} > \frac{a_6}{a_4} > \frac{a_8}{a_6} > \cdots .$$

Wallis assumed that, in like manner,

$$\frac{a_2}{a_1} > \frac{a_3}{a_2} > \frac{a_4}{a_3} > \cdots ,$$

and considered each inequality in turn.

Starting with

$$\frac{a_2}{a_1} > \frac{a_3}{a_2} \quad \text{or} \quad \frac{\square}{1} > \frac{1 \times 3/2}{\square} ,$$

Wallis obtained

$$\square > \sqrt{\frac{3}{2}} .$$

Then

$$\frac{a_3}{a_2} > \frac{a_4}{a_3} \quad \text{or} \quad \frac{1 \times 3/2}{\square} > \frac{\square \times 4/3}{1 \times 3/2}$$

yields

$$\square < \frac{3}{2} \times \sqrt{\frac{3}{4}} . \tag{A.2}$$

The next such result is

$$\square > \frac{3}{2} \times \frac{3}{4} \times \sqrt{\frac{5}{4}} .$$

If we write (A.2) as

$$\square < \frac{3}{2} \times \frac{3}{4} \times \sqrt{\frac{4}{3}} ,$$

then we can merge the last two inequalities to see that

$$\frac{3}{2} \times \frac{3}{4} \times \sqrt{\frac{5}{4}} < \square < \frac{3}{2} \times \frac{3}{4} \times \sqrt{\frac{4}{3}} .$$

Continuing in this vein, we obtain

$$\frac{3}{2} \times \frac{3}{4} \times \frac{5}{4} \times \frac{5}{6} \times \sqrt{\frac{6}{5}} < \square < \frac{3}{2} \times \frac{3}{4} \times \frac{5}{4} \times \frac{5}{6} \times \sqrt{\frac{5}{4}}.$$

As we carry on, the two square roots approach 1, forcing the value $\square$ ever closer to

$$\frac{3}{2} \times \frac{3}{4} \times \frac{5}{4} \times \frac{5}{6} \times \frac{7}{6} \times \frac{7}{8} \times \cdots.$$

Because $1/\square = \pi/4$, we have demonstrated the validity of (2.11), named after the thinker who so creatively discovered it.

APPENDIX **B**

**Newton’s original
generalized binomial
theorem**

Section 2.7 presented the generalized binomial theorem without any hint of its origins. This appendix provides that story, which illustrates the fearless persistence with which mathematicians like Isaac Newton applied themselves to unsolved problems.

Table B.1 reorganizes John Wallis’s Table A.1 from Appendix A. Each entry in the table (to the right of the dashed vertical line) is the coefficient of the x^m term in the expansion of the binomial expression $(1 + x)^k$. Newton extended the table’s columns to the left of the dashed line, allowing k to take negative values. Because each entry is the sum

Table B.1: Each entry is the coefficient of the x^m term upon expanding $(1+x)^k$.

		k						
		−2	−1					
				0	1	2	3	4
m	0	1	1	1	1	1	1	1
	1	−2	−1	0	1	2	3	4
	2	3	1	0	0	1	3	6
	3	−4	−1	0	0	0	1	4
	4	5	1	0	0	0	0	1
	5	−6	−1	0	0	0	0	0

of the one directly to its left and the one above that, Newton completed columns $k = -1$ and $k = -2$ as shown. As a result, he conjectured that one could expand binomials like

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots \quad \text{and}$$

$$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + 5x^4 - \dots$$

that have negative exponents.

Now while Wallis interpolated the sequences that appear as rows to the right of the dashed line in Table B.1, introducing columns for $k = 1/2, 3/2, 5/2, \dots$, Newton discovered a pattern in the entries of the table that allowed him to create a column for any fractional k . Any collection of columns in the first two rows of Table B.1 may be represented by the template

$$\begin{array}{ccccccc} \dots & 1 & 1 & 1 & 1 & 1 & 1 & \dots \\ & a-2 & a-1 & a & a+1 & a+2 & a+3 & \end{array}$$

(where the entry a need not be aligned with any particular entry in the second row of the table.) Newton hoped to interpolate the second row. Knowing that its entries would no longer differ by 1, he altered the template to read

$$\begin{array}{ccccccc} \dots & b & b & b & b & b & b & \dots \\ & a-2b & a-b & a & a+b & a+2b & a+3b & \end{array}$$

to preserve the additive pattern (namely, that each entry is the sum of the one directly to its left and the one above that) while allowing the difference between the entries in the second row to be a constant b other than 1. Then, extending to a third row, Newton wrote

$$\begin{array}{ccccccc} & b & & b & & b & & b \\ & a-2b & & a-b & & a & & a+b & & a+2b & & a+3b \\ -2a+3b+c & & -a+b+c & & c & & a+c & & 2a+b+c & & 3a+3b+c \end{array}$$

using a new constant c and the additive pattern to fill in the row. This process may generate further rows, but we will stop at row three and show how this tableau helped Newton interpolate the rows of Table B.1.

The columns of Table B.1 give the coefficients in the expansion of $(1+x)^k$, where k is an integer. Newton aimed to allow k to be fractional.

Table B.2: The coefficients in the expansion of $(1+x)^{1/2}$ will appear in the spots marked $\square$.

		k									
		$-3/2$		$-1/2$	0	$1/2$	1	$3/2$	2	$5/2$	3
m		-2		-1							
	0	1		1		1	$\square$	1		1	
	1					0	$\square$	1		2	
	2	-2		-1							
	2	3	$\circ$	1	$\circ$	0	$\square$	0	$\circ$	1	$\circ$
	3					0	$\square$	0		0	
	3	-4		-1							
	4	5		1		0	$\square$	0		0	
	5					0	$\square$	0		0	
		-6		-1							

Let's follow his approach as we expand $(1+x)^{1/2}$. First, insert a column between each column of Table B.1 as in Table B.2. The numbers that belong where the $\square$ symbols appear are the desired coefficients.

Let's target the $\square$ in the row for $m = 2$, where the symbol $\circ$ marks the other missing coefficients for other values of k . Overlay row 3 of Newton's tableau onto row $m = 2$ of Table B.2 so that $a + c$ aligns with the 0 in column $k = 0$. From the matching

$$\begin{array}{ccccccc}
 -2a + 3b + c & -a + b + c & c & a + c & 2a + b + c & 3a + 3b + c & \\
 \circ & & 1 & \circ & 0 & \square & 0
 \end{array}$$

that results, we obtain the equations

$$-a + b + c = 1$$

$$a + c = 0$$

$$3a + 3b + c = 0$$

with solution $a = -3/8, b = 1/4, c = 3/8$. Thus,

$$\square = 2a + b + c = -1/8$$

is the coefficient of x^2 in the expansion of $(1+x)^{1/2}$.

Calculating the coefficient of x^3 in the expansion of $(1 + x)^{1/2}$ requires that we expand Newton's tableau by a row and a column, and overlay the new row onto row $m = 3$ of Table B.2. If we align the new variable d directly under c in the tableau, then we can obtain the resulting system of equations

$$a - b - c + d = -1$$

$$c + d = 0$$

$$3a + b + 3c + d = 0$$

$$6a - 10b - 3c + d = -4$$

with solution $a = -1/4, b = 1/8, c = 5/16, d = -5/16$. Thus,

$$\square = a + 2c + d = 1/16$$

is the coefficient of x^3 in the expansion of $(1 + x)^{1/2}$. Newton produced the expansion

$$(1 + x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4 + \frac{7}{256}x^5 - \dots \quad (\text{B.1})$$

by repeatedly expanding his tableau and solving systems of equations just as we have done.

To expand $(1 + x)^{1/3}$, we must insert two columns between each column of Table B.1 before we overlay Newton's tableau. Doing this yields the expansion

$$(1 + x)^{1/3} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 - \frac{10}{243}x^4 + \frac{22}{729}x^5 - \dots$$

after quite a bit of work.

Newton's method provokes some questions. Can we insert as many blank columns as we like? Does it matter how we overlay the pattern onto the table? Can't we include infinitely many equations in the systems we solve? Perhaps because of these questions, Newton checked his results. An algebraic check of (B.1) involves squaring both sides to see if the infinite expression on the right squares to $(1 + x)$, and indeed it appears to do just that.

Newton's technique, while both imaginative and effective, requires a good deal of algebra. He sought and found a shortcut, the very same

we use today. Studying the coefficients in expansions like (B.1), Newton realized that multiplying together the first $m + 1$ factors of

$$1 \times \frac{k}{1} \times \frac{k-1}{2} \times \frac{k-2}{3} \times \frac{k-3}{4} \times \dots$$

produces the coefficient of x^m in the expansion of $(1+x)^k$. This much-streamlined technique lies at the heart of what we call the *generalized binomial theorem*. A complete proof that properly addresses the skepticisms (listed in the previous paragraph) took over 100 years to emerge.

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Extra help

[1] Because the pentagon is regular, we know that isosceles $\triangle AED$ is congruent to isosceles $\triangle BAE$. Thus all of $\angle ABE, \angle AEB, \angle EAD$, and $\angle EDA$ are equal. Let α be the common measure of these angles. Now $\angle EMD = 2\alpha$ because, as an exterior angle of $\triangle AME$, it equals the sum of the two opposite interior angles. If we can show that $\angle MBC = 2\alpha$ as well, then we will know that AD is parallel to BC .

The interior angles of a regular pentagon are all 108° , so $\angle MED = 108^\circ - \alpha$. Summing the angles of $\triangle MED$, we obtain $2\alpha + \alpha + 108^\circ - \alpha = 180^\circ$, so $\alpha = 36^\circ$. Thus $\angle MED = 108^\circ - 36^\circ = 72^\circ$, which equals 2α , making $\triangle MED$ isosceles (so that $MD = 1$) and $\angle MBC = 72^\circ$ just like $\angle EMD$. So AD is parallel to BC as desired.

[2] To solve

$$AD(AD - 1) = 1$$

for AD , we may let $x = AD$ and write

$$x(x - 1) = 1 \quad \text{or} \quad x^2 - x = 1.$$

We complete the square by adding $1/4$ to both sides, obtaining

$$x^2 - x + 1/4 = 5/4, \quad \text{which factors as} \quad (x - 1/2)^2 = 5/4.$$

Taking the square root of both sides and solving for x leads us to the desired answer.

[3] We must first establish that point G in Figure 1.6 does in fact lie on diagonal BD . Place the rectangle $ABCD$ on coordinate axes so that B lies at the origin, and BC lies along the horizontal axis. The slope of line BD equals

$$\frac{CD}{BC} = \frac{1}{\varphi}$$

and the slope of line BG equals

$$\frac{FG}{BF} = \frac{\varphi - 1}{1}.$$

These slopes are equal by (1.2), so G lies on BD . Now we have shown earlier that rectangles $ABCD$ and $FCDE$ are similar, so the constructions within $FCDE$ are all proportional to those within $ABCD$. Repeating the argument above (forever) leads to the conclusion that ω lies on the intersection of BD and CE .

[4] The first identity $\varphi = \varphi$ gives us a starting point, and the second identity $\varphi^2 = \varphi + 1$ comes from (1.4). Adding these equations yields

$$\varphi + \varphi^2 = 2\varphi + 1,$$

whereupon we use (1.7) to write the left side as φ^3 . Now we may in like manner add $\varphi^2 = \varphi + 1$ and $\varphi^3 = 2\varphi + 1$ to reach $\varphi^4 = 3\varphi + 2$, and continue as long as we wish.

[5] Some additional details that pertain to (1.9):

$$\begin{aligned} B\omega &= \sqrt{\frac{(3\varphi + 2)(3 - \varphi)}{(\varphi + 2)(3 - \varphi)}} \\ &= \sqrt{\frac{7\varphi + 6 - 3\varphi^2}{\varphi + 6 - \varphi^2}} \\ &= \sqrt{\frac{7\varphi + 6 - 3(\varphi + 1)}{\varphi + 6 - (\varphi + 1)}} \quad (\text{because } \varphi^2 = \varphi + 1) \\ &= \sqrt{\frac{4\varphi + 3}{5}}. \end{aligned}$$

[6] Multiply inside the radical by 1 (in the form of $(1/4) \times 4$) to obtain

$$AM = \sqrt{(1/4) \times 4(\varphi^2 - (1/2)^2)} = \sqrt{(1/4)(4\varphi^2 - 1)},$$

and then bring $1/4$ outside the radical as $1/2$.

[7] The proper way to write

$$S_\infty = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \cdots}}} \quad (2)$$

is

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 1 + \frac{1}{1 + \frac{1}{\ddots + \frac{1}{1+1}}} \quad (3)$$

where there are n plus signs. Now we can apply basic properties of limits to move the limit inside the continued fraction. These properties allow us to write

$$\lim_{n \rightarrow \infty} S_n = 1 + \frac{1}{\lim_{n \rightarrow \infty} 1 + \frac{1}{\ddots + \frac{1}{1+1}}} \quad (4)$$

where there are $n - 1$ plus signs in the limit on the right. That limit equals the one on the right side of (3), so if we let

$$x = \lim_{n \rightarrow \infty} S_n,$$

we may write (4) as

$$x = 1 + \frac{1}{x}. \quad (5)$$

Looking back at (1.5), we see that x is the golden ratio φ . Thus, we conclude not only that φ is the limiting ratio of the ratio of consecutive Fibonacci numbers, but also that

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \cdots}}} \quad (6)$$

is a continued fraction for φ .

[8] We are assuming that $\varphi = m/n$, so from

$$1 < \varphi < 2 \quad \text{we can write} \quad 1 < \frac{m}{n} < 2.$$

Multiplying through by n and then subtracting n from each of the three expressions yields

$$0 < m - n < n.$$

[9] The statement “If φ is rational, then $\sqrt{5}$ is rational,” is called an *implication*, and takes the form “If P , then Q .” Such a claim is false only when P is true but Q is false. To see this, consider the implication “If your final grade is 90–100, then you get an A.” One of four outcomes is possible:

Your final grade is 90–100 and you get an A.

Your final grade is below 90 and you do not get an A.

Your final grade is below 90 and you get an A.

Your final grade is 90–100 and you do not get an A.

No one would challenge the implication if either of the first two outcomes occur. Nor should we claim that the implication is false if the third outcome should happen, although we might be a bit bewildered by the generosity. Only the final outcome shows the implication to be a lie, and this is where P is true but Q is false.

Now we know by the equation $2\varphi - 1 = \sqrt{5}$ that if φ is rational (call this P), then $\sqrt{5}$ is rational (call this Q). This implication is a true one, so we cannot have P true when Q is false. All we must do, then, is prove that $\sqrt{5}$ is not rational, making Q false, and we will therefore know that P must also be false.

[10] Substitute $p = q + r$ into $p/q = q/r$ to obtain

$$\frac{q+r}{q} = \frac{q}{r} \quad \text{or} \quad q^2 - rq = r^2.$$

We may use the quadratic formula to solve for q (using the coefficients $a = 1$ and $b = -r$ and $c = -r^2$) or, as we do here, we may complete the square. Add $r^2/4$ to each side to reach

$$q^2 - rq + \frac{r^2}{4} = r^2 + \frac{r^2}{4}$$

and then factor the left while adding the right to see that

$$\left(q - \frac{r}{2}\right)^2 = \frac{5r^2}{4}.$$

Finding the square root of each side and simplifying yields

$$q = \frac{r}{2} \pm \frac{\sqrt{5r^2}}{2} = r \cdot \frac{1 \pm \sqrt{5}}{2}.$$

Because q is a positive length, we may discard the negative solution and conclude that $q = \varphi r$.

Now from the original proportion we get $pr = q^2$ and with $q = \varphi r$ we find that $p = \varphi q$.

[11] In showing that an angle inscribed in a semicircle must be a right angle, we will make an assumption about parallel lines: if lines AB and DE are parallel (as in Figure 1), and a line AC cuts them, then angles CAQ and ACD are equal. In the same way, angles CBQ and BCE are equal. This intuitive assumption can be proved, but we will not do so here.

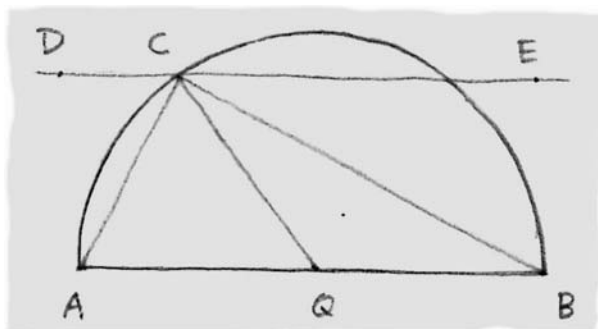


Figure 1

As in the figure, let Q be the center of a semicircle with diameter AB and choose a point C on the semicircle. Construct sides AC and BC of $\triangle ABC$ along with radius QC . Draw DE parallel to AB through C . We wish to show that $\angle ACB$ is a right angle. We are assuming (as described above) that

$$\begin{aligned}\angle CAQ &= \angle ACD \quad \text{and} \\ \angle CBQ &= \angle BCE.\end{aligned}\tag{7}$$

Now

$$\angle ACD + \angle ACB + \angle BCE = 180^\circ$$

because these angles lie on a straight line, and therefore

$$\angle CAQ + \angle ACB + \angle CBQ = 180^\circ\tag{8}$$

when we make the replacements indicated in (7). As an aside, (8) indicates that the angles in a triangle sum to 180° .

Because both $\triangle AQC$ and $\triangle BQC$ are isosceles, we know that

$$\begin{aligned}\angle CAQ &= \angle ACQ \quad \text{and} \\ \angle CBQ &= \angle BCQ.\end{aligned}$$

Substituting these results into (8) yields

$$\angle ACQ + \angle ACB + \angle BCQ = 180^\circ.$$

But $\angle ACB$ clearly equals the sum of the other two on the left side of this equation, so we may write

$$2\angle ACB = 180^\circ,$$

from which the desired conclusion immediately follows.

[12] Here we quickly deal with the claim that the arithmetic geometric mean inequality

$$\frac{1}{n}(a_1 + a_2 + \cdots + a_n) \geq (a_1 a_2 \cdots a_n)^{1/n}$$

achieves equality when $a_1 = a_2 = \cdots = a_n$. Let a denote the common value of the n numbers. Then the right side of the inequality is

$$(a a \cdots a)^{1/n} \quad \text{or} \quad (a^n)^{1/n}$$

because there are n copies of a in the parentheses. This expression simplifies to a , as does the left side

$$\frac{1}{n}(a + a + \cdots + a) \quad \text{or} \quad \frac{1}{n}(na)$$

when there are n copies of a .

[13] We reach (2.2) from

$$(QM)^2 + \left(\frac{a}{2}\right)^2 = 1^2 \quad \text{and} \quad QM + MC = 1$$

by solving the first equation for $(QM)^2$:

$$(QM)^2 = 1 - \frac{a^2}{4} = \sqrt{\frac{1}{4}(4 - a^2)} = \frac{1}{2}\sqrt{4 - a^2},$$

and then using the second equation to replace QM :

$$(1 - MC)^2 = \frac{1}{2}\sqrt{4 - a^2}.$$

Finally, take the square root of each side, and solve for MC .

[14] The Pythagorean theorem applied to $\triangle AMC$ yields

$$b^2 = \left(\frac{a}{2}\right)^2 + (MC)^2,$$

whereupon we may use (2.2) to replace MC , then simplify:

$$\begin{aligned} b^2 &= \frac{a^2}{4} + \left(1 - \frac{1}{2}\sqrt{4 - a^2}\right)^2 \\ &= \frac{a^2}{4} + 1 - \sqrt{4 - a^2} + \frac{1}{4}(4 - a^2) \\ &= 2 - \sqrt{4 - a^2}. \end{aligned}$$

[15] We would use (2.1) with $n = 12$ to find the area of a 24-gon. Because a is the length of the side of an n -gon, we would use (2.4) and relabel b as $a = \sqrt{2 - \sqrt{3}}$. Thus

$$\begin{aligned}\text{area of a 24-gon} &= \frac{an}{2} \\ &= \frac{\left(\sqrt{2 - \sqrt{3}}\right) \cdot 12}{2} \approx 3.10583.\end{aligned}$$

The 24-gon approximates $\pi \approx 3.14159$ more closely than does a 12-gon, which has area 3. If we sought an even closer approximation to π by finding the area of a 48-gon, we would first use (2.4) to find that

$$b^2 = 2 - \sqrt{4 - a^2}$$

or

$$\begin{aligned}b &= \sqrt{2 - \sqrt{4 - \left(\sqrt{2 - \sqrt{3}}\right)^2}} \\ &= \sqrt{2 - \sqrt{2 + \sqrt{3}}}\end{aligned}$$

is the length of one side of a 24-gon. This is the value we would use for a when we looped back to (2.1) to find the area of a 48-gon.

[16] To establish (2.5), let

$$S_n = 1 + x + x^2 + x^3 + \cdots + x^n$$

and multiply both sides by $(1 - x)$ to obtain

$$\begin{aligned}(1 - x)S_n &= (1 - x)(1 + x + x^2 + x^3 + \cdots + x^n) \\ &= 1 - x + x - x^2 + x^2 - \cdots - x^n + x^n - x^{n+1} \\ &= 1 - x^{n+1}.\end{aligned}$$

Dividing by $(1 - x)$ and taking the limit yields

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{1 - x^{n+1}}{1 - x} = \frac{1}{1 - x}$$

so long as $x \in (-1, 1)$.

[17] We have constructed the angle α so that $\tan \alpha = 1/5$, so by (2.9) we see that

$$\begin{aligned}\tan(\alpha + \alpha) &= \frac{\tan \alpha + \tan \alpha}{1 - \tan \alpha \tan \alpha} \\ &= \frac{1/5 + 1/5}{1 - (1/5)^2} \\ &= \frac{2/5}{24/25} \\ &= \frac{5}{12}.\end{aligned}$$

The second application of the identity follows similarly:

$$\begin{aligned}\tan(2\alpha + 2\alpha) &= \frac{\tan 2\alpha + \tan 2\alpha}{1 - \tan 2\alpha \cdot \tan 2\alpha} \\ &= \frac{5/12 + 5/12}{1 - (5/12)^2} \\ &= \frac{120}{119}.\end{aligned}$$

[18] Because $\tan \pi/4 = 1$ we can write

$$\frac{120}{119} = \frac{1 + \tan(\angle BQE)}{1 - \tan(\angle BQE)}$$

and then cross-multiply to reach

$$120(1 - \tan(\angle BQE)) = 119(1 + \tan(\angle BQE)).$$

Distributing and rearranging terms brings us to

$$239 \tan(\angle BQE) = 1,$$

from which the result $\tan(\angle BQE) = 1/239$ follows.

[19] The integral

$$\int_0^1 (1 - x^2)^{1/2} dx$$

gives us the area of a quarter of a unit circle because the curve

$$y = (1 - x^2)^{1/2} \quad \text{can be written} \quad x^2 + y^2 = 1,$$

which is the equation of a unit circle, and then integrating from $x = 0$ to $x = 1$ finds the area in just the first quadrant. We would then evaluate the integral via the substitution $x = \sin \alpha$ and the identity

$$\cos^2 \alpha = \frac{\cos(2\alpha) + 1}{2}.$$

[20] On the interval $x \in [0, \pi/2]$, we know that

$$0 \leq \sin x \leq 1,$$

and multiplying through by $\sin x$ gives us

$$0 \leq \sin^2 x \leq \sin x.$$

These two double inequalities combine to give us

$$\sin^2 x \leq \sin x \leq 1.$$

Now we multiply through by $\sin^{2n} x$ to reach the desired double inequality

$$\sin^{2n+2} x \leq \sin^{2n+1} x \leq \sin^{2n} x.$$

At no point do we change the directions of the inequalities, because we never multiplied by a negative number.

[21] For example, if $n = 1$ then we are flipping the coin two times, looking for the probability that we will flip one head and one tail. The $2^2 = 4$ possible outcomes are

$$HH, HT, TH, TT$$

and two of the four are the desired outcomes. So the probability P_2 of getting one heads and one tails in two flips is

$$P_2 = \frac{2}{4} = \frac{1}{2}.$$

Now if $n = 2$ so that we are flipping four times, then direct counting of the desired outcomes becomes more tedious, and we look for a more robust approach. We wish to count the number of ways that we can flip two heads and two tails. If we choose two of the four flips to be heads, then the remaining two flips must be tails, so all that we must do is count the number of ways that we can assign two of the four flips to be heads. The binomial coefficient

$$\binom{4}{2} = \frac{4!}{2!2!} = 6$$

does the job. By the same argument, the number of ways that we can assign n of $2n$ flips to be heads is

$$\binom{2n}{n} = \frac{(2n)!}{n!n!}.$$

[22] The claim

$$n! 2^n = 2 \times 4 \times 6 \times \cdots \times 2n$$

merely reflects a shuffling of the factors on the left. For example, when $n = 3$ we have

$$3!2^3 = (1 \cdot 2 \cdot 3) \cdot (2 \cdot 2 \cdot 2) = (1 \cdot 2) \cdot (2 \cdot 2) \cdot (3 \cdot 2) = 2 \cdot 4 \cdot 6.$$

[23] We reach (2.18) from

$$\frac{\pi}{2} \asymp \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \cdots \times \frac{2n}{2n-1} \times \frac{2n}{2n+1}$$

by ignoring the 1 in the first fraction and factoring $(2n+1)$ from the denominator, allowing us to write the expression in the form

$$\frac{2 \times 2 \times 4 \times 4 \times \cdots \times 2n \times 2n}{3 \times 3 \times 5 \times 5 \times \cdots \times (2n-1) \times (2n-1)} \cdot \frac{1}{2n+1}.$$

We may express this as

$$\left[\frac{2 \times 4 \times \cdots \times (2n)}{3 \times 5 \times \cdots \times (2n-1)} \right]^2 \frac{1}{2n+1}.$$

[24] Substituting $\pi/2$ for x in (2.20) yields

$$\frac{\sin \pi/2}{\pi/2} = \left(1 - \frac{1}{2}\right) \left(1 + \frac{1}{2}\right) \left(1 - \frac{1}{4}\right) \left(1 + \frac{1}{4}\right) \left(1 - \frac{1}{6}\right) \cdots$$

or, using $\sin \pi/2 = 1$ and then inverting both sides,

$$\frac{\pi}{2} = \frac{2}{1} \times \frac{2}{3} \times \frac{4}{3} \times \frac{4}{5} \times \frac{6}{5} \times \frac{6}{7} \times \cdots.$$

[25] Make the substitution

$$A = \frac{1}{(-\pi)(\pi)(-2\pi)(2\pi)(-3\pi)(3\pi) \cdots}$$

in (2.21) and divide by x to reach

$$\frac{\sin x}{x} = \frac{(x-\pi)(x+\pi)(x-2\pi)(x+2\pi)(x-3\pi)(x+3\pi) \cdots}{(-\pi)(\pi)(-2\pi)(2\pi)(-3\pi)(3\pi) \cdots}.$$

Pairing factors from numerator to denominator yields

$$\frac{\sin x}{x} = \frac{x-\pi}{-\pi} \cdot \frac{x+\pi}{\pi} \cdot \frac{x-2\pi}{-2\pi} \cdot \frac{x+2\pi}{2\pi} \cdot \frac{x-3\pi}{-3\pi} \cdot \frac{x+3\pi}{3\pi} \cdots$$

and then, dividing through each fraction by its denominator, we reach

$$\frac{\sin x}{x} = \left(1 - \frac{x}{\pi}\right) \left(1 + \frac{x}{\pi}\right) \left(1 - \frac{x}{2\pi}\right) \left(1 + \frac{x}{2\pi}\right) \left(1 - \frac{x}{3\pi}\right) \cdots.$$

[26] When $n = 0$, the result is clear. When $n = 1$, the right side of (2.24) equals

$$\begin{aligned}\frac{1}{m+1} - \frac{1}{m+2} &= \frac{m+2 - (m+1)}{(m+1)(m+2)} \\ &= \frac{1}{(m+1)(m+2)}.\end{aligned}$$

And when $n = 2$, we get

$$\frac{1}{m+1} - \frac{2}{m+2} + \frac{2}{2(m+3)},$$

which upon finding a least common denominator is equivalent to

$$\frac{(m+2)(m+3) - 2(m+1)(m+3) + (m+1)(m+2)}{(m+1)(m+2)(m+3)}.$$

The numerator simplifies to 2 once we expand and collect like terms. In the text, we write the numerator as $1 \cdot 2$ because of the factorial pattern that appears across the numerators.

[27] Euler provided the following derivation of the series for $y = \arcsin x$. The differential is

$$dy = \frac{dx}{\sqrt{1-x^2}} = (1-x^2)^{-1/2} dx,$$

and applying the generalized binomial theorem (2.23), Euler reached

$$\begin{aligned}dy &= \left(1 + \frac{(-\frac{1}{2})}{1}(-x^2) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{1 \cdot 2}(-x^2)^2 \right. \\ &\quad \left. + \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{1 \cdot 2 \cdot 3}(-x^2)^3 + \dots \right) dx \\ &= \left(1 + \frac{1}{2}x^2 + \frac{1 \cdot 3}{2 \cdot 4}x^4 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^6 + \dots \right) dx.\end{aligned}$$

Integrating yields

$$y = x + \frac{1}{2 \cdot 3}x^3 + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5}x^5 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 7}x^7 + \dots$$

and because $y = \arcsin x$, we are finished.

[28] We can illustrate the two implications by example. When $n = 2$, we have

$$\begin{aligned}f(x) &= \frac{x^2(p-qx)^2}{2!} \\ &= \frac{x^2(p^2 - 2pqx + q^2x^2)}{2!} \\ &= \frac{p^2}{2}x^2 - pqx^3 + \frac{q^2}{2}x^4.\end{aligned}$$

So the terms range in degree from $n = 2$ to $2n = 4$ as promised. Now

$$f'(x) = p^2x - 3pqx^2 + 2q^2x^3 \quad \text{and}$$

$$f''(x) = p^2 - 6pqx + 6q^2x^2.$$

The text claims that the first $n - 1$ derivatives will equal zero when evaluated at $x = 0$, and we see that this is true here because every term in $f'(x)$ contains a power of x . Second, the text claims that the denominator $2!$ in $f(x)$ will be fully canceled in $f''(x)$, which we see is indeed the case.

[29] For

$$f(x) = \frac{x^n(p - qx)^n}{n!}$$

we have

$$\begin{aligned} f((p/q) - x) &= \frac{((p/q) - x)^n (p - q((p/q) - x))^n}{n!} \\ &= \frac{((1/q)(p - qx))^n (qx)^n}{n!} \\ &= \frac{(p - qx)^n x^n}{n!} \\ &= f(x). \end{aligned}$$

[30] To show that $p^n/n!$ (which is positive) can be made as small as we wish (in particular, for our purposes, we wish it to be less than 1), let's write

$$0 < \frac{p^n}{n!} = \frac{p^p}{p!} \cdot \frac{p}{p+1} \cdot \frac{p}{p+2} \cdot \dots \cdot \frac{p}{n}.$$

Each fraction on the right (after the first two) is smaller than $p/(p+1)$, so we may write

$$0 < \frac{p^n}{n!} < \frac{p^p}{p!} \cdot \left(\frac{p}{p+1} \right)^{n-p}$$

and then choose n large enough to force the expression on the right below 1.

[31] First, note that no matter how we split the number 10 into n parts, the arithmetic mean of the parts is constant. To see this, split a positive number M into n parts, as in

$$a_1 + a_2 + \dots + a_n = M,$$

and multiply by $1/n$ to get

$$\frac{1}{n}(a_1 + a_2 + \dots + a_n) = \frac{M}{n}.$$

On the left we have the arithmetic mean of the n parts, while the right is a constant that is independent of how we chose those parts.

Now the arithmetic geometric mean inequality states that

$$\frac{1}{n}(a_1 + a_2 + \cdots + a_n) \geq (a_1 a_2 \cdots a_n)^{1/n}$$

with equality precisely when $a_1 = a_2 = \cdots = a_n$, so if (as we just observed) the left side is constant, then the product on the right will be maximized when the inequality is an equality. This happens when all of the parts are equal.

[32] Rearrange the factors in

$$\left(1 + \frac{x}{n}\right)^n = 1 + \frac{n}{1} \cdot \frac{x}{n} + \frac{n}{1} \cdot \frac{n-1}{2} \cdot \frac{x^2}{n^2} + \frac{n}{1} \cdot \frac{n-1}{2} \cdot \frac{n-2}{3} \cdot \frac{x^3}{n^3} + \cdots$$

to reach

$$\left(1 + \frac{x}{n}\right)^n = 1 + \frac{n}{n} \cdot \frac{x}{1!} + \frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{x^2}{2!} + \frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdot \frac{x^3}{3!} + \cdots$$

Divide each term in each fraction (that contains n) by n to obtain

$$\left(1 + \frac{x}{n}\right)^n = 1 + \frac{x}{1!} + \frac{1 - (1/n)}{1} \cdot \frac{x}{2!} + \frac{1 - (1/n)}{1} \cdot \frac{1 - (2/n)}{1} \cdot \frac{x}{3!} + \cdots$$

As n approaches infinity, all terms on the right in parentheses vanish.

[33] We will show that

$$\frac{d}{dx} e^x = e^x$$

by first proving that

$$\frac{d}{dx} \ln x = \frac{1}{x}.$$

The limit definition of the derivative

$$\frac{d}{dx} f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

applied to $f(x) = \ln x$ lets us begin

$$\frac{d}{dx} \ln x = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h},$$

and we may combine the logarithms in the numerator to write

$$\frac{d}{dx} \ln x = \lim_{h \rightarrow 0} \left[\frac{1}{h} \ln \left(\frac{x+h}{x} \right) \right].$$

For a strictly strategic purpose, we choose to write $1/h$ as $(1/x)(x/h)$, and so

$$\frac{d}{dx} \ln x = \lim_{h \rightarrow 0} \left[\left(\frac{1}{x} \cdot \frac{x}{h} \right) \ln \left(1 + \frac{h}{x} \right) \right].$$

The fraction $1/x$ does not change as h approaches zero, so it may be brought outside the limit as a constant, and its partner x/h pulls up into the exponent; hence,

$$\frac{d}{dx} \ln x = \frac{1}{x} \lim_{h \rightarrow 0} \left[\ln \left(1 + \frac{h}{x} \right)^{x/h} \right].$$

The natural logarithm function is one-to-one and continuous, so we may exchange it with the limit to reach

$$\frac{d}{dx} \ln x = \frac{1}{x} \ln \left[\lim_{h \rightarrow 0} \left(1 + \frac{h}{x} \right)^{x/h} \right]. \quad (9)$$

We will recognize the limit on the right if we let $n = x/h$, so that $n \rightarrow \infty$ as $h \rightarrow 0$ and the limit becomes

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n.$$

Section 3.4 offers a proof (based on the arithmetic geometric mean inequality) that this limit equals e . Thus from (9) we see that

$$\frac{d}{dx} \ln x = \frac{1}{x} \ln e = \frac{1}{x}.$$

With this established, let $y = e^x$ so that $x = \ln y$ and use implicit differentiation (with respect to x) on the latter to get

$$1 = \frac{1}{y} \cdot \frac{dy}{dx} \quad \text{or} \quad \frac{dy}{dx} = y,$$

and because $y = e^x$, we have our desired result

$$\frac{d}{dx} e^x = e^x.$$

[34] We are proving Bernoulli's inequality:

Claim. If m is a natural number and $y \geq -1$, then

$$(1 + y)^m \geq 1 + my.$$

Our induction proof begins by showing that the claim holds for $m = 1$ (and we did this in the section itself). Now we assume that the claim is true for a fixed natural number m and show that via this assumption we can reach the next case:

To show: If $m + 1$ is a natural number and $y \geq -1$, then

$$(1 + y)^{m+1} \geq 1 + (m + 1)y.$$

Starting with the left side, we may write

$$\begin{aligned}(1 + y)^{m+1} &= (1 + y)^m(1 + y) \\ &\geq (1 + my)(1 + y)\end{aligned}$$

by first factoring and then by appealing to our assumption that the claim holds for m . It is here that we require $(1 + y) \geq 0$ (so $y \geq -1$) in order that the inequality be true. Now we may expand the last expression to see that

$$\begin{aligned}(1 + y)^{m+1} &\geq 1 + (m + 1)y + my^2 \\ &\geq 1 + (m + 1)y,\end{aligned}$$

which is what we set out to show.

[35] From $x \geq 0$ we know that $x/e \geq 0$, and then subtracting 1 gives us

$$\frac{x}{e} - 1 \geq -1 \quad \text{or} \quad \frac{x - e}{e} \geq -1.$$

[36] We may write

$$e^{(x-e)/e} \geq 1 + \frac{x - e}{e} \quad \text{as} \quad e^{x/e} e^{-1} \geq \frac{e + x - e}{e} = xe$$

and then multiply through by e to reach

$$e^{x/e} \geq x.$$

Taking the natural logarithm of both sides, we obtain

$$\frac{x}{e} \geq \ln x \quad \text{or} \quad \frac{1}{e} \geq \frac{\ln x}{x}.$$

Because $x > 0$, we do not need to reverse the inequality in this last step. Now exponentiating both sides leads to the desired result.

[37] I recommend that you use technology to take a look at the first few members of the family of curves

$$y_n = \frac{1}{n!} x^n e^{-x},$$

namely,

$$y_0 = \frac{1}{0!} x^0 e^{-x} = e^{-x},$$

$$y_1 = \frac{1}{1!} x^1 e^{-x} = x e^{-x},$$

$$y_2 = \frac{1}{2!} x^2 e^{-x} = \frac{x^2 e^{-x}}{2},$$

and so on. Seeing the family of curves may help you follow the forthcoming argument.

[38] Recall that we are only interested in determining if the derivative

$$\frac{d}{dx} x^n e^{-x} = x^{n-1} e^{-x} (n - x)$$

equals zero on the interval $x \in (0, 1)$. The first factor x^{n-1} can only equal zero if $x = 0$, and the second factor e^{-x} can never equal zero. The third factor $(n - x)$ is always positive because n is a natural number. Because none of the three factors can equal zero on the interval $x \in (0, 1)$, we conclude that the curve has no critical points (and therefore no maximums) on that interval.

[39] We calculate

$$a_1 = \int_0^1 x e^{-x} dx$$

using integration by parts with

$$u = x \quad \text{so} \quad du = dx,$$

$$dv = e^{-x} dx \quad \text{so} \quad v = -e^{-x},$$

so

$$a_1 = [-x e^{-x}]_0^1 + \int_0^1 e^{-x} dx = -\frac{1}{e} + \int_0^1 e^{-x} dx.$$

[40] For example, suppose that $e = 19/7$. Then we would split series (3.17) at the $1/7!$ term:

$$\begin{aligned} e &= \sum_{k=0}^7 \frac{1}{k!} + \sum_{k=8}^{\infty} \frac{1}{k!} \\ &= \left(\frac{1}{0!} + \frac{1}{1!} + \cdots + \frac{1}{7!} \right) + \left(\frac{1}{8!} + \frac{1}{9!} + \cdots \right). \end{aligned}$$

In preparation for what follows in the argument, note that multiplying each term by $7!$ turns each term in the first group to an integer.

[41] The geometric series

$$\frac{1}{q+1} + \frac{1}{(q+1)^2} + \frac{1}{(q+1)^3} + \cdots$$

has leading term $a = 1/(q+1)$ and common ratio $r = 1/(q+1)$. Thus the series sums to

$$S = \frac{a}{1-r} = \frac{\frac{1}{q+1}}{1 - \frac{1}{q+1}} = \frac{1}{q+1-1} = \frac{1}{q}.$$

[42] To see why

$$\int_1^n \ln x \, dx \approx \ln 1 + \ln 2 + \cdots + \ln n,$$

redraw Figure 3.1 but use rectangles in place of the trapezoids.

[43] Exponentiating

$$L - 1 \asymp n \ln n - n - \ln n! + (1/2) \ln n$$

yields

$$e^{L-1} \asymp e^{n \ln n - n - \ln n! + (1/2) \ln n}$$

and the right side can be written

$$\frac{(e^{\ln n})^n (e^{\ln n})^{1/2}}{e^n e^{\ln n!}} \quad \text{or} \quad \frac{n^n \sqrt{n}}{n! e^n}.$$

Thus,

$$e^{L-1} \asymp \frac{n^n \sqrt{n}}{n! e^n}.$$

[44] The expression

$$\frac{1}{\sqrt{\pi(n+1/2)}} \cdot \frac{n}{\sqrt{2n}}$$

can be written

$$\frac{n}{\sqrt{2\pi n(n+1/2)}} = \sqrt{\frac{n^2}{2\pi n^2 + \pi n}},$$

and then we can divide through each term by n^2 to reach

$$\sqrt{\frac{1}{2\pi + \pi/n}}.$$

Letting $n \rightarrow \infty$ gives us $1/\sqrt{2\pi}$.

[45] The step

$$\lim_{n \rightarrow \infty} S_{2n} = \lim_{n \rightarrow \infty} S_n$$

is justified by a theorem from the theory of limits: If a sequence converges to a limit, then every subsequence converges to the same limit. The sequence $\{S_{2n}\}$ simply picks out the even-numbered entries of $\{S_n\}$ and is therefore a subsequence.

[46] From (3.24) we may write

$$e^{L-1} = \frac{1}{\sqrt{2\pi}}$$

and, taking the natural logarithm of each side, we find that

$$L - 1 = \ln(\sqrt{2\pi})^{-1} \quad \text{or} \quad L = 1 - \ln(\sqrt{2\pi}).$$

[47] Let Q denote the infinite tail end of the right side of

$$\frac{1}{1 + \frac{1}{e-1}} = \frac{1}{1 + \frac{1}{1 + \frac{2}{1 + \frac{3}{2 + \frac{3}{3 + \cdots}}}}}$$

so that we have

$$\frac{1}{1 + \frac{1}{e-1}} = \frac{1}{1 + \frac{1}{1+Q}} \quad \text{where} \quad Q = \frac{2}{2 + \frac{3}{3 + \cdots}}.$$

Rewriting this expression as

$$\frac{1}{1 + \frac{1}{e-1}} = \frac{1}{1 + \frac{1}{(2+Q)-1}}$$

shows that $e = 2 + Q$, which yields the desired result.

[48] Expanding

$$(u+v)^3 = 6(u+v) + 10$$

yields

$$u^3 + 3u^2v + 3uv^2 + v^3 = 6u + 6v + 10.$$

Subtracting $6u$ and $6v$ gives us

$$u^3 + v^3 + 3u^2v - 6u + 3uv^2 - 6v = 10$$

or

$$u^3 + v^3 + (3uv - 6)u + (3uv - 6)v = 10,$$

and factoring $(3uv - 6)$ from the appropriate terms gives us the desired result.

[49] Substituting the first of the two solutions $u = (5 + \sqrt{17})^{1/3}$ into $u^3 + v^3 = 10$ gives us

$$\left((5 + \sqrt{17})^{1/3}\right)^3 + v^3 = 10 \quad \text{or} \quad v^3 = 10 - (5 + \sqrt{17}),$$

from which we conclude that

$$v = (5 - \sqrt{17})^{1/3}.$$

In the same way, starting with the second solution $u = (5 - \sqrt{17})^{1/3}$ leads to $v = (5 + \sqrt{17})^{1/3}$. So because $x = u + v$, we only get one solution for x , namely

$$x = (5 + \sqrt{17})^{1/3} + (5 - \sqrt{17})^{1/3}.$$

[50] As an example of depressing a cubic, let's rid the polynomial

$$4x^3 + 8x^2 + x + 2$$

of its x^2 term by first dividing by the leading coefficient 4 to obtain

$$x^3 + 2x^2 + \frac{1}{4}x + \frac{1}{2}$$

and then identifying from this that $b = 2$. So to shift the curve appropriately, we substitute $x - 2/3$ for x and get

$$\left(x - \frac{2}{3}\right)^3 + 2\left(x - \frac{2}{3}\right)^2 + \frac{1}{4}\left(x - \frac{2}{3}\right) + \frac{1}{2} \quad \text{or} \quad x^3 - \frac{13}{12}x + \frac{25}{27}$$

as desired. Now we can use del Ferro's technique to find the zeros of this shifted cubic polynomial, and shift them by $2/3$ to get the zeros of the original cubic polynomial.

[51] Multiply both sides of

$$3p > \frac{a^3 - 2q}{a}$$

by a (thus reversing the inequality because $a < 0$) to get

$$3ap < a^3 - 2q.$$

From $p = a^2$ we have $ap = a^3$ and so

$$3ap < ap - 2q \quad \text{or} \quad 2ap < -2q.$$

Dividing by -2 yields

$$-ap > q$$

and we square both sides to reach

$$a^2 p^2 > q^2.$$

Using $p = a^2$ again, we soon reach the desired result

$$q^2 - p^3 < 0.$$

[52] My thanks to Chenchen Zhao and Ben Oltsik for showing me this alternative solution. In it, we will use the customary symbol i to represent $\sqrt{-1}$. We wish to simplify

$$(2 + 2i)^{1/3} + (2 - 2i)^{1/3}.$$

Let

$$x = (2 + 2i)^{1/3}, \text{ so } x^3 = 2 + 2i, \text{ and}$$

$$y = (2 - 2i)^{1/3}, \text{ so } y^3 = 2 - 2i.$$

The question becomes: what is $x + y$? This sum appears as a factor on the right of the identity

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2). \quad (10)$$

On the left,

$$x^3 + y^3 = 2 + 2i + 2 - 2i = 4,$$

while on the right,

$$xy = (2 + 2i)^{1/3}(2 - 2i)^{1/3} = ((2 + 2i)(2 - 2i))^{1/3} = (4 - (-4))^{1/3} = 2,$$

and

$$x^2 = ((2 + 2i)^2)^{1/3} = (4 + 8i - 4)^{1/3} = 2i^{1/3},$$

$$y^2 = ((2 - 2i)^2)^{1/3} = (4 - 8i - 4)^{1/3} = -2i^{1/3}.$$

Thus, from (10) we have

$$4 = (x + y)(2i^{1/3} - 2 - 2i^{1/3}),$$

or $x + y = -2$ as Bombelli found.

[53] From Figure 4.4, we have

$$\cos \theta = \frac{a}{r} \quad \text{or} \quad a = r \cos \theta,$$

$$\sin \theta = \frac{b}{r} \quad \text{or} \quad b = r \sin \theta,$$

so

$$a + bi = r \cos \theta + r \sin \theta \cdot i = r(\cos \theta + i \sin \theta).$$

[54] Before we prove anything about how to perform multiplication in the complex plane, we will observe what happens for an example. As a vector in the complex plane, the number $1 + i$ has magnitude

$$|1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

and creates angle $\pi/4$ with the horizontal axis. When we square $1 + i$ we get

$$(1 + i)(1 + i) = 1 + 2i - 1 = 2i.$$

The complex number $2i$ has magnitude 2 and angle $\pi/2$. It is no coincidence that

$$|1 + i| \cdot |1 + i| = \sqrt{2} \cdot \sqrt{2} = 2 = |2i|, \text{ and}$$

$$\angle(1 + i) + \angle(1 + i) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} = \angle(2i).$$

Let's prove that we add the angles of two complex numbers when we find their product. The angle θ in Figure 4.4 can be expressed

$$\theta = \arctan b/a.$$

Suppose we have a second complex number $c + di$ with its angle $\alpha = \arctan d/c$. Then

$$\begin{aligned} \theta + \alpha &= \arctan b/a + \arctan d/c \\ &= \arctan \left(\frac{ad + bc}{ac - bd} \right), \end{aligned} \tag{11}$$

using the identity

$$\arctan \alpha + \arctan \beta = \arctan \left(\frac{\alpha + \beta}{1 - \alpha\beta} \right)$$

for the last step. Now

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i \tag{12}$$

makes the angle

$$\arctan \left(\frac{ad + bc}{ac - bd} \right)$$

with the horizontal axis, and this matches with (11). Thus, when we multiply two complex numbers, we add the angles of their corresponding vectors.

Now why do we multiply the lengths of $a + bi$ and $c + di$ when we find their product? From Figure 4.4 and the Pythagorean theorem, we see that

$$|a + bi| \cdot |c + di| = (a^2 + b^2)^{1/2} (c^2 + d^2)^{1/2} \tag{13}$$

and from (12) we have

$$\begin{aligned}
 |(a + bi)(c + di)| &= [(ac - bd)^2 + (ad + bc)^2]^{1/2} \\
 &= [(ac)^2 + (bd)^2 + (ad)^2 + (bc)^2]^{1/2} \\
 &= [(a^2 + b^2)(c^2 + d^2)]^{1/2},
 \end{aligned}$$

from which we can reach (13). So we multiply the magnitudes of two complex numbers when we find their product, rather than add as we do with the angles.

[55] From $\cos(\pi/6) = \sqrt{3}/2$ and the identity

$$\cos\left(\frac{A}{2}\right) = \sqrt{\frac{1 + \cos A}{2}}$$

we know that

$$\sqrt{2} \cos(\pi/12) = \sqrt{2} \cdot \sqrt{\frac{1 + \sqrt{3}/2}{2}} = \sqrt{\frac{2 + \sqrt{3}}{2}} = \frac{\sqrt{4 + 2\sqrt{3}}}{2}. \quad (14)$$

Now (as an aside) we note that

$$(1 + \sqrt{3})(1 + \sqrt{3}) = 4 + 2\sqrt{3} \quad \text{so} \quad 1 + \sqrt{3} = \sqrt{4 + 2\sqrt{3}},$$

which lets us write the last expression in (14) as

$$\frac{1 + \sqrt{3}}{2}$$

as desired. The result for $\sqrt{2} \sin(\pi/12)$ follows similarly.

Historical note. An expression like

$$\sqrt{4 + 2\sqrt{3}}$$

is called a *nested radical*, and writing it in a simpler form is possible only in certain circumstances. Even then, the process is a bit tricky. Because nested radicals appear in a variety of problems of historical importance (such as Liu Hui's approximation of π in section 2.1, or the solution (4.5) to the depressed cubic), simplifying nested radicals has attracted attention for centuries. Progress continues to this day; for example, in 1992, **Susan Landau** (United States, born 1954) discovered the first algorithm that can simplify a general nested radical.

[56] We use the half-angle identity

$$\sin\left(\frac{A}{2}\right) = \pm \sqrt{\frac{1 - \cos A}{2}} \quad \text{with} \quad A = \frac{17\pi}{6}$$

and, because A is in the second quadrant, we choose the negative version. So

$$\begin{aligned}\sin\left(\frac{17\pi}{12}\right) &= -\sqrt{\frac{1 - \cos(17\pi/6)}{2}} \\ &= -\sqrt{\frac{1 - \cos(5\pi/6)}{2}} \\ &= -\sqrt{\frac{1 + (\sqrt{3}/2)}{2}} \\ &= -\frac{1}{2}\sqrt{2 + \sqrt{3}}.\end{aligned}$$

Similarly, using the half-angle identity for cosine, we find that

$$\cos\left(\frac{17\pi}{12}\right) = -\frac{1}{2}\sqrt{2 - \sqrt{3}}.$$

Thus,

$$\begin{aligned}\sqrt{2}(\cos(17\pi/12) + i\sin(17\pi/12)) \\ &= \sqrt{2}\left(-\frac{1}{2}\sqrt{2 - \sqrt{3}} - \frac{i}{2}\sqrt{2 + \sqrt{3}}\right) \\ &= -\frac{\sqrt{2}}{2}\left(\sqrt{2 - \sqrt{3}} + i\sqrt{2 + \sqrt{3}}\right).\end{aligned}$$

[57] Add the fractions on the right to get

$$\frac{(A + Bi)(1 - ix) + (C + Di)(1 + ix)}{(1 + ix)(1 - ix)}.$$

Expanding and collecting like terms in the numerator yields

$$\frac{[(A + C) + (B + D)i] + [(B - D) + (C - A)i]x}{(1 + ix)(1 - ix)}. \quad (15)$$

We need to find A, B, C, D so that the numerator equals 1, or equivalently,

$$[1 + 0i] + [0 + 0i]x. \quad (16)$$

Matching the coefficients within brackets in (15) and (16), we obtain the system of equations

$$\begin{array}{ll}A + C = 1 & B + D = 0 \\ C - A = 0 & B - D = 0,\end{array}$$

which has solutions $A = C = 1/2$ and $B = D = 0$.

[58] Substituting $z = \tan \alpha$ and $dz = \sec^2 \alpha \, d\alpha$ in

$$\theta = i \int \frac{dz}{\sqrt{1+z^2}} \quad \text{gives us} \quad \theta = i \int \frac{\sec^2 \alpha \, d\alpha}{\sqrt{1+\tan^2 \alpha}}$$

and then the identity $\sec^2 \alpha = 1 + \tan^2 \alpha$ leads to

$$\theta = i \int \sec \alpha \, d\alpha.$$

[59] Multiply within the logarithm of

$$\theta = i \ln (\sqrt{1-y^2} - iy)$$

by $(\sqrt{1-y^2} + iy)$ over itself to get

$$\theta = i \ln \left(\frac{(\sqrt{1-y^2} - iy)(\sqrt{1-y^2} + iy)}{\sqrt{1-y^2} + iy} \right).$$

The numerator, if expanded, simplifies to 1. Thus we may write

$$\begin{aligned} \theta &= i \ln \left[(\sqrt{1-y^2} + iy)^{-1} \right] \\ &= -i \ln (\sqrt{1-y^2} + iy). \end{aligned}$$

[60] The substitution $x = \pi/2$ in

$$e^{i\theta} = \cos \theta + i \sin \theta$$

yields

$$e^{i(\pi/2)} = \cos (\pi/2) + i \sin (\pi/2) = 0 + i, \quad \text{so} \quad e^{i(\pi/2)} = i. \quad (17)$$

Taking the natural logarithm of each side, and then solving for π , gives us

$$\pi = \frac{2 \ln i}{i},$$

whereas raising each side of (17) to the power i leads to

$$e^{i^2(\pi/2)} = i^i \quad \text{or} \quad i^i = e^{-\pi/2}.$$

[61] Squaring and multiplying by x^2 converts

$$\frac{y}{x} = \sqrt{4\varphi + 3} \quad \text{to} \quad y^2 = x^2(4\varphi + 3),$$

and using $x^2 + y^2 = 1$ to replace y^2 gives us

$$1 - x^2 = x^2(4\varphi + 3) \quad \text{or} \quad 1 = x^2(4\varphi + 4) = 1 = 4x^2(\varphi + 1).$$

From (1.8) we can write

$$1 = 4x^2\varphi^2 \quad \text{or} \quad x = \frac{1}{2\varphi}$$

as desired. From

$$y = \frac{\sqrt{4\varphi + 3}}{2\varphi} = \sqrt{\frac{4\varphi + 3}{4\varphi^2}}$$

we may simplify by multiplying both numerator and denominator by $2 - \varphi$ to get

$$y = \sqrt{\frac{(4\varphi + 3)(2 - \varphi)}{(4\varphi^2)(2 - \varphi)}} = \sqrt{\frac{\varphi + 2}{4}} = \frac{\sqrt{\varphi + 2}}{2}.$$

[62] The half-angle identity for cosine is

$$\cos\left(\frac{\alpha}{2}\right) = \sqrt{\frac{\cos \alpha + 1}{2}},$$

so with $\alpha = 2\pi/5$ we get

$$\cos\left(\frac{\pi}{5}\right) = \sqrt{\frac{\cos(2\pi/5) + 1}{2}}.$$

We know that $\cos(2\pi/5) = 1/(2\varphi)$ from (4.26), and so, substituting and simplifying, we find that

$$\begin{aligned} \cos\left(\frac{\pi}{5}\right) &= \sqrt{\frac{1/(2\varphi) + 1}{2}} \\ &= \sqrt{\frac{1 + 2\varphi}{4\varphi}} \\ &= \sqrt{\frac{\varphi^3}{4\varphi}} \quad \text{from (1.8)} \\ &= \varphi/2. \end{aligned}$$

[63] To fill in the $p = 1, k = 3$ entry of Table A.1, we would find the area under the curve $y = (1 - x)^3$ inside the unit square. As in (A.1), we expand:

$$\begin{aligned} \text{area under } (1 - x)^3 &= \text{area under } (1 - 3x + 3x^2 - x^3) \\ &= 1 - 3 \cdot \frac{1}{2} + 3 \cdot \frac{1}{3} - \frac{1}{4} \\ &= \frac{1}{4}. \end{aligned}$$

So we put the reciprocal of $1/4$ in the appropriate spot in the table.

As another example, we fill in the $p = 2$, $k = 2$ entry by calculating

$$\begin{aligned}\text{area under } (1 - x^{1/2})^2 &= \text{area under } (1 - 2x^{1/2} + x) \\ &= 1 - 2 \cdot \frac{1}{3/2} + \frac{1}{2} \\ &= \frac{1}{6}.\end{aligned}$$

The reciprocal 6 therefore belongs in the corresponding spot in the table.

[64] For example, we use Wallis's equation

$$\frac{\# \text{ dots total}}{\# \text{ dots along base}} = \frac{1}{2}(n + 1)$$

with $n = 5/2$ (where n is the number of dots along the base) to find that

$$\frac{\# \text{ dots total}}{5/2} = \frac{1}{2} \left(\frac{5}{2} + 1 \right) \quad \text{and so} \quad \# \text{ dots total} = \frac{35}{8}.$$

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